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Replicon Compositions and Methods of Using Same for the Treatment of Diseases

Abstract

The present invention embraces compositions comprising at least two RNA replicons (self-amplifying RNA vectors (saRNAs or rRNAs)) that can be replicated by a replicase of a self-replicating virus, e.g., a replicase of alphavirus origin. Of the at least two replicons, at least one of which optionally comprises an open reading frame encoding for the RNA-dependent RNA polymerase or replicase that is able to replicate each of the at least two replicons. Further, each replicon comprises an open reading frame encoding for different antigens of interest, e.g., different antigens derived from the same or from different pathogenic organisms, for example the glycoprotein and nucleoprotein of Ebola virus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Phase filing of International Patent Application No. PCT/EP2023/061483, filed May 2, 2023 and which claims priority to International Patent Application No. PCT/EP2022/051735, filed May 2, 2022, the entireties of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention embraces compositions comprising at least two RNA replicons (self-amplifying RNA vectors (saRNAs)) that can be replicated by an RNA-dependent RNA polymerase (replicase) of a self-replicating virus, e.g., a replicase of alphavirus origin. Each replicon comprises an open reading frame encoding for different antigens of interest, e.g., different antigens derived from the same or from different pathogenic organisms, for example the glycoprotein and nucleoprotein of Ebola virus. The compositions also comprise a replicase that is able to replicate each replicon. The replicase can be provided by being encoded by another open reading frame comprised in one or both replicons and/or by another RNA molecule having an open reading frame encoding the replicase but which third RNA molecule is not able to be replicated by the encoded replicase (non-replicative RNA).

BACKGROUND

[0003] Recently, mRNA-based vaccines proved their immunogenicity in clinical studies to combat the Covid-19 epidemic. These RNA vaccines are highly effective and induce very strong T cell immune responses and high levels of neutralizing antibodies (Walsh et al., 2020, N Engl J Med

383:2439-2450; Sahin et al., 2020, *Nature* 586:594-599). These approved RNA vaccines require 30 to 100 µg RNA per dose, and two consecutive doses spaced by several weeks (prime-boost regimen). This culminates in 60 to 200 g RNA needed to immunize 1 million people. A dose reduction to less than 1 µg would therefore have great impact on the production time needed to supply the population with a vaccine against a novel pathogen. [0004] A vaccine approach under investigation that promises to achieve a significant dose reduction is to use self-amplifying RNA (saRNA). saRNA can be engineered from alphaviral genomes by replacing alphaviral structural genes with antigens against which an immune response is desired. saRNA can encode the alphaviral replicase which harbors all enzymatic function to replicate the saRNA molecule, thus leading to an amplification of the input vaccine amount.

[0005] Alphaviruses are typical representatives of enveloped positive-stranded RNA viruses. The hosts of alphaviruses include a wide range of organisms, comprising insects, fish and mammals, such as domesticated animals and humans. Alphaviruses replicate in the cytoplasm of infected cells (for review of the alphaviral life cycle see José et al., 2009, *Future Microbiol.* 4:837-856). The total genome length of many alphaviruses typically ranges between 11,000 and 12,000 nucleotides, and the genomic RNA typically has a 5'-cap, and a 3' poly(A) tail. The genome of alphaviruses encodes non-structural proteins (involved in transcription, modification and replication of viral RNA and in protein modification) and structural proteins (forming the virus particle). There are typically two open reading frames (ORFs) in the genome. The four non-structural proteins (nsP1-nsP4) are typically encoded together by a first ORF beginning near the 5' terminus of the genome, while alphavirus structural proteins are encoded together by a second ORF which is found downstream of the first ORF and extends near the 3' terminus of the genome. Typically, the first ORF is larger than the second ORF, the ratio being roughly 2:1.

[0006] In cells infected by an alphavirus, only the non-structural proteins are translated from the genomic RNA, while the structural proteins are translatable from a subgenomic transcript, which is an RNA molecule that resembles eukaryotic messenger RNA (mRNA; Gould et al., 2010, *Antiviral Res.* 87:111-124). Following infection, i.e., at early stages of the viral life cycle, the (+) stranded genomic RNA directly acts like a messenger RNA for the translation of the open reading frame encoding the non-structural poly-protein (nsP1234). In some alphaviruses, there is an opal stop codon between the coding sequences of nsP3 and nsP4: polyprotein P123, containing nsP1, nsP2, and nsP3, is produced when translation terminates at the opal stop codon, and polyprotein P1234, containing in addition nsP4, is produced upon readthrough of this opal codon (Strauss & Strauss, 1994, *Microbiol. Rev.* 58:491-562; Rupp et al., 2015, *J. Gen. Virology* 96:2483-2500). nsP1234 is autoproteolytically cleaved into the fragments nsP123 and nsP4. The polypeptides nsP123 and nsP4 associate to form the (-) strand replicase complex that transcribes (-) stranded RNA, using the (+) stranded genomic RNA as template. Typically, at later stages, the nsP123 fragment is completely cleaved into individual proteins nsP1, nsP2 and nsP3 (Shirako & Strauss, 1994, *J. Virol.* 68:1874-1885). All four proteins assemble to form the (+) strand replicase complex that synthesizes new (+) stranded genomes and subgenomic RNA, using the (-) stranded complement of genomic RNA as template (Kim et al., 2004, *Virology* 323:153-163, Vasiljeva et al., 2003, *J. Biol. Chem.* 278:41636-41645).

[0007] In infected cells, subgenomic RNA as well as new genomic RNA is provided with a 5'-cap by nsP1 (Pettersson et al., 1980, *Eur. J. Biochem.* 105:435-443; Rozanov et al., 1992, *J. Gen. Virology* 73:2129-2134), and provided with a poly-adenylate [poly(A)] tail by nsP4 (Rubach et al., 2009, *Virology* 384:201-208). Thus, both subgenomic RNA and genomic RNA resemble messenger RNA (mRNA).

[0008] The synthesis of alphaviral RNA is also regulated by cis-acting RNA elements, including four conserved sequence elements (CSEs; Strauss & Strauss, 1994, *Microbiol. Rev.* 58:491-562; and Frolov, 2001, *RNA* 7:1638-1651). The alphavirus genome comprises four CSEs which are understood to be important for viral RNA replication in the host cell. CSE 1, located at or near the 5' end of the virus genome, is believed to function as a promoter for (+) strand synthesis from (-) strand templates. CSE 2, located downstream of CSE 1 but still close to the 5' end of the genome within the coding sequence for nsP1 is thought to act as a promoter for initiation of (-) strand synthesis from a genomic RNA template (note that the subgenomic RNA transcript, which does not comprise CSE 2, does not function as a template for (-) strand synthesis). CSE 3 is located in the junction region between the coding sequence for the non-structural and structural proteins and acts as core promoter for the efficient transcription of the subgenomic transcript. Finally, CSE 4, which is located just upstream of the poly(A) sequence in the 3' untranslated region of the alphavirus genome, is understood to function as a core promoter for initiation of (-) strand synthesis (José et al., 2009, *Future Microbiol.* 4:837-856). CSE 4 and the poly(A) tail of the alphavirus are understood to function together for efficient (-) strand synthesis (Hardy & Rice, 2005, *J. Virol.* 79:4630-4639).

[0009] Alphavirus-derived vectors have been proposed for delivery of foreign genetic information into target cells or target organisms. In simple approaches, the open reading frame encoding alphaviral structural proteins is replaced by an open reading frame encoding a protein of interest. Alphavirus-based trans-replication systems rely on alphavirus nucleotide sequence elements on two separate nucleic acid molecules (RNA molecules): one RNA molecule encodes a viral replicase (typically as poly-protein nsP1234), and the other RNA molecule is capable of being replicated by said replicase in trans (hence the designation trans-replication and/or nano-transreplicon system). Trans-replication requires the presence of both these nucleic acid molecules in a given host cell. The nucleic acid molecule capable of being replicated by the replicase in trans must comprise certain alphaviral sequence elements to allow recognition and RNA synthesis by the alphaviral replicase.

[0010] The outbreak of the Ebola virus (EBOV) in West Africa from 2014 to 2016 as well as the currently ongoing SARS-CoV-2 pandemic teach us that emerging and re-emerging viruses can suddenly turn into a global calamity that can require millions of vaccine doses within a few months. Due to increasing globalization and access to rural areas, the incidence of emerging and re-emerging pathogens is increasing (Murray et al., 2015, *Proc. Natl. Acad. Sci. USA* 112:12746-12751).

[0011] Thus, there remains an urgent need for vaccines that can be quickly developed and allow for the treatment of diseases such as infectious diseases. The present invention fulfills such need.

SUMMARY

[0012] The present invention generally relates to compositions comprising at least two replicable RNA molecules (replicons), each comprising a first open-reading frame (ORF) encoding at least one peptide or protein comprising an antigen or epitope suitable to induce an immune response against the antigen or epitope when administered to a subject; wherein the at least one peptide or protein encoded by one of the replicable RNA molecules is different from the at least one peptide or protein encoded by the other replicable RNA molecule, and, optionally, wherein at least one of the replicable RNA molecules further comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules in the composition. Optionally, the replicase capable of replicating in cis or in trans the replicable RNA molecules in the composition can be encoded by another (third) RNA molecule comprised within the composition, which RNA molecule can be replicated by the encoded replicase, but preferably is not able to be replicated by the encoded replicase.

[0013] The present invention is based, in part, on the observation that administering a non-DNA-based vaccine composition of at least two antigens from a pathogenic organism encoded by at least two replicons can induce a specific antibody response, as well as CD4+ and/or CD8+ T cell responses to the antigens.

[0014] In one aspect, the present invention is directed to a composition comprising at least two replicable RNA molecules, each comprising a first open-reading frame (ORF) encoding at least one peptide or protein comprising an antigen or epitope suitable to induce an immune response against the antigen or epitope when administered to a subject; wherein the at least one peptide or protein encoded by one of the replicable RNA molecules is different from the at least one peptide or protein encoded by the other replicable RNA molecule, and optionally wherein at least one of the replicable RNA molecules further comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules.

[0015] As used in certain embodiments, the term "different", in the context of the different peptides or proteins encoded by the first and second replicable RNA molecules, does not encompass the same peptide or protein derived from different strains of a microorganism even though the amino acid sequences of the encoded peptides or proteins may be different. Thus, in certain embodiments, the encoded peptides or proteins are not

considered “different” solely due to amino acid sequence differences. In an embodiment, the different peptides or proteins are not the same peptide or protein with amino acid sequence differences. In an embodiment, the different peptides or proteins have a different function or are derived from peptides or proteins having different functions, for example, a viral nucleoprotein and a viral protein expressed on the surface of the virus, such as a glycoprotein. In an embodiment, the different peptides or proteins are encoded by separate open reading frames in the genomes of the pathogen, e.g., virus.

[0016] The labeling of the first and second open reading frame does not necessarily indicate that the first open reading frame is 5' to the second open reading frame in the replicable RNA molecule. In an embodiment, the “second” open reading frame encoding the replicase in the replicable RNA molecule is 5' of the open reading frame encoding the at least one peptide or protein.

[0017] In an embodiment, the composition can further comprise a third RNA molecule encoding the replicase capable of replicating in cis or in trans the replicable RNA molecules and/or the third RNA molecule. In an embodiment, the third RNA molecule is replicable. In an embodiment, the third RNA molecule is not replicable, i.e., is a non-replicative RNA molecule. In an embodiment, the composition comprises the at least two replicable RNA molecules, each of which does not encode the replicase, and a third non-replicative RNA molecule encoding the replicase. In an embodiment, the non-replicative RNA is a mRNA.

[0018] In an embodiment, the replicable RNA molecule in the composition can comprise an internal ribosome entry site (IRES) which controls expression of the first ORF encoding the protein or peptide comprising an antigen or epitope and/or can comprise an internal ribosome entry site (IRES) which controls expression of a second ORF, e.g., encoding the replicase. In various embodiments, the IRES can be insensitive to cellular stress and/or the IRES can be insensitive to interferons, preferably type I interferons and/or the IRES can be a cellular or viral IRES, preferably a viral IRES, for example, the IRES can be derived from viruses selected from the group consisting of picornaviruses, flaviviruses or dicistroviruses. In an embodiment, the IRES can be derived from a picornavirus or a dicistrovirus, preferably a dicistrovirus. In an embodiment, the IRES can be a type IV IRES.

[0019] In an embodiment, expression controlled by the IRES can be independent of IRES trans-acting factors and/or expression controlled by the IRES can be independent of cellular translation initiation factors. In an embodiment, expression controlled by the IRES can be independent of phosphorylation of eukaryotic initiation factor 2 (eIF2).

[0020] In an embodiment, both replicable RNA molecules can comprise a second ORF encoding the replicase. In an embodiment, neither replicable RNA molecule comprises an ORF encoding the replicase.

[0021] In an embodiment, any of the RNA molecules, whether replicative or non-replicative, can comprise a 5' cap for driving translation of the replicase and/or for driving translation of the peptide or protein comprising an antigen or epitope. The 5' cap can be a natural 5' cap or a 5' cap analog.

[0022] In an embodiment, at least one replicable RNA can comprise a 5' replication recognition sequence which is characterized in that at least one initiation codon is removed compared to a native alphavirus 5' replication recognition sequence. In an embodiment, all but one initiation codons are removed. In an embodiment, the 5' replication recognition sequence can comprise a sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus, wherein the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus can be characterized in that it comprises the removal of at least one initiation codon compared to the native viral sequence. In an embodiment, the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus can be characterized in that it comprises the removal of at least the native start codon of the open reading frame of a non-structural protein from a self-replicating virus. In an embodiment, the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus can be characterized in that it comprises the removal of at least one initiation codon other than the native start codon of the open reading frame of a non-structural protein from a self-replicating virus. In an embodiment, the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus can be characterized in that it is free of initiation codons. In an embodiment, at least one nucleotide change can be introduced which compensates for nucleotide pairing disruptions within at least one stem loop introduced by the removal of at least one initiation codon.

[0023] In an embodiment, the open reading frame encoding a functional non-structural protein from a self-replicating virus (replicase) does not overlap with the 5' replication recognition sequence. In an embodiment, the first or second ORF can be downstream from the 5' replication recognition sequence and upstream from the IRES.

[0024] In an embodiment, the replicable RNA can comprise a subgenomic promoter controlling production of subgenomic RNA comprising the first ORF encoding the protein or peptide. The subgenomic RNA can be a transcription product of an RNA-dependent RNA polymerase (replicase) derived from the functional non-structural protein from a self-replicating virus. Optionally, the protein or peptide can be expressed from the subgenomic RNA as a template. In an embodiment, the first ORF encoding the protein or peptide controlled by the subgenomic promoter can be downstream from the second ORF encoding the replicase. In an embodiment, the subgenomic promoter can overlap with the second ORF.

[0025] In an embodiment, at least one of the replicable RNA molecules can comprise a 3' replication recognition sequence. In an embodiment, the 5' and/or 3' replication recognition sequences and the subgenomic promoter can be derived from a self-replicating virus, preferably the same self-replicating virus species.

[0026] In an embodiment, the replicable RNA molecules can be replicated by an RNA-dependent RNA polymerase derived from the functional non-structural protein from a self-replicating virus. The self-replicating virus can be an alphavirus, for example, selected from the group consisting of Venezuelan equine encephalitis complex viruses, Eastern equine encephalitis complex viruses, Western equine encephalitis complex viruses, Chikungunya virus, Semliki Forest virus complex viruses, Sindbis virus, Barmah Forest virus, Middelburg virus and Ndumu virus.

[0027] In an embodiment, at least one of the RNA molecules can comprise a 3' poly(A) sequence.

[0028] In an embodiment, the first and/or second ORF can be flanked by a 5' untranslated region (UTR) and/or 3'UTR. In an embodiment, the 5' UTR and/or 3' UTR is/are not native to the alphavirus from which the replicase is derived. In an embodiment, the 5' UTR and/or 3' UTR is/are native to the alphavirus from which the replicase is derived.

[0029] In an embodiment where more than one RNA molecule in the composition encodes a replicase, the replicases can be derived from different alphaviruses and thus the encoding replicase sequences in the RNA molecules are different. In an embodiment where more than one RNA molecule in the composition encodes a replicase, the replicases can be the same and thus the encoding replicase sequences in the RNA molecules can be the same. In an embodiment, where more than one RNA molecule in the composition encodes a replicase, the replicases can be variants derived from the same alphavirus. In an embodiment, the replicase is derived from Semliki Forest Virus (SFV) or is derived from Venezuelan equine encephalitis virus (VEEV). In an embodiment, the replicase comprises non-structural proteins nsP1, nsP2, nsP3 and nsP4.

[0030] In an embodiment, at least one, if not both of the replicable RNA molecules does not comprise an open reading frame for an intact alphavirus structural protein.

[0031] In an embodiment, the replicable RNA molecules comprise, in a 5' to 3' order, a 5' cap, 5' UTR, the open reading frame encoding the replicase, an IRES, an open reading encoding the antigen, a 3' UTR and a poly-A sequence. In this embodiment, both open reading frames can be directly translated, i.e., no daughter RNAs need be produced in order for the open reading frame encoding the antigen to be translated. In an embodiment, the replicable RNA molecules comprise, in a 5' to 3' order, a 5' cap, 5' UTR, the open reading frame encoding the replicase, a subgenomic promoter, an open reading encoding the antigen, a 3' UTR and a poly-A sequence.

[0032] An exemplary poly-A sequence is depicted in SEQ ID NO:78. In an embodiment, a poly-A sequence useful in the RNA molecules described herein is one that is at least 75%, 80%, 85%, 90%, 95%, 98% or 99% homologous to SEQ ID NO:78. Exemplary 5' UTR sequences are depicted in

SEQ ID NOS:74 and 75. In an embodiment, a 5' UTR sequence useful in the RNA molecules described herein is one that is at least 75%, 80%, 85%, 90%, 95%, 98% or 99% homologous to SEQ ID NO:74 or 75. Exemplary 3' UTR sequences are depicted in SEQ ID Nos:76 and 77. In an embodiment, a 3' UTR sequence useful in the RNA molecules described herein is one that is at least 75%, 80%, 85%, 90%, 95%, 98% or 99% homologous to SEQ ID NO:76 or 77. An exemplary subgenomic promoter is depicted in SEQ ID NO:73. In an embodiment, a subgenomic promoter useful in the RNA molecules described herein is one that is at least 85%, 90%, 95%, 98% or 99% homologous to SEQ ID NO:73. It is envisioned that the RNA molecules described herein can have any combination of functional 5' UTR, 3' UTR, poly-A and subgenomic sequences. Thus, in certain embodiments, the RNA molecules described herein can have any combination of the exemplified 5' UTR, 3' UTR, poly-A and subgenomic sequences.

[0033] In an embodiment, the antigen or epitope of the encoded protein or peptide is a or is derived from a bacterial, viral, parasitical or fungal antigen. In an embodiment, the protein or peptide encoded by the first replicable RNA and the protein or peptide encoded by the second replicable RNA can be both obtained or derived from the same bacterium, virus, parasite or fungus. In an embodiment, the protein or peptide encoded by the first replicable RNA and the protein or peptide encoded by the second replicable RNA can be obtained or derived from different strains of the same bacterium, virus, parasite or fungus, respectively or can be obtained or derived from different pathogenic organisms, for example, different viruses. In an embodiment, the peptide or protein encoded by the first replicable RNA and the peptide or protein encoded by the second replicable RNA are not the same peptide or protein derived from different strains of an organism. For example, the peptide or protein encoded by the first replicable RNA and the peptide or protein encoded by the second replicable RNA are not each, for example, influenza hemagglutinin (HA) proteins from two different strains of influenza. In an embodiment, the epitope of the encoded protein can be a T cell epitope.

[0034] In an embodiment, the protein or peptide encoded by the first replicable RNA is a surface expressed protein or peptide, and the protein or peptide encoded by the second replicable RNA is not a surface expressed protein or peptide. The surface expressed and non-surface expressed proteins or peptides can be obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or can be obtained or derived from different pathogenic organisms, for example, different viruses. In an embodiment, the protein or peptide encoded by the first replicable RNA is not a surface expressed protein or peptide, and the protein or peptide encoded by the second replicable RNA also is not a surface expressed protein or peptide and is different from that encoded by the first replicable RNA. The different non-surface expressed proteins or peptides can be obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or can be obtained or derived from different pathogenic organisms, for example, different viruses. In an embodiment, the protein or peptide encoded by the first replicable RNA is a surface expressed protein or peptide, and the protein or peptide encoded by the second replicable RNA also is a surface expressed protein or peptide and is different from that encoded by the first replicable RNA. The surface expressed proteins or peptides can be obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or can be obtained or derived from different pathogenic organisms, for example, different viruses.

[0035] In an embodiment, the surface expressed protein is expressed on the surface of a virus/viral particle or wherein, where the virus is an enveloped virus, the surface expressed protein is expressed on the surface of the viral envelope, for example, the surface expressed protein is a viral capsid protein or a viral envelope or glycoprotein.

[0036] In an embodiment, the protein or peptide that is not a surface expressed protein or peptide is a viral matrix protein, a viral nucleoprotein, or a viral capsid protein where the virus is an enveloped virus.

[0037] In an embodiment, the protein or peptide encoded by the first replicable RNA is a viral glycoprotein and the protein or peptide encoded by the second replicable RNA is a viral nucleoprotein, wherein the glycoprotein and the nucleoprotein can be obtained or derived from the same virus, optionally from the same strain of the same virus.

[0038] In an embodiment, the surface expressed protein or peptide can be the glycoprotein GP of the Ebola virus. In an embodiment, the non-surface expressed protein or peptide can be the matrix protein VP40 or the nucleoprotein NP of the Ebola virus. The amino acid sequence of an exemplary Ebola virus GP protein is depicted in SEQ ID NO:89 and the amino acid sequence of an exemplary Ebola virus NP protein is depicted in SEQ ID NO:91.

[0039] In an embodiment, the surface expressed protein or peptide can be the glycoprotein Gc of the CCHFV virus. In an embodiment, the non-surface expressed protein or peptide can be the nucleoprotein NP of the CCHFV virus. The amino acid sequence of an exemplary CCHFV virus Gc protein is depicted in SEQ ID NO:93 and the amino acid sequence of an exemplary CCHFV virus NP protein is depicted in SEQ ID NO:95.

[0040] In an embodiment, the surface expressed protein or peptide can be the spike (S) protein of the MERS CoV virus. In an embodiment, the non-surface expressed protein or peptide can be the nucleoprotein NP of the MERS CoV virus. The amino acid sequence of an exemplary MERS CoV virus S protein is depicted in SEQ ID NO:97 and the amino acid sequence of an exemplary MERS CoV virus NP protein is depicted in SEQ ID NO:99.

[0041] In an embodiment, the amino acid sequence of the surface expressed protein or peptide or the amino acid sequence of the non-surface expressed protein or peptide can be one that is at least 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identical to the respective SEQ ID NOS set forth above.

[0042] In an embodiment, the composition comprises a first replicable RNA molecule encoding the Ebola virus GP protein having the amino acid sequence depicted in SEQ ID NO:89 and a second replicable RNA molecule encoding the Ebola virus NP protein having the amino acid sequence depicted in SEQ ID NO:91. In an embodiment, the composition comprises a first replicable RNA molecule encoding the CCHFV virus Gc protein having the amino acid sequence depicted in SEQ ID NO:93 and a second replicable RNA molecule encoding the CCHFV NP protein having the amino acid sequence depicted in SEQ ID NO:95. In an embodiment, the composition comprises a first replicable RNA molecule encoding the MERS CoV S protein having the amino acid sequence depicted in SEQ ID NO:97 and a second replicable RNA molecule encoding the MERS CoV virus NP protein having the amino acid sequence depicted in SEQ ID NO:99. Optionally, the first or second replicable RNA molecule encodes the replicase. The composition optionally comprises a third RNA molecule, which may or may not be replicable, and which encodes the RNA replicase. The third RNA molecule can be a non-replicable mRNA.

[0043] In a preferred embodiment, the first and second replicable RNA molecules in the composition do not encode the replicase and the composition further comprise an mRNA encoding the replicase which is able to replicate the first and second replicable RNA molecules in trans.

[0044] In an embodiment, the induced immune response against the antigens or epitopes can be an antibody response against the antigens or epitopes. In an embodiment, the induced immune response against the antigens or epitopes can be an increase in the activity of CD4⁺ T cells and/or CD8⁺ T cells. In an embodiment, the induced immune response against the antigens or epitopes can be an increase in the activity of both CD4⁺ and CD8⁺ T cells. In an embodiment, the induced immune response against the antigens or epitopes can be an antibody response against the antigens or epitopes and an increase in the activity of both CD4⁺ and CD8⁺ T cells.

[0045] In an embodiment, the protein or peptide encoded by the first replicable RNA can be an Ebola virus protein or fragment thereof or an epitope of the Ebola virus protein, and the protein or peptide encoded by the second replicable RNA can be a different Ebola virus protein or fragment thereof or an epitope of the different Ebola virus protein. The protein or peptide encoded by at least one replicable RNA molecule can be a structural Ebola virus protein selected from the group consisting of glycoprotein (GP), nucleoprotein (NP), polymerase cofactor (VP35), VP40, transcription factor (VP30), VP24 or RNA-dependent RNA polymerase (L), or a fragment thereof or an epitope of the Ebola virus structural protein. The epitope can be a T cell epitope. In an embodiment, the protein or peptide encoded by the first replicable RNA can be a viral surface protein, such as Ebola virus (EBOV) GP, CCHFV Gc-TM, or MERS-CoV S1, and the protein or peptide encoded by the second replicable RNA can be a protein from the nucleoprotein complex, such as Ebola virus NP, CCHFV NP, or MERS-CoV NP. The protein or peptide can comprise a T cell epitope.

[0046] In an embodiment, the protein or peptide encoded by at least one replicable RNA molecule can be expressed as a fusion protein, for example, the protein or peptide can be fused to a targeting or secretory motif.

[0047] In an embodiment, the first of the at least two replicable RNA molecules comprised in the composition encodes a protein or epitope thereof of a strain of a pathogen, such as the Ebola virus and the second of the at least two replicable RNA molecules comprised in the composition encodes the same or a different protein or epitope thereof from a different strain of the pathogen. For example, one replicable RNA molecule encodes the GP protein or an epitope thereof of Ebola virus subtype Zaire and the second replicable molecule encodes the GP protein or epitope thereof of Ebola virus strain Sudan. In an embodiment, the first of the at least two replicable RNA molecules comprised in the composition encodes several proteins or epitopes thereof from a strain of a pathogen and the second of the at least two RNA molecules comprised in the composition encodes several proteins or epitopes thereof from a different strain of the same pathogen. In an embodiment, the pathogen is Ebola virus or Crimean Congo hemorrhagic fever virus (CCHFV). Exemplary Ebola virus strains include Zaire, Sudan, Tal Forest and Bundibugyo. Exemplary Zaire strains include *H. sapiens*-wt/SLE/2014/Makona-EM095B (GenBank: KM034551.1), *H. sapiens*-wt/GIN/2014/Makona-Gueckedou-07 (GenBank: 10660347.2), *H. sapiens*-tc/COD/1976/Yambuku-Mayinga (GenBank NC_002549.1), Zaire ebola virus strain EBO-0003 (GenBank: MZ605321.1), and Zaire ebola virus strain EBO-0002 (GenBank: MZ605320.1). Other stains include those involved in the 2013-2016 EBOV epidemic in Guinea, Ueberla and Sierra Leone in West Africa, the sequences of which are available at the European Nucleotide Archive under accession number PRJEB43650 and at github.com under PFHVG/EBOVsequencing.

[0048] In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Ebola virus structural GP protein or a fragment thereof, or an epitope of the GP protein. The GP protein can be derived or obtained from the Ebola virus subtype Zaire, virus strain *H. sapiens*-wt/SLE/2014/Makona-EM095B (GenBank: KM034551.1). In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Ebola virus structural NP protein or a fragment thereof, or an epitope of the NP protein. The NP antigen can be derived or obtained from the Ebola virus subtype Zaire, virus strain *H. sapiens*-wt/GIN/2014/Makona-EM096 (GenBank: KM034551.1). In an embodiment, the GP amino acid sequence of Ebola virus sequence *H. sapiens*-wt/GIN/2014/Makona-Gueckedou-C07 (Gen Bank: K1660347.2) can be mutated at position 82 compared to C07 wild type, for example an alanine to valine substitution (Ala82Val).

[0049] In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Ebola virus structural GP protein or a fragment thereof, or an epitope of the GP protein. The GP protein can be derived or obtained from the Ebola virus sequence *H. sapiens*-wt/GIN/2014/Makona-Gueckedou-C07 (Gen Bank: KJ660347.2). In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Ebola virus structural NP protein or a fragment thereof, or an epitope of the NP protein. The NP antigen can be derived or obtained from the Ebola virus sequence *H. sapiens*-wt/GIN/2014/Makona-Gueckedou-C07 (Gen Bank: K1660347.2). In an embodiment, the NP amino acid sequence of Ebola virus subtype Zaire, virus strain *H. sapiens*-wt/GIN/2014/Makona-EM096 (GenBank: KM034551.1) can be mutated at position 111 compared to EM096 wild type, for example, an alanine to cysteine substitution at (Ala111Cys).

[0050] In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Crimean Congo hemorrhagic fever virus (CCHFV) structural GP protein or a fragment thereof, or an epitope of the GP protein. The GP protein can be derived or obtained from the Crimean Congo hemorrhagic fever virus strain Afg09-2990 sequence (Gen Bank: HM452306.1). In an embodiment, the protein or peptide encoded by at least one of the replicable RNA molecules can be the Crimean Congo hemorrhagic fever virus (CCHFV) structural NP protein or a fragment thereof, or an epitope of the NP protein. The NP antigen can be derived or obtained from the Crimean Congo hemorrhagic fever virus (CCHFV) strain Afg09-2990 sequence (Gen Bank: HM452305.1).

[0051] In an embodiment, at least one, preferably both, replicable RNA molecules can be codon-optimized, for example, by exchanging only the worst (least optimal) triplets for human usage, and optionally without increasing the overall GC content of the molecules.

[0052] In an embodiment, the protein or peptide encoded by the first replicable RNA molecule can be the Ebola virus structural GP protein or a fragment thereof, or an epitope of the GP protein, and the protein or peptide encoded by the second replicable RNA molecule can be the Ebola virus structural NP protein or a fragment thereof, or an epitope of the NP protein. In an embodiment, the amino acid sequence of an exemplary Ebola virus glycoprotein is depicted in SEQ ID NO:89. In an embodiment, the amino acid sequence of an exemplary Ebola virus nucleoprotein is depicted in SEQ ID NO:91. In an embodiment, the nucleotide sequence of a replicable RNA encoding an exemplary Ebola virus glycoprotein is depicted in SEQ ID NO:79. In an embodiment, the nucleotide sequence of a replicable RNA encoding an exemplary Ebola virus nucleoprotein is depicted in SEQ ID NO:80.

[0053] In an embodiment, the ratio of the number of first replicable RNA molecules to the number of second replicable RNA molecules in the composition (molar ratio) can vary from an approximately equal number of molecules. In an embodiment, the ratio of the first to second replicable RNA molecules can fall in the range from about 20:1 to about 1:20, about 15:1 to about 1:15, about 10:1 to about 1:10, or about 5:1 to about 1:5. In an embodiment, the ratio can be about 10:1, about 9:1, about 8:1, about 7:1, about 6:1, about 5:1, about 4:1, about 3:1, about 2:1, about 1:1, about 1:2, about 1:3, about 1:4, about 1:5, about 1:6, about 1:7, about 1:8, about 1:9, about 1:10.

[0054] In an embodiment, the composition further comprises a reagent capable of forming particles with the replicable RNA molecules, for example, the reagent can be a lipid or polyalkyleneimine. In various embodiments, the lipid can comprise a cationic headgroup, and/or the lipid can be a pH-responsive lipid, and/or the lipid can be a PEGylated-lipid. In an embodiment, the reagent can be conjugated to polysarcosine.

[0055] In an embodiment, the particles formed from the replicable RNA molecules and the reagent can be polymer-based polyplexes (PLX) or lipid nanoparticles (LNP), wherein the LNP is preferably a lipoplex (LPX) or a liposome. In an embodiment, the particle can further comprise at least one phosphatidylserine. In an embodiment, the particles can be nanoparticles, in which (i) the number of positive charges in the nanoparticles does not exceed the number of negative charges in the nanoparticles and/or (ii) the nanoparticles have a neutral or net negative charge and/or (iii) the charge ratio of positive charges to negative charges in the nanoparticles is 1.4:1 or less and/or (iv) the zeta potential of the nanoparticles is 0 or less. In an embodiment, the charge ratio of positive charges to negative charges in the nanoparticles can be between 1.4:1 and 1:8, preferably between 1.2:1 and 1:4.

[0056] In an embodiment, the nanoparticles can comprise at least one lipid, preferably comprise at least one cationic lipid, optionally wherein the positive charges are contributed by the at least one cationic lipid and the negative charges are contributed by the replicable RNA molecules.

[0057] In an embodiment, the nanoparticles can further comprise at least one helper lipid, wherein, for example, the helper lipid can be a neutral lipid. In various embodiments, the at least one cationic lipid can comprise 1,2-di-O-octadecenyl-3-trimethylammonium propane (DOTMA), 1,2-dioleoyloxy-3-dimethylaminopropane (DODMA), and/or 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP). In various embodiments, the at least one helper lipid can comprise 1,2-di-(9Z-octadecenoyl)-sn-glycero-3-phosphoethanolamine (DOPE), cholesterol (Chol), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), and/or 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC). In various embodiments, the molar ratio of the at least one cationic lipid to the at least one helper lipid can be from 10:0 to 3:7, preferably 9:1 to 3:7, 4:1 to 1:2, 4:1 to 2:3, 7:3 to 1:1, or 2:1 to 1:1, preferably about 1:1.

[0058] In an embodiment, the nanoparticles can be lipoplexes comprising DODMA and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DODMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2. In an embodiment, the nanoparticles can be lipoplexes comprising DODMA and Cholesterol in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DODMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2. In an embodiment, the nanoparticles can be lipoplexes comprising DODMA and DSPC in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DODMA to

negative charges in the RNA is 1.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1:1:2 and even more preferably about 1.2:2. In an embodiment, the nanoparticles are lipoplexes comprising DODMA:Cholesterol:DOPE:PEGcerC16 in a molar ratio of 40:48:10:2.

[0059] In an embodiment, the nanoparticles can be lipoplexes comprising DOTMA and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2. In an embodiment, the nanoparticles can be lipoplexes comprising DOTMA and Cholesterol in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2. In an embodiment, the nanoparticles can be lipoplexes comprising DOTAP and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2. [0060] In an embodiment, the reagent can comprise a lipid and the particles formed can be LNPs which are complexed with and/or encapsulate the replicable RNA molecules. In an embodiment, the reagent can comprise a lipid and the particles formed can be vesicles encapsulating the replicable RNA molecules, preferably unilamellar liposomes.

[0061] In an embodiment, the reagent is polyalkyleneimine, and, for example, the molar ratio of the number of nitrogen atoms (N) in the polyalkyleneimine to the number of phosphor atoms (P) in the replicable RNA molecules (N:P ratio) can be 2.0 to 15.0, preferably 6.0 to 12.0 and/or the ionic strength of the composition can be 50 mM or less, preferably wherein the concentration of monovalent cationic ions can be 25 mM or less and the concentration of divalent cationic ions can be 20 μ M or less.

[0062] In an embodiment, the particles formed can be polyplexes.

[0063] In an embodiment, the polyalkyleneimine can comprise the following general formula (I):

##STR00001## [0064] wherein [0065] R is H, an acyl group or a group comprising the following general formula (II):

##STR00002## [0066] wherein R.sub.1 is H or a group comprising the following general formula (III):

##STR00003## [0067] n, m, and l are independently selected from integers from 2 to 10; and [0068] p, q, and r are integers, wherein the sum of p, q, and r is such that the average molecular weight of the polymer is 1.5.Math.10.sup.2 to 10.sup.7 Da, preferably 5000 to 10.sup.5 Da, more preferably 10000 to 40000 Da, more preferably 15000 to 30000 Da, even more preferably 20000 to 25000 Da. In an embodiment, n, m, and l can be independently selected from 2, 3, 4, and 5, preferably from 2 and 3 and/or R.sub.1 can be H. In an embodiment, R can be H or an acyl group.

[0069] In an embodiment, the polyalkyleneimine can comprise polyethylenimine and/or polypropylenimine, preferably polyethyleneimine. In an embodiment, at least 92% of the N atoms in the polyalkyleneimine can be protonatable.

[0070] In an embodiment, the composition can further comprise one or more peptide-based adjuvants, wherein peptide-based adjuvants optionally comprise immune regulatory molecules, such as cytokines, lymphokines and/or co-stimulatory molecules. In an embodiment, the composition can further comprise one or more additives, wherein the additives optionally are selected from the group consisting of buffering substances, saccharides, stabilizers, cryoprotectants, lyoprotectants, and chelating agents. The buffering substances can comprise at least one selected from the group consisting of 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), 2-(N-morpholino)ethanesulfonic add (MES), 3-morpholino-2-hydroxypropanesulfonic acid (MOPSO), acetic acid, acetate buffers and analogues, phosphoric acid and phosphate buffers, and citric acid and citrate buffers. The saccharides can comprise at least one selected from the group consisting of monosaccharides, disaccharides, trisaccharides, oligosaccharides, and polysaccharides preferably from glucose, trehalose, and saccharose. The cryoprotectants can comprise at least one selected from the group consisting of glycols, such as ethylene glycol, propylene glycol, and glycerol. In an embodiment, the chelating agent can comprise EDTA.

[0071] In an embodiment, the replicable RNA molecules in the composition are present in a relative ratio from 6:1 to 1:6. In an embodiment, the replicable RNA molecules in the composition are present in a relative ratio from 3:1 to 1:3. In an embodiment, the replicable RNA molecules in the composition are present in a relative ratio from 2:1 to 1:2. In an embodiment, the replicable RNA molecules in the composition are present in a ratio of 1:1. In an embodiment, the ratio is determined by molecular weight of the RNA molecules.

[0072] In an embodiment, the inventive composition of at least two replicable RNA molecules can be a vaccine.

[0073] In an aspect, the present invention is directed to a pharmaceutical composition comprising the inventive composition of at least two replicable RNA molecules and a pharmaceutically acceptable carrier. In various embodiments, the pharmaceutical composition can be formulated for intradermal, intranasal, intrapulmonary, subcutaneous, and/or intramuscular administration, such as by injection.

[0074] In an aspect, the present invention is directed to the use of the inventive composition of at least two replicable RNA molecules in therapy, such as inducing an immune response or vaccination. In an aspect, the present invention is directed to the use of the inventive composition of at least two replicable RNA molecules in a method for inducing an immune response specific for the encoded proteins or peptides in a subject, preferably wherein the subject is a mammal, more preferably wherein the mammal is a human, said method comprising administering the pharmaceutical composition of the invention.

[0075] In an aspect, the present invention is directed to a method for inducing an immune response specific for at least two antigens or epitopes in a subject comprising administering the pharmaceutical composition of the invention to the subject, preferably wherein the subject is a mammal, more preferably wherein the mammal is a human.

[0076] In an embodiment, the immune response can comprise the activation of T cells and/or B cells, preferably wherein the activated T cells comprise T helper cells and cytotoxic T cells. In an embodiment, the immune response can comprise activation of antigen-specific T helper cells, optionally wherein the T helper cells proliferate, release T cell cytokines, mediate the growth and/or activation of antigen specific cytotoxic T cells. In an embodiment, the immune response can comprise activation of antigen specific T helper cells, wherein the T helper cells stimulate B cell proliferation, antibody class switching, production and/or secretion of neutralizing antibodies.

[0077] In an aspect, the present invention is directed to a method for producing at least two proteins or peptides of interest in a cell comprising inoculating the pharmaceutical composition of the invention into the cell. In an aspect, the present invention is directed to a method for producing at least two proteins or peptides of interest in a subject comprising administering the pharmaceutical composition of the invention to the subject. In an embodiment, the method comprises administering to a subject a first replicable RNA molecule comprises the nucleotide sequence depicted in SEQ ID NO:27 and a second replicable RNA molecule comprising the nucleotide sequence depicted in SEQ IN NO:29. In an embodiment, the method comprises administering to a subject the replicable RNA molecule depicted in SEQ ID NO:30 and the replicable RNA molecule depicted in SEQ ID NO:31.

[0078] In an aspect, the present invention is directed to a method for the treatment or prevention of a bacterial, viral, parasitical or fungal infection in a subject, said method comprising administering to the subject a composition comprising at least two replicable RNA molecules, each comprising a first open-reading frame (ORF) encoding at least one peptide or protein comprising an antigen or epitope suitable to induce an immune response against the bacterium, virus, parasite or fungus, respectively; wherein the at least one peptide or protein encoded by one of the replicable RNA molecules is different from the at least one peptide or protein encoded by the other replicable RNA molecule, and wherein at least one of the replicable RNA molecules further comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules.

[0079] In an embodiment, the method comprises administering to a subject a first replicable RNA molecule comprises the nucleotide sequence depicted in SEQ ID NO:27 and a second replicable RNA molecule comprising the nucleotide sequence depicted in SEQ IN NO:29. In an embodiment, the method comprises administering to a subject the replicable RNA molecule depicted in SEQ ID NO:30 and the replicable RNA

molecule depicted in SEQ ID NO:31.

[0080] In an embodiment, the immune response is a specific immune response against the bacterium, virus, parasite or fungus, respectively and/or the immune response lessens the severity of one or more symptoms of the infection.

[0081] In an embodiment, the infection can be a viral infection, optionally wherein the infection is an Ebola virus infection.

[0082] In an embodiment, the method of treatment involves only a single administration of the composition or the method of treatment comprises multiple administrations of the composition. In an embodiment, the method can further comprise administering a booster dose of the pharmaceutical composition of the invention.

DETAILED DESCRIPTION

[0083] Although the present invention is described in detail below, it is to be understood that this invention is not limited to the particular methodologies, protocols and reagents described herein as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

[0084] Preferably, the terms used herein are defined as described in “A multilingual glossary of biotechnological terms: (IUPAC Recommendations)”, H. G. W. Leuenberger, B. Nagel, and H. Kölbl, Eds., Helvetica Chimica Acta, CH-4010 Basel, Switzerland, (1995).

[0085] The practice of the present invention will employ, unless otherwise indicated, conventional methods of chemistry, biochemistry, cell biology, immunology, and recombinant DNA techniques which are explained in the literature in the field (cf., e.g., Molecular Cloning: A Laboratory Manual, 2nd Edition, J. Sambrook et al. eds., Cold Spring Harbor Laboratory Press, Cold Spring Harbor 1989).

[0086] In the following, the elements of the present invention will be described. These elements are listed with specific embodiments; however, it should be understood that they may be combined in any manner and in any number to create additional embodiments. The variously described examples and preferred embodiments should not be construed to limit the present invention to only the explicitly described embodiments. This description should be understood to disclose and encompass embodiments which combine the explicitly described embodiments with any number of the disclosed and/or preferred elements. Furthermore, any permutations and combinations of all described elements in this application should be considered disclosed by this description unless the context indicates otherwise.

[0087] The term “about” means approximately or nearly, and in the context of a numerical value or range set forth herein preferably means $\pm 10\%$ of the numerical value or range recited or claimed.

[0088] The terms “a” and “an” and “the” and similar reference used in the context of describing the invention (especially in the context of the claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it was individually recited herein.

[0089] All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”), provided herein is intended merely to better illustrate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

[0090] Unless expressly specified otherwise, the term “comprising” is used in the context of the present document to indicate that further members may optionally be present in addition to the members of the list introduced by “comprising”. It is, however, contemplated as a specific embodiment of the present invention that the term “comprising” encompasses the possibility of no further members being present, i.e., for the purpose of this embodiment “comprising” is to be understood as having the meaning of “consisting of”.

[0091] Indications of relative amounts of a component characterized by a generic term are meant to refer to the total amount of all specific variants or members covered by said generic term. If a certain component defined by a generic term is specified to be present in a certain relative amount, and if this component is further characterized to be a specific variant or member covered by the generic term, it is meant that no other variants or members covered by the generic term are additionally present such that the total relative amount of components covered by the generic term exceeds the specified relative amount; more preferably no other variants or members covered by the generic term are present at all.

[0092] Several documents are cited throughout the text of this specification. Each of the documents cited herein (including all patents, patent applications, scientific publications, manufacturers specifications, instructions, etc.), whether supra or infra, are hereby incorporated by reference in their entirety. Nothing herein is to be construed as an admission that the present invention was not entitled to antedate such disclosure.

[0093] Terms such as “reduce” or “inhibit” as used herein means the ability to cause an overall decrease, preferably of 5% or greater, 10% or greater, 20% or greater, more preferably of 50% or greater, and most preferably 75% or greater, in the level. The term “inhibit” or similar phrases includes a complete or essentially complete inhibition, i.e., a reduction to zero or essentially to zero.

[0094] Terms such as “increase” or “enhance” preferably relate to an increase or enhancement by about at least 10%, preferably at least 20%, preferably at least 30%, more preferably at least 40%, more preferably at least 50%, even more preferably at least 80%, and most preferably at least 100%.

[0095] The term “net charge” refers to the charge on a whole object, such as a compound or particle.

[0096] An ion having an overall net positive charge is a cation, while an ion having an overall net negative charge is an anion. Thus, according to the invention, an anion is an ion with more electrons than protons, giving it a net negative charge; and a cation is an ion with fewer electrons than protons, giving it a net positive charge.

[0097] Terms as “charged”, “net charge”, “negatively charged” or “positively charged”, with reference to a given compound or particle, refer to the electric net charge of the given compound or particle when dissolved or suspended in water at pH 7.0.

[0098] The term “nucleic add” according to the invention also comprises a chemical derivatization of a nucleic add on a nucleotide base, on the sugar or on the phosphate, and nucleic adds containing non-natural nucleotides and nucleotide analogs. In some embodiments, the nucleic acid is a deoxyribonucleic acid (DNA) or a ribonucleic acid (RNA). In general, a nucleic add molecule or a nucleic acid sequence refers to a nucleic acid which is preferably deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). According to the invention, nucleic adds comprise genomic DNA, cDNA, mRNA, viral RNA, recombinantly prepared and chemically synthesized molecules. According to the invention, a nucleic acid may be in the form of a single-stranded or double-stranded and linear or covalently closed circular molecule.

[0099] According to the invention “nucleic acid sequence” refers to the sequence of nucleotides in a nucleic acid, e.g.; a ribonucleic acid (RNA) or a deoxyribonucleic acid (DNA). The term may refer to an entire nucleic add molecule (such as to the single strand of an entire nucleic add molecule) or to a part (e.g. a fragment) thereof.

[0100] According to the present invention, the term “RNA” or “RNA molecule” relates to a molecule which comprises ribonucleotide residues and which is preferably entirely or substantially composed of ribonucleotide residues. The term “ribonucleotide” relates to a nucleotide with a hydroxyl group at the 2'-position of a β -D-ribofuranosyl group.

[0101] The term “RNA” comprises double-stranded RNA, single stranded RNA, isolated RNA such as partially or completely purified RNA, essentially pure RNA, synthetic RNA, and recombinantly generated RNA such as modified RNA which differs from naturally occurring RNA by addition, deletion, substitution and/or alteration of one or more nucleotides.

[0102] Such alterations can include addition of non-nucleotide material, such as to the end(s) of an RNA or internally, for example at one or more nucleotides of the RNA. Nucleotides in RNA molecules can also comprise non-standard nucleotides, such as non-naturally occurring nucleotides or

chemically synthesized nucleotides or deoxynucleotides.

[0103] These altered RNAs can be referred to as analogs, particularly analogs of naturally occurring RNAs.

[0104] According to the invention, RNA may be single-stranded or double-stranded. In some embodiments of the present invention, single-stranded RNA is preferred. The term “single-stranded RNA” generally refers to an RNA molecule to which no complementary nucleic acid molecule (typically no complementary RNA molecule) is associated. Single-stranded RNA may contain self-complementary sequences that allow parts of the RNA to fold back and to form secondary structure motifs including without limitation base pairs, stems, stem loops and bulges. Single-stranded RNA can exist as minus strand [(-) strand] or as plus strand [(+) strand]. The (+) strand is the strand that comprises or encodes genetic information. The genetic information may be for example a polynucleotide sequence encoding a protein. When the (+) strand RNA encodes a protein, the (+) strand may serve directly as template for translation (protein synthesis). The (-) strand is the complement of the (+) strand. In the case of double-stranded RNA, (+) strand and (-) strand are two separate RNA molecules, and both these RNA molecules associate with each other to form a double-stranded RNA (“duplex RNA”).

[0105] The term “stability” of RNA relates to the “half-life” of RNA. “Half-life” relates to the period of time which is needed to eliminate half of the activity, amount, or number of molecules. In the context of the present invention, the half-life of an RNA is indicative for the stability of said RNA. The half-life of RNA may influence the “duration of expression” of the RNA. It can be expected that RNA having a long half-life will be expressed for an extended time period.

[0106] The term “translation efficiency” relates to the amount of translation product provided by an RNA molecule within a particular period of time.

[0107] “Fragment”, with reference to a nucleic acid sequence, relates to a part of a nucleic acid sequence, i.e.; a sequence which represents the nucleic acid sequence shortened at the 5'- and/or 3'-end(s). Preferably, a fragment of a nucleic acid sequence comprises at least 80%, preferably at least 90%, 95%, 96%, 97%, 98%, or 99% of the nucleotide residues from said nucleic acid sequence. In the present invention those fragments of RNA molecules are preferred which retain RNA stability and/or translational efficiency. “Fragment”, with reference to an amino acid sequence (peptide or protein), relates to a part of an amino acid sequence, i.e. a sequence which represents the amino acid sequence shortened at the N-terminus and/or C-terminus. A fragment shortened at the C-terminus (N-terminal fragment) is obtainable, e.g., by translation of a truncated open reading frame that lacks the 3'-end of the open reading frame. A fragment shortened at the N-terminus (C-terminal fragment) is obtainable, e.g., by translation of a truncated open reading frame that lacks the 5'-end of the open reading frame, as long as the truncated open reading frame comprises a start codon that serves to initiate translation. A fragment of an amino acid sequence comprises e.g. at least 1%, at least 2%, at least 3%, at least 4%, at least 5%, at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90% of the amino acid residues from an amino acid sequence.

[0108] The term “variant” with respect to, for example, nucleic acid and amino acid sequences, according to the invention includes any variants, in particular mutants, viral strain variants, splice variants, conformations, isoforms, allelic variants, species variants and species homologs, in particular those which are naturally present. An allelic variant relates to an alteration in the normal sequence of a gene, the significance of which is often unclear. Complete gene sequencing often identifies numerous allelic variants for a given gene. With respect to nucleic acid molecules, the term “variant” includes degenerate nucleic acid sequences, wherein a degenerate nucleic acid according to the invention is a nucleic acid that differs from a reference nucleic acid in codon sequence due to the degeneracy of the genetic code. A species homolog is a nucleic acid or amino acid sequence with a different species of origin from that of a given nucleic acid or amino acid sequence. A virus homolog is a nucleic acid or amino acid sequence with a different virus of origin from that of a given nucleic acid or amino acid sequence. For example, hemagglutinin derived from one strain of influenza and hemagglutinin derived from another strain of influenza which do not have an identical amino acid sequence are variants of each other. In another example, the same bacterial protein derived from two different bacterial strains and which do not have identical amino acid sequences are variants of each other.

[0109] Nucleic acid variants include single or multiple nucleotide deletions, additions, mutations, substitutions and/or insertions in comparison with the reference nucleic acid. Deletions include removal of one or more nucleotides from the reference nucleic acid. Addition variants comprise 5'- and/or 3-terminal fusions of one or more nucleotides, such as 1, 2, 3, 5, 10, 20, 30, 50, or more nucleotides. In the case of substitutions, at least one nucleotide in the sequence is removed and at least one other nucleotide is inserted in its place (such as transversions and transitions).

[0110] Mutations include abasic sites, crosslinked sites, and chemically altered or modified bases. Insertions include the addition of at least one nucleotide into the reference nucleic acid.

[0111] According to the invention, “nucleotide change” can refer to single or multiple nucleotide deletions, additions, mutations, substitutions and/or insertions in comparison with the reference nucleic acid. In some embodiments, a “nucleotide change” is selected from the group consisting of a deletion of a single nucleotide, the addition of a single nucleotide, the mutation of a single nucleotide, the substitution of a single nucleotide and/or the insertion of a single nucleotide, in comparison with the reference nucleic acid. According to the invention, a nucleic acid variant can comprise one or more nucleotide changes in comparison with the reference nucleic acid.

[0112] Variants of specific nucleic acid sequences preferably have at least one functional property of said specific sequences and preferably are functionally equivalent to said specific sequences, e.g., nucleic acid sequences exhibiting properties identical or similar to those of the specific nucleic acid sequences.

[0113] As described below, some embodiments of the present invention are characterized, inter alia, by nucleic acid sequences that are homologous to other nucleic acid sequences. These homologous sequences are variants of other nucleic acid sequences.

[0114] Preferably the degree of identity between a given nucleic acid sequence and a nucleic acid sequence which is a variant of said given nucleic acid sequence will be at least 70%, preferably at least 75%, preferably at least 80%, more preferably at least 85%, even more preferably at least 90% or most preferably at least 95%, 96%, 97%, 98% or 99%. The degree of identity is preferably given for a region of at least about 30, at least about 50, at least about 70, at least about 90, at least about 100, at least about 150, at least about 200, at least about 250, at least about 300, or at least about 400 nucleotides. In preferred embodiments, the degree of identity is given for the entire length of the reference nucleic acid sequence.

[0115] “Sequence similarity” indicates the percentage of amino acids that either are identical or that represent conservative amino acid substitutions. “Sequence identity” between two polypeptide or nucleic acid sequences indicates the percentage of amino acids or nucleotides that are identical between the sequences.

[0116] The term “% identical” is intended to refer, in particular, to a percentage of nucleotides which are identical in an optimal alignment between two sequences to be compared, with said percentage being purely statistical, and the differences between the two sequences may be randomly distributed over the entire length of the sequence and the sequence to be compared may comprise additions or deletions in comparison with the reference sequence, in order to obtain optimal alignment between two sequences. Comparisons of two sequences are usually carried out by comparing said sequences, after optimal alignment, with respect to a segment or “window of comparison”, in order to identify local regions of corresponding sequences. The optimal alignment for a comparison may be carried out manually or with the aid of the local homology algorithm by Smith and Waterman, 1981, *Adv. Math.* 2:482, with the aid of the local homology algorithm by Needleman and Wunsch, 1970, *J. Mol. Biol.* 48:443, and with the aid of the similarity search algorithm by Pearson and Lipman, 1988, *Proc. Natl. Acad. Sci. USA* 85:2444 or with the aid of computer programs using said algorithms (GAP, BESTFIT, FASTA, BLAST P, BLAST N and TFASTA in Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Drive, Madison, Wis.).

[0117] Percentage identity is obtained by determining the number of identical positions in which the sequences to be compared correspond, dividing this number by the number of positions compared and multiplying this result by 100.

[0118] For example, the BLAST program “BLAST 2 sequences” which is available on the website

[0119] A nucleic add is “capable of hybridizing” or “hybridizes” to another nucleic acid if the two sequences are complementary with one another. A nucleic add is “complementary” to another nucleic acid if the two sequences are capable of forming a stable duplex with one another. According to the invention, hybridization is preferably carried out under conditions which allow specific hybridization between polynucleotides (stringent conditions).

[0120] Stringent conditions are described, for example, in *Molecular Cloning: A Laboratory Manual*, J. Sambrook et al., Editors, 2nd Edition, Cold Spring Harbor Laboratory press, Cold Spring Harbor, New York, 1989 or *Current Protocols in Molecular Biology*, F. M. Ausubel et al., Editors, John Wiley & Sons, Inc., New York and refer, for example, to hybridization at 65° C. in hybridization buffer (3.5×SSC, 0.02% Ficoll, 0.02% polyvinylpyrrolidone, 0.02% bovine serum albumin, 2.5 mM NaH₂PO₄ (pH 7), 0.5% SDS, 2 mM EDTA). SSC is 0.15 M sodium chloride/0.15 M sodium citrate, pH 7. After hybridization, the membrane to which the DNA has been transferred is washed, for example, in 2×SSC at room temperature and then in 0.1-0.5×SSC/0.1×SDS at temperatures of up to 68° C. A percent complementarity indicates the percentage of contiguous residues in a nucleic acid molecule that can form hydrogen bonds (e.g., Watson-Crick base pairing) with a second nucleic acid sequence (e.g., 5, 6, 7, 8, 9, 10 out of 10 being 50%, 60%, 70%, 80%, 90%, and 100% complementary). “Perfectly complementary” or “fully complementary” means that all the contiguous residues of a nucleic acid sequence will hydrogen bond with the same number of contiguous residues in a second nucleic acid sequence. Preferably, the degree of complementarity according to the invention is at least 70%, preferably at least 75%, preferably at least 80%, more preferably at least 85%, even more preferably at least 90% or most preferably at least 95%, 96%, 97%, 98% or 99%. Most preferably, the degree of complementarity according to the invention is 100%.

[0121] The term “derivative” comprises any chemical derivatization of a nucleic add on a nucleotide base, on the sugar or on the phosphate. The term “derivative” also comprises nucleic acids which contain nucleotides and nucleotide analogs not occurring naturally. Preferably, a derivatization of a nucleic acid increases its stability.

[0122] A “nucleic add sequence which is derived from a nucleic add sequence” refers to a nucleic acid which is a variant of the nucleic add from which it is derived. Preferably, a sequence which is a variant with respect to a specific sequence, when it replaces the specific sequence in an RNA molecule retains RNA stability and/or translational efficiency.

[0123] “nt” is an abbreviation for nucleotide; or for nucleotides, preferably consecutive nucleotides in a nucleic acid molecule.

[0124] According to the invention, the term “codon” refers to a base triplet in a coding nucleic acid that specifies which amino add will be added next during protein synthesis at the ribosome.

[0125] The terms “transcription” and “transcribing” relate to a process during which a nucleic acid molecule with a particular nucleic add sequence (the “nucleic add template”) is read by an RNA polymerase so that the RNA polymerase produces a single-stranded RNA molecule. During transcription, the genetic information in a nucleic acid template is transcribed. The nucleic acid template may be DNA; however, e.g.; in the case of transcription from an alphaviral nucleic add template, the template is typically RNA. Subsequently, the transcribed RNA may be translated into protein. According to the present invention, the term “transcription” comprises “in vitro transcription”, wherein the term “in vitro transcription” relates to a process wherein RNA, in particular mRNA, is in vitro synthesized in a cell-free system. Preferably, cloning vectors are applied for the generation of transcripts. These cloning vectors are generally designated as transcription vectors and are according to the present invention encompassed by the term “vector”. The cloning vectors are preferably plasmids. According to the present invention, RNA preferably is in vitro transcribed RNA (IVT-RNA) and may be obtained by in vitro transcription of an appropriate DNA template. The promoter for controlling transcription can be any promoter for any RNA polymerase. A DNA template for in vitro transcription may be obtained by cloning of a nucleic acid, in particular cDNA, and introducing it into an appropriate vector for in vitro transcription. The cDNA may be obtained by reverse transcription of RNA.

[0126] The single-stranded nucleic acid molecule produced during transcription typically has a nucleic acid sequence that is the complementary sequence of the template.

[0127] According to the invention, the terms “template” or “nucleic add template” or “template nucleic acid” generally refer to a nucleic add sequence that may be replicated or transcribed.

[0128] “Nucleic add sequence transcribed from a nucleic acid sequence” and similar terms refer to a nucleic acid sequence, where appropriate as part of a complete RNA molecule, which is a transcription product of a template nucleic add sequence. Typically, the transcribed nucleic add sequence is a single-stranded RNA molecule.

[0129] “3' end of a nucleic add” refers according to the invention to that end which has a free hydroxy group. In a diagrammatic representation of double-stranded nucleic acids, in particular DNA, the 3' end is always on the right-hand side. “5' end of a nucleic acid” refers according to the invention to that end which has a free phosphate group. In a diagrammatic representation of double-strand nucleic acids, in particular DNA, the 5' end is always on the left-hand side.

TABLE-US-00001 5' end 5'-P-NNNNNNN-OH-3' 3' end 3'-HO-NNNNNNN-P-5'

[0130] “Upstream” describes the relative positioning of a first element of a nucleic acid molecule with respect to a second element of that nucleic add molecule, wherein both elements are comprised in the same nucleic acid molecule, and wherein the first element is located nearer to the 5' end of the nucleic acid molecule than the second element of that nucleic acid molecule. The second element is then said to be “downstream” of the first element of that nucleic acid molecule. An element that is located “upstream” of a second element can be synonymously referred to as being located “S” of that second element. For a double-stranded nucleic acid molecule, indications like “upstream” and “downstream” are given with respect to the (+) strand.

[0131] According to the invention, “functional linkage” or “functionally linked” relates to a connection within a functional relationship. A nucleic acid is “functionally linked” if it is functionally related to another nucleic add sequence. For example, a promoter is functionally linked to a coding sequence if it influences transcription of said coding sequence.

[0132] Functionally linked nucleic acids are typically adjacent to one another, where appropriate separated by further nucleic acid sequences, and, in particular embodiments, are transcribed by RNA polymerase to give a single RNA molecule (common transcript).

[0133] In particular embodiments, a nucleic add is functionally linked according to the invention to expression control sequences which may be homologous or heterologous with respect to the nucleic acid.

[0134] The term “expression control sequence” comprises according to the invention promoters, ribosome-binding sequences and other control elements which control transcription of a gene or translation of the derived RNA. In particular embodiments of the invention, the expression control sequences can be regulated. The precise structure of expression control sequences may vary depending on the species or cell type but usually includes 5'-untranscribed and 5'- and 3'-untranslated sequences involved in initiating transcription and translation, respectively. More specifically, 5'-untranscribed expression control sequences include a promoter region which encompasses a promoter sequence for transcription control of the functionally linked gene. Expression control sequences may also include enhancer sequences or upstream activator sequences. An expression control sequence of a DNA molecule usually includes 5'-untranscribed and 5'- and 3'-untranslated sequences such as TATA box, capping sequence, CAAT sequence and the like. An expression control sequence of alphaviral RNA may include a subgenomic promoter and/or one or more conserved sequence element(s). A specific expression control sequence according to the present invention is a subgenomic promoter of an alphavirus, as described herein.

[0135] The nucleic acid sequences specified herein, in particular transcribable and coding nucleic acid sequences, may be combined with any expression control sequences, in particular promoters, which may be homologous or heterologous to said nucleic acid sequences, with the term

“homologous” referring to the fact that a nucleic acid sequence is also functionally linked naturally to the expression control sequence, and the term “heterologous” referring to the fact that a nucleic acid sequence is not naturally functionally linked to the expression control sequence.

[0136] A transcribable nucleic acid sequence, in particular a nucleic acid sequence coding for a peptide or protein, and an expression control sequence are “functionally” linked to one another, if they are covalently linked to one another in such a way that transcription or expression of the transcribable and in particular coding nucleic acid sequence is under the control or under the influence of the expression control sequence. If the nucleic acid sequence is to be translated into a functional peptide or protein, induction of an expression control sequence functionally linked to the coding sequence results in transcription of said coding sequence, without causing a frame shift in the coding sequence or the coding sequence being unable to be translated into the desired peptide or protein.

[0137] The term “promoter” or “promoter region” refers to a nucleic acid sequence which controls synthesis of a transcript, e.g. a transcript comprising a coding sequence, by providing a recognition and binding site for RNA polymerase.

[0138] The promoter region may include further recognition or binding sites for further factors involved in regulating transcription of said gene. A promoter may control transcription of a prokaryotic or eukaryotic gene. A promoter may be “inducible” and initiate transcription in response to an inducer, or may be “constitutive” if transcription is not controlled by an inducer. An inducible promoter is expressed only to a very small extent or not at all, if an inducer is absent. In the presence of the inducer, the gene is “switched on” or the level of transcription is increased.

[0139] This is usually mediated by binding of a specific transcription factor. A specific promoter according to the present invention is a subgenomic promoter, e.g., of an alphavirus, as described herein. Other specific promoters are genomic plus-strand or negative-strand promoters, e.g., of an alphavirus.

[0140] The term “core promoter” refers to a nucleic acid sequence that is comprised by the promoter. The core promoter is typically the minimal portion of the promoter required to properly initiate transcription. The core promoter typically includes the transcription start site and a binding site for RNA polymerase.

[0141] A “polymerase” generally refers to a molecular entity capable of catalyzing the synthesis of a polymeric molecule from monomeric building blocks. An “RNA polymerase” is a molecular entity capable of catalyzing the synthesis of an RNA molecule from ribonucleotide building blocks. A “DNA polymerase” is a molecular entity capable of catalyzing the synthesis of a DNA molecule from deoxy ribonucleotide building blocks. For the case of DNA polymerases and RNA polymerases, the molecular entity is typically a protein or an assembly or complex of multiple proteins.

[0142] Typically, a DNA polymerase synthesizes a DNA molecule based on a template nucleic acid, which is typically a DNA molecule. Typically, an RNA polymerase synthesizes an RNA molecule based on a template nucleic acid, which is either a DNA molecule (in that case the RNA polymerase is a DNA-dependent RNA polymerase, DdRP), or is an RNA molecule (in that case the RNA polymerase is an RNA-dependent RNA polymerase, RdRP).

[0143] An “RNA-dependent RNA polymerase” or “RdRP”, is an enzyme that catalyzes the transcription of RNA from an RNA template. In the case of alphaviral RNA-dependent RNA polymerase, sequential synthesis of (–) strand complement of genomic RNA and of (+) strand genomic RNA leads to RNA replication. RNA-dependent RNA polymerase is thus synonymously referred to as “RNA replicase” or simply “replicase”. In nature, RNA-dependent RNA polymerases are typically encoded by all RNA viruses except retroviruses. Typical representatives of viruses encoding an RNA-dependent RNA polymerase are alphaviruses.

[0144] According to the present invention, “RNA replication” generally refers to an RNA molecule synthesized based on the nucleotide sequence of a given RNA molecule (template RNA molecule). The RNA molecule that is synthesized may be, e.g., identical or complementary to the template RNA molecule. In general, RNA replication may occur via synthesis of a DNA intermediate, or may occur directly by RNA-dependent RNA replication mediated by an RNA-dependent RNA polymerase (RdRP). In the case of alphaviruses, RNA replication does not occur via a DNA intermediate, but is mediated by a RNA-dependent RNA polymerase (RdRP): a template RNA strand (first RNA strand)—or a part thereof—serves as template for the synthesis of a second RNA strand that is complementary to the first RNA strand or to a part thereof. The second RNA strand—or a part thereof—may in turn optionally serve as a template for synthesis of a third RNA strand that is complementary to the second RNA strand or to a part thereof. Thereby, the third RNA strand is identical to the first RNA strand or to a part thereof. Thus, RNA-dependent RNA polymerase is capable of directly synthesizing a complementary RNA strand of a template, and of indirectly synthesizing an identical RNA strand (via a complementary intermediate strand).

[0145] According to the invention, the term “template RNA” refers to RNA that can be transcribed or replicated by an RNA-dependent RNA polymerase.

[0146] According to the invention, the term “gene” refers to a particular nucleic acid sequence which is responsible for producing one or more cellular products and/or for achieving one or more intercellular or intracellular functions.

[0147] More specifically, said term relates to a nucleic acid section (typically DNA; but RNA in the case of RNA viruses) which comprises a nucleic acid coding for a specific protein or a functional or structural RNA molecule.

[0148] An “isolated molecule” as used herein, is intended to refer to a molecule which is substantially free of other molecules such as other cellular material. The term “isolated nucleic acid” means according to the invention that the nucleic acid has been (i) amplified in situ, for example by polymerase chain reaction (PCR), (ii) recombinantly produced by cloning, (iii) purified, for example by cleavage and gel-electrophoretic fractionation, or (iv) synthesized, for example by chemical synthesis. An isolated nucleic acid is a nucleic acid available to manipulation by recombinant techniques.

[0149] The term “vector” is used here in its most general meaning and comprises any intermediate vehicles for a nucleic acid which, for example, enable said nucleic acid to be introduced into prokaryotic and/or eukaryotic host cells and, where appropriate, to be integrated into a genome. Such vectors are preferably replicated and/or expressed in the cell. Vectors comprise plasmids, phagemids, virus genomes, and fractions thereof.

[0150] The term “recombinant” in the context of the present invention means “made through genetic engineering”. Preferably, a “recombinant object” such as a recombinant cell in the context of the present invention is not occurring naturally.

[0151] The term “naturally occurring” as used herein refers to the fact that an object can be found in nature. For example, a peptide or nucleic acid that is present in an organism (including viruses) and can be isolated from a source in nature and which has not been intentionally modified by man in the laboratory is naturally occurring. The term “found in nature” means “present in nature” and includes known objects as well as objects that have not yet been discovered and/or isolated from nature, but that may be discovered and/or isolated in the future from a natural source.

[0152] According to the invention, the term “expression” is used in its most general meaning and comprises production of RNA and/or protein. It also comprises partial expression of nucleic acids. Furthermore, expression may be transient or stable. With respect to RNA, the term “expression” or “translation” relates to the process in the ribosomes of a cell by which a strand of coding RNA (e.g. messenger RNA) directs the assembly of a sequence of amino acids to make a peptide or protein.

[0153] According to the invention, the term “mRNA” means “messenger-RNA” and relates to a transcript which is typically generated by using a DNA template and encodes a peptide or protein. Typically, mRNA comprises a 5'-UTR, a protein coding region, a 3'-UTR, and a poly(A) sequence. mRNA may be generated by in vitro transcription from a DNA template. The in vitro transcription methodology is known to the skilled person. For example, there is a variety of in vitro transcription kits commercially available. According to the invention, mRNA may be modified by stabilizing modifications and capping.

[0154] According to the invention, the terms “poly(A) sequence” or “poly(A) tail” refer to an uninterrupted or interrupted sequence of adenylate residues which is typically located at the 3' end of an RNA molecule. An uninterrupted sequence is characterized by consecutive adenylate residues. In nature, an uninterrupted poly(A) sequence is typical. While a poly(A) sequence is normally not encoded in eukaryotic DNA, but is attached during

eukaryotic transcription in the cell nucleus to the free 3' end of the RNA by a template-independent RNA polymerase after transcription, the present invention encompasses poly(A) sequences encoded by DNA.

[0155] According to the invention, the term “primary structure”, with reference to a nucleic acid molecule, refers to the linear sequence of nucleotide monomers.

[0156] According to the invention, the term “secondary structure”, with reference to a nucleic acid molecule, refers to a two-dimensional representation of a nucleic acid molecule that reflects base pairings; e.g., in the case of a single-stranded RNA molecule particularly intramolecular base pairings. Although each RNA molecule has only a single polynucleotide chain, the molecule is typically characterized by regions of (intramolecular) base pairs. According to the invention, the term “secondary structure” comprises structural motifs including without limitation base pairs, stems, stem loops, bulges, loops such as interior loops and multi-branch loops. The secondary structure of a nucleic acid molecule can be represented by a two-dimensional drawing (planar graph), showing base pairings (for further details on secondary structure of RNA molecules, see Auber et al., 2006; J. Graph Algorithms Appl. 10:329-351).

[0157] As described herein, the secondary structure of certain RNA molecules is relevant in the context of the present invention.

[0158] According to the invention, secondary structure of a nucleic acid molecule, particularly of a single-stranded RNA molecule, is determined by prediction using the web server for RNA secondary structure prediction

(<http://ma.urmc.rochester.edu/RNAstatureWeb/Servers/Predict1/Predict1.html>). Preferably, according to the invention, “secondary structure”, with reference to a nucleic acid molecule, specifically refers to the secondary structure determined by said prediction. The prediction may also be performed or confirmed using MFOLD structure prediction (<http://unafold.ma.albany.edu/?q=mfold>).

[0159] According to the invention, a “base pair” is a structural motif of a secondary structure wherein two nucleotide bases associate with each other through hydrogen bonds between donor and acceptor sites on the bases. The complementary bases, A:U and G:C, form stable base pairs through hydrogen bonds between donor and acceptor sites on the bases; the A:U and G:C base pairs are called Watson-Crick base pairs. A weaker base pair (called Wobble base pair) is formed by the bases G and U (G:U). The base pairs A:U and G:C are called canonical base pairs. Other base pairs like G:U (which occurs fairly often in RNA) and other rare base-pairs (e.g. A:C; U:U) are called non-canonical base pairs.

[0160] According to the invention, “nucleotide pairing” refers to two nucleotides that associate with each other so that their bases form a base pair (canonical or non-canonical base pair, preferably canonical base pair, most preferably Watson-Crick base pair).

[0161] According to the invention, the terms “stem loop” or “hairpin” or “hairpin loop”, with reference to a nucleic acid molecule, all interchangeably refer to a particular secondary structure of a nucleic acid molecule, typically a single-stranded nucleic acid molecule, such as single-stranded RNA. The particular secondary structure represented by the stem loop consists of a consecutive nucleic acid sequence comprising a stem and a (terminal) loop, also called hairpin loop, wherein the stem is formed by two neighbored entirely or partially complementary sequence elements; which are separated by a short sequence (e.g. 3-10 nucleotides), which forms the loop of the stem-loop structure.

[0162] The two neighbored entirely or partially complementary sequences may be defined as, e.g., stem loop elements stem 1 and stem 2. The stem loop is formed when these two neighbored entirely or partially reverse complementary sequences, e.g. stem loop elements stem 1 and stem 2, form base-pairs with each other, leading to a double stranded nucleic acid sequence comprising an unpaired loop at its terminal ending formed by the short sequence located between stem loop elements stem 1 and stem 2. Thus, a stem loop comprises two stems (stem 1 and stem 2), which—at the level of secondary structure of the nucleic acid molecule—form base pairs with each other, and which—at the level of the primary structure of the nucleic acid molecule—are separated by a short sequence that is not part of stem 1 or stem 2. For illustration, a two-dimensional representation of the stem loop resembles a lollipop-shaped structure. The formation of a stem-loop structure requires the presence of a sequence that can fold back on itself to form a paired double strand; the paired double strand is formed by stem 1 and stem 2. The stability of paired stem loop elements is typically determined by the length, the number of nucleotides of stem 1 that are capable of forming base pairs (preferably canonical base pairs, more preferably Watson-Crick base pairs) with nucleotides of stem 2, versus the number of nucleotides of stem 1 that are not capable of forming such base pairs with nucleotides of stem 2 (mismatches or bulges). According to the present invention, the optimal loop length is 3-10 nucleotides, more preferably 4 to 7, nucleotides, such as 4 nucleotides, 5 nucleotides, 6 nucleotides or 7 nucleotides. If a given nucleic acid sequence is characterized by a stem loop, the respective complementary nucleic acid sequence is typically also characterized by a stem loop. A stem loop is typically formed by single-stranded RNA molecules. For example, several stem loops are present in the 5' replication recognition sequence of alphaviral genomic RNA.

[0163] According to the invention, “disruption” or “disrupt”, with reference to a specific secondary structure of a nucleic acid molecule (e.g., a stem loop) means that the specific secondary structure is absent or altered. Typically, a secondary structure may be disrupted as a consequence of a change of at least one nucleotide that is part of the secondary structure. For example, a stem loop may be disrupted by change of one or more nucleotides that form the stem, so that nucleotide pairing is not possible.

[0164] According to the invention, “compensates for secondary structure disruption” or “compensating for secondary structure disruption” refers to one or more nucleotide changes in a nucleic acid sequence; more typically it refers to one or more second nucleotide changes in a nucleic acid sequence, which nucleic acid sequence also comprises one or more first nucleotide changes, characterized as follows: while the one or more first nucleotide changes, in the absence of the one or more second nucleotide changes, cause a disruption of the secondary structure of the nucleic acid sequence, the co-occurrence of the one or more first nucleotide changes and the one or more second nucleotide changes does not cause the secondary structure of the nucleic acid to be disrupted. Co-occurrence means presence of both the one or more first nucleotide changes and of the one or more second nucleotide changes. Typically, the one or more first nucleotide changes and the one or more second nucleotide changes are present together in the same nucleic acid molecule. In a specific embodiment, one or more nucleotide changes that compensate for secondary structure disruption is/are one or more nucleotide changes that compensate for one or more nucleotide pairing disruptions. Thus, in one embodiment, “compensating for secondary structure disruption” means “compensating for nucleotide pairing disruptions”, i.e. one or more nucleotide pairing disruptions, for example one or more nucleotide pairing disruptions within one or more stem loops. The one or more one or more nucleotide pairing disruptions may have been introduced by the removal of at least one initiation codon. Each of the one or more nucleotide changes that compensates for secondary structure disruption is a nucleotide change, which can each be independently selected from a deletion, an addition, a substitution and/or an insertion of one or more nucleotides. In an illustrative example, when the nucleotide pairing A:U has been disrupted by substitution of A to C (C and U are not typically suitable to form a nucleotide pair); then a nucleotide change that compensates for nucleotide pairing disruption may be substitution of U by G, thereby enabling formation of the C:G nucleotide pairing. The substitution of U by G thus compensates for the nucleotide pairing disruption. In an alternative example, when the nucleotide pairing A:U has been disrupted by substitution of A to C; then a nucleotide change that compensates for nucleotide pairing disruption may be substitution of C by A, thereby restoring formation of the original A:U nucleotide pairing. In general, in the present invention, those nucleotide changes compensating for secondary structure disruption are preferred which do neither restore the original nucleic acid sequence nor create novel AUG triplets. In the above set of examples, the U to G substitution is preferred over the C to A substitution.

[0165] According to the invention, the term “tertiary structure”, with reference to a nucleic acid molecule, refers to the three-dimensional structure of a nucleic acid molecule, as defined by the atomic coordinates.

[0166] According to the invention, a nucleic acid such as RNA, e.g., rRNA, may encode a peptide or protein. Accordingly, a transcribable nucleic acid sequence or a transcript thereof may contain an open reading frame (ORF) encoding a peptide or protein.

[0167] According to the invention, the term “nucleic acid encoding a peptide or protein” means that the nucleic acid, if present in the appropriate environment, preferably within a cell, can direct the assembly of amino acids to produce the peptide or protein during the process of translation. Preferably, coding RNA according to the invention is able to interact with the cellular translation machinery allowing translation of the coding RNA to yield a peptide or protein.

[0168] According to the invention, the term “peptide” comprises oligo- and polypeptides and refers to substances which comprise two or more, preferably 3 or more, preferably 4 or more, preferably 6 or more, preferably 8 or more, preferably 10 or more, preferably 13 or more, preferably 16 or more, preferably 20 or more, and up to preferably 50, preferably 100 or preferably 150, consecutive amino acids linked to one another via peptide bonds. The term “protein” refers to large peptides, preferably peptides having at least 151 amino acids, but the terms “peptide” and “protein” are used herein usually as synonyms.

[0169] The terms “peptide” and “protein” comprise, according to the invention, substances which contain not only amino acid components but also non-amino acid components such as sugars and phosphate structures, and also comprise substances containing bonds such as ester, thioether or disulfide bonds.

[0170] According to the invention, the terms “initiation codon” and “start codon” synonymously refer to a codon (base triplet) of an RNA molecule that is potentially the first codon that is translated by a ribosome. Such codon typically encodes the amino acid methionine in eukaryotes and a modified methionine in prokaryotes. The most common initiation codon in eukaryotes and prokaryotes is AUG. Unless specifically stated herein that an initiation codon other than AUG is meant, the terms “initiation codon” and “start codon”, with reference to an RNA molecule, refer to the codon AUG. According to the invention, the terms “initiation codon” and “start codon” are also used to refer to a corresponding base triplet of a deoxyribonucleic acid, namely the base triplet encoding the initiation codon of an RNA. If the initiation codon of messenger RNA is AUG, the base triplet encoding the AUG is ATG. According to the invention, the terms “initiation codon” and “start codon” preferably refer to a functional initiation codon or start codon, i.e., to an initiation codon or start codon that is used or would be used as a codon by a ribosome to start translation. There may be AUG codons in an RNA molecule that are not used as codons by a ribosome to start translation, e.g., due to a short distance of the codons to the cap. These codons are not encompassed by the term functional initiation codon or start codon.

[0171] According to the invention, the terms “start codon of the open reading frame” or “initiation codon of the open reading frame” refer to the base triplet that serves as initiation codon for protein synthesis in a coding sequence, e.g., in the coding sequence of a nucleic acid molecule found in nature. In an RNA molecule, the start codon of the open reading frame is often preceded by a 5' untranslated region (5'-UTR), although this is not strictly required.

[0172] According to the invention, the terms “native start codon of the open reading frame” or “native initiation codon of the open reading frame” refer to the base triplet that serves as initiation codon for protein synthesis in a native coding sequence. A native coding sequence may be, e.g., the coding sequence of a nucleic acid molecule found in nature. In some embodiments, the present invention provides variants of nucleic acid molecules found in nature, which are characterized in that the native start codon (which is present in the native coding sequence) has been removed (so that it is not present in the variant nucleic acid molecule).

[0173] According to the invention, “first AUG” means the most upstream AUG base triplet of a messenger RNA molecule, preferably the most upstream AUG base triplet of a messenger RNA molecule that is used or would be used as a codon by a ribosome to start translation. Accordingly, “first ATG” refers to the ATG base triplet of a coding DNA sequence that encodes the first AUG. In some instances, the first AUG of a mRNA molecule is the start codon of an open reading frame, i.e., the codon that is used as start codon during ribosomal protein synthesis.

[0174] According to the invention, the terms “comprises the removal” or “characterized by the removal” and similar terms, with reference to a certain element of a nucleic acid variant, mean that said certain element is not functional or not present in the nucleic acid variant, compared to a reference nucleic acid molecule. Without limitation, a removal can consist of deletion of all or part of the certain element, of substitution of all or part of the certain element, or of alteration of the functional or structural properties of the certain element. The removal of a functional element of a nucleic acid sequence requires that the function is not exhibited at the position of the nucleic acid variant comprising the removal. For example, an RNA variant characterized by the removal of a certain initiation codon requires that ribosomal protein synthesis is not initiated at the position of the RNA variant characterized by the removal. The removal of a structural element of a nucleic acid sequence requires that the structural element is not present at the position of the nucleic acid variant comprising the removal. For example, a RNA variant characterized by the removal of a certain AUG base triplet, i.e., of a AUG base triplet at a certain position, may be characterized, e.g., by deletion of part or all of the certain AUG base triplet (e.g., Δ AUG), or by substitution of one or more nucleotides (A, U, G) of the certain AUG base triplet by any one or more different nucleotides, so that the resulting nucleotide sequence of the variant does not comprise said AUG base triplet. Suitable substitutions of one nucleotide are those that convert the AUG base triplet into a GUG, CUG or UUG base triplet, or into a AAG, ACG or AGG base triplet, or into a AUA, AUC or AUU base triplet. Suitable substitutions of more nucleotides can be selected accordingly.

[0175] According to the invention, the term “self-replicating virus” includes RNA viruses capable of replicating autonomously in a host cell. Self-replicating viruses may have a single-stranded RNA (ssRNA) genome and include alphaviruses, flaviviruses, measles viruses (MV) and rhabdoviruses. Alphaviruses and flaviviruses possess a genome of positive polarity, whereas the genome of measles viruses (MV) and rhabdoviruses is negative strand ssRNA. Typically, a self-replicating virus is a virus with a (+) stranded RNA genome which can be directly translated after infection of a cell, and this translation provides an RNA-dependent RNA polymerase which then produces both antisense and sense transcripts from the infected RNA. In the following, the invention is illustrated by referring to alphavirus-derived vectors as an example of self-replicating virus-derived vectors. However, it is to be understood that the present invention is not limited to alphavirus-derived vectors.

[0176] According to the invention, the term “alphavirus” is to be understood broadly and includes any virus particle that has characteristics of alphaviruses. Characteristics of alphavirus include the presence of a (+) stranded RNA which encodes genetic information suitable for replication in a host cell, including RNA polymerase activity. Further characteristics of many alphaviruses are described, e.g., in Strauss & Strauss, 1994, Microbiol. Rev. 58:491-562.

[0177] The term “alphavirus” includes alphavirus found in nature, as well as any variant or derivative thereof. In some embodiments, a variant or derivative is not found in nature.

[0178] In one embodiment, the alphavirus is an alphavirus found in nature. Typically, an alphavirus found in nature is infectious to any one or more eukaryotic organisms, such as an animal (including a vertebrate such as a human, and an arthropod such as an insect). An alphavirus found in nature is preferably selected from the group consisting of the following: Barmah Forest virus complex (comprising Barmah Forest virus); Eastern equine encephalitis complex (comprising seven antigenic types of Eastern equine encephalitis virus); Middelburg virus complex (comprising Middelburg virus); Ndumu virus complex (comprising Ndumu virus); Semliki Forest virus complex (comprising Bebaru virus, Chikungunya virus, Mayaro virus and its subtype Una virus, O'Nyong Nyong virus, and its subtype Igbo-Ora virus, Ross River virus and its subtypes Bebaru virus, Getah virus, Sagiya virus, Semliki Forest virus and its subtype Me Tri virus); Venezuelan equine encephalitis complex (comprising Cabassou virus, Everglades virus, Mosso das Pedras virus, Mucambo virus, Paramana virus, Pixuna virus, Rio Negro virus, Trocara virus and its subtype Bijou Bridge virus, Venezuelan equine encephalitis virus); Western equine encephalitis complex (comprising Aura virus, Babanki virus, Kyzylgach virus, Sindbis virus, Ockelbo virus, Whataroa virus, Buggy Creek virus, Fort Morgan virus, Highlands J virus, Western equine encephalitis virus); and some unclassified viruses including Salmon pancreatic disease virus; Sleeping Disease virus; Southern elephant seal virus; Tonate virus. More preferably, the alphavirus is selected from the group consisting of Semliki Forest virus complex (comprising the virus types as indicated above, including Semliki Forest virus), Western equine encephalitis complex (comprising the virus types as indicated above, including Sindbis virus), Eastern equine encephalitis virus (comprising the virus types as indicated above), Venezuelan equine encephalitis complex (comprising the virus types as indicated above, including Venezuelan equine encephalitis virus).

[0179] In a further preferred embodiment, the alphavirus is Semliki Forest virus. In an alternative further preferred embodiment, the alphavirus is Sindbis virus. In an alternative further preferred embodiment, the alphavirus is Venezuelan equine encephalitis virus.

[0180] In some embodiments of the present invention, the alphavirus is not an alphavirus found in nature. Typically, an alphavirus not found in

nature is a variant or derivative of an alphavirus found in nature, that is distinguished from an alphavirus found in nature by at least one mutation in the nucleotide sequence, i.e., the genomic RNA. The mutation in the nucleotide sequence may be selected from an insertion, a substitution or a deletion of one or more nucleotides, compared to an alphavirus found in nature. A mutation in the nucleotide sequence may or may not be associated with a mutation in a polypeptide or protein encoded by the nucleotide sequence. For example, an alphavirus not found in nature may be an attenuated alphavirus. An attenuated alphavirus not found in nature is an alphavirus that typically has at least one mutation in its nucleotide sequence by which it is distinguished from an alphavirus found in nature, and that is either not infectious at all, or that is infectious but has a lower disease-producing ability or no disease-producing ability at all. As an illustrative example, TC83 is an attenuated alphavirus that is distinguished from the Venezuelan equine encephalitis virus (VEEV) found in nature (McKinney et al., 1963, *Am. J. Trop. Med. Hyg.* 12:597-603).

[0181] Members of the alphavirus genus may also be classified based on their relative clinical features in humans: alphaviruses associated primarily with encephalitis, and alphaviruses associated primarily with fever, rash, and polyarthrititis.

[0182] The term “alphaviral” means found in an alphavirus, or originating from an alphavirus or derived from an alphavirus, e.g., by genetic engineering.

[0183] According to the invention, “SFV” stands for Semliki Forest virus. According to the invention, “SIN” or “SINV” stands for Sindbis virus. According to the invention, “VEE” or “VEEV” stands for Venezuelan equine encephalitis virus.

[0184] According to the invention, the term “of an alphavirus” refers to an entity of origin from an alphavirus. For illustration, a protein of an alphavirus may refer to a protein that is found in alphavirus and/or to a protein that is encoded by alphavirus; and a nucleic acid sequence of an alphavirus may refer to a nucleic acid sequence that is found in alphavirus and/or to a nucleic acid sequence that is encoded by alphavirus. Preferably, a nucleic acid sequence “of an alphavirus” refers to a nucleic acid sequence “of the genome of an alphavirus” and/or “of genomic RNA of an alphavirus”.

[0185] According to the invention, the term “alphaviral RNA” refers to any one or more of alphaviral genomic RNA (i.e., (+) strand), complement of alphaviral genomic RNA (i.e., (–) strand), and the subgenomic transcript (i.e., (+) strand), or a fragment of any thereof.

[0186] According to the invention, “alphavirus genome” refers to genomic (+) strand RNA of an alphavirus.

[0187] According to the invention, the term “native alphavirus sequence” and similar terms typically refer to a (e.g., nucleic acid) sequence of a naturally occurring alphavirus (alphavirus found in nature). In some embodiments, the term “native alphavirus sequence” also includes a sequence of an attenuated alphavirus.

[0188] According to the invention, the term “5′ replication recognition sequence” preferably refers to a continuous nucleic acid sequence, preferably a ribonucleic acid sequence, that is identical or homologous to a 5′ fragment of a genome of a self-replicating virus, such as an alphavirus genome. The “5′ replication recognition sequence” is a nucleic acid sequence that can be recognized by a replicase such as an alphaviral replicase. The term 5′ replication recognition sequence includes native 5′ replication recognition sequences as well as functional equivalents thereof, such as, e.g., functional variants of a 5′ replication recognition sequence of a self-replicating virus found in nature, e.g., alphavirus found in nature. According to the invention, functional equivalents include derivatives of 5′ replication recognition sequences characterized by the removal of at least one initiation codon as described herein. The 5′ replication recognition sequence is required for synthesis of the (–) strand complement of alphavirus genomic RNA, and is required for synthesis of (+) strand viral genomic RNA based on a (–) strand template. A native 5′ replication recognition sequence typically encodes at least the N-terminal fragment of nsP1; but does not comprise the entire open reading frame encoding nsP1234. In view of the fact that a native 5′ replication recognition sequence typically encodes at least the N-terminal fragment of nsP1, a native 5′ replication recognition sequence typically comprises at least one initiation codon, typically AUG. In one embodiment, the 5′ replication recognition sequence comprises conserved sequence element 1 of an alphavirus genome (CSE 1) or a variant thereof and conserved sequence element 2 of an alphavirus genome (CSE 2) or a variant thereof. The 5′ replication recognition sequence is typically capable of forming four stem loops (SL), i.e. SL1, SL2, SL3, SL4. The numbering of these stem loops begins at the 5′ end of the 5′ replication recognition sequence.

[0189] The term “conserved sequence element” or “CSE” refers to a nucleotide sequence found in alphavirus RNA. These sequence elements are termed “conserved” because orthologs are present in the genome of different alphaviruses, and orthologous CSEs of different alphaviruses preferably share a high percentage of sequence identity and/or a similar secondary or tertiary structure. The term CSE includes CSE 1, CSE 2, CSE 3 and CSE 4.

[0190] According to the invention, the terms “CSE 1” or “44-nt CSE” synonymously refer to a nucleotide sequence that is required for (+) strand synthesis from a (–) strand template. The term “CSE 1” refers to a sequence on the (+) strand; and the complementary sequence of CSE 1 (on the (–) strand) functions as a promoter for (+) strand synthesis. Preferably, the term CSE 1 includes the most 5′ nucleotide of the alphavirus genome. CSE 1 typically forms a conserved stem-loop structure. Without wishing to be bound to a particular theory, it is believed that, for CSE 1, the secondary structure is more important than the primary structure, i.e., the linear sequence. In genomic RNA of the model alphavirus Sindbis virus, CSE 1 consists of a consecutive sequence of 44 nucleotides, which is formed by the most 5′44 nucleotides of the genomic RNA (Strauss & Strauss, 1994, *Microbiol. Rev.* 58:491-562).

[0191] According to the invention, the terms “CSE 2” or “51-nt CSE” synonymously refer to a nucleotide sequence that is required for (–) strand synthesis from a (+) strand template. The (+) strand template is typically alphavirus genomic RNA or an RNA replicon (note that the subgenomic RNA transcript, which does not comprise CSE 2, does not function as a template for (–) strand synthesis). In alphavirus genomic RNA, CSE 2 is typically localized within the coding sequence for nsP1. In genomic RNA of the model alphavirus Sindbis virus, the 51-nt CSE is located at nucleotide positions 155-205 of genomic RNA (Frolov et al., 2001, *RNA*, vol. 7, pp. 1638-1651). CSE 2 forms typically two conserved stem loop structures. These stem loop structures are designated as stem loop 3 (SL3) and stem loop 4 (SL4) because they are the third and fourth conserved stem loop, respectively, of alphavirus genomic RNA, counted from the 5′ end of alphavirus genomic RNA. Without wishing to be bound to a particular theory, it is believed that, for CSE 2, the secondary structure is more important than the primary structure, i.e. the linear sequence.

[0192] According to the invention, the terms “CSE 3” or “junction sequence” synonymously refer to a nucleotide sequence that is derived from alphaviral genomic RNA and that comprises the start site of the subgenomic RNA. The complement of this sequence in the (–) strand acts to promote subgenomic RNA transcription. In alphavirus genomic RNA, CSE 3 typically overlaps with the region encoding the C-terminal fragment of nsP4 and extends to a short non-coding region located upstream of the open reading frame encoding the structural proteins.

[0193] According to the invention, the terms “CSE 4” or “19-nt conserved sequence” or “19-nt CSE” synonymously refer to a nucleotide sequence from alphaviral genomic RNA, immediately upstream of the poly(A) sequence in the 3′ untranslated region of the alphavirus genome. CSE 4 typically consists of 19 consecutive nucleotides. Without wishing to be bound to a particular theory, CSE 4 is understood to function as a core promoter for initiation of (–) strand synthesis (José et al., 2009, *Future Microbiol.* 4:837-856); and/or CSE 4 and the poly(A) tail of the alphavirus genomic RNA are understood to function together for efficient (–) strand synthesis (Hardy & Rice, 2005, *J. Virol.* 79:4630-4639).

[0194] According to the invention, the term “subgenomic promoter” or “SGP” refers to a nucleic acid sequence upstream (5′) of a nucleic acid sequence (e.g., coding sequence), which controls transcription of said nucleic acid sequence by providing a recognition and binding site for RNA polymerase, typically RNA-dependent RNA polymerase, in particular functional alphavirus non-structural protein. The SGP may include further recognition or binding sites for further factors. A subgenomic promoter is typically a genetic element of a positive strand RNA virus, such as an alphavirus. A subgenomic promoter of alphavirus is a nucleic acid sequence comprised in the viral genomic RNA. The subgenomic promoter is generally characterized in that it allows initiation of the transcription (RNA synthesis) in the presence of an RNA-dependent RNA polymerase, e.g., functional alphavirus non-structural protein. An RNA (–) strand, i.e., the complement of alphaviral genomic RNA, serves as a template for synthesis of a (+) strand subgenomic transcript, and synthesis of the (+) strand subgenomic transcript is typically initiated at or near the subgenomic promoter.

The term “subgenomic promoter” as used herein, is not confined to any particular localization in a nucleic acid comprising such subgenomic promoter. In some embodiments, the SGP is identical to CSE 3 or overlaps with CSE 3 or comprises CSE 3.

[0195] The terms “subgenomic transcript” or “subgenomic RNA” synonymously refer to an RNA molecule that is obtainable as a result of transcription using a RNA molecule as template (“template RNA”), wherein the template RNA comprises a subgenomic promoter that controls transcription of the subgenomic transcript. The subgenomic transcript is obtainable in the presence of an RNA-dependent RNA polymerase, in particular functional alphavirus non-structural protein. For instance, the term “subgenomic transcript” may refer to the RNA transcript that is prepared in a cell infected by an alphavirus, using the (–) strand complement of alphavirus genomic RNA as template. However, the term “subgenomic transcript”, as used herein, is not limited thereto and also includes transcripts obtainable by using heterologous RNA as template. For example, subgenomic transcripts are also obtainable by using the (–) strand complement of SGP-containing replicons according to the present invention as template. Thus, the term “subgenomic transcript” may refer to an RNA molecule that is obtainable by transcribing a fragment of alphavirus genomic RNA, as well as to an RNA molecule that is obtainable by transcribing a fragment of a replicon according to the present invention.

[0196] The term “autologous” is used to describe anything that is derived from the same subject. For example, “autologous cell” refers to a cell derived from the same subject introduction of autologous cells into a subject is advantageous because these cells overcome the immunological barrier which otherwise results in rejection.

[0197] The term “allogeneic” is used to describe anything that is derived from different individuals of the same species.

[0198] Two or more individuals are said to be allogeneic to one another when the genes at one or more loci are not identical.

[0199] The term “syngeneic” is used to describe anything that is derived from individuals or tissues having identical genotypes, i.e., identical twins or animals of the same inbred strain, or their tissues or cells.

[0200] The term “heterologous” is used to describe something consisting of multiple different elements. As an example, the introduction of one individual's cell into a different individual constitutes a heterologous transplant. A heterologous gene is a gene derived from a source other than the subject.

[0201] Cells that may be used in the methods for identifying sequence changes are any appropriate cell in which the replicable RNA, with or without any nucleotide modifications, can be replicated and/or translated. The cell may be a mammalian cell, for example, a human cell. The cell may constitutively express a replicase which recognizes the sequences present in the replicable RNA for replication or may transiently express such replicase.

[0202] The following provides specific and/or preferred variants of the individual features of the invention. The present invention also contemplates as particularly preferred embodiments those embodiments, which are generated by combining two or more of the specific and/or preferred variants described for two or more of the features of the present invention.

[0203] RNA replicon A replicable RNA molecule (rRNA) is an RNA that is able to be replicated by an RNA-dependent RNA polymerase (replicase) by virtue of comprising nucleotide sequences that can be recognized by the replicase such that the RNA is replicated. The rRNA does not necessarily encode the replicase, such that rRNAs can be replicated in cis (by the encoded replicase) or in trans (by a replicase provided in another manner, e.g., a separate replicase encoding nucleic acid, such as an mRNA). The terms “RNA replicon”, “replicon” and “replicable RNA molecule” can be used interchangeably.

[0204] In an embodiment, the replicable RNA (rRNA) molecule comprises a modified regulatory region of a self-replicating single-stranded positive-sense virus comprising sequence changes compared to a reference modified regulatory region, which sequence changes restore or improve the function of the rRNA molecule that comprises at least one modified nucleotide. These changes may be identified by the methods described herein for identifying such sequence changes. In an embodiment, the modified regulatory region is an alphavirus regulatory region, e.g., a 5′ or a 3′ regulatory region. In an embodiment, the 5′ regulatory region is the VEEV alphaviral 5′ regulatory region.

[0205] In an embodiment, an RNA replicon may comprise an internal ribosome entry site (IRES) and an open reading frame encoding a functional non-structural protein from a self-replicating virus, wherein the IRES controls expression of the functional non-structural protein, e.g., a replicase. Preferably, the RNA replicon contains sequence elements allowing replication by the functional non-structural protein. In one embodiment, the self-replicating virus is an alphavirus and the sequence elements allowing replication by the functional non-structural protein are derived from an alphavirus.

[0206] Alphavirus replicases have a capping enzyme function, and, typically, genomic as well as subgenomic (+) stranded RNAs are capped. The 5′-cap serves to protect mRNA from degradation, and to direct the ribosomal subunits as well as cellular factors to the mRNA in order to form a ribonucleoprotein complex on the mRNA that then can start translation from a nearby start codon. This complex process is extensively described in the literature (Jackson et al., 2010, Nat Rev Mol Biol; Vol 10:113-127). Despite the very elaborated and efficient mechanism of cap dependent translation, cells have means to initiate translation fully or partially independently from the 5′ cap (Thompson 2012; Trends in Microbiology 20:558-566). Thereby, in situations of cellular stress that lead to a global down regulation of cap-dependent translation, the cells may still express selected genes preferentially, often with the help of an IRES.

[0207] Viruses also evolved different means to exploit the cells machinery for translation of the viral genes. Since a viral infection is often sensed by the cell which leads to cellular antiviral response (interferon response; stress response), many viruses also make use of cap-independent translation, especially RNA viruses. Cap independent translation ensure an advantage for the viral RNA translation upon cellular stress response giving the viruses the opportunity to fulfil their life cycle and be released from infected cells.

[0208] Internal ribosomal entry sites (IRESs) are RNA sequences forming appropriate secondary structures that attract the pre-initiation complex near to a translational start codon, AUG or others. Four classes of IRESs are described in literature that share common features. Prototypic IRESs are the poliovirus IRES (Type I), the encephalomyocarditis virus (EMCV) IRES (Type II), the hepatitis C virus (HCV) IRES (Type III) and the IRES found in the intergenic regions of dicistroviruses (Type IV) (Thompson, 2012; Trends in Microbiology 20:558-566; Lozano et al., 2018; Open Biology 8:180155).

[0209] Type I to III IRESs have in common that they initiate translation at AUG start codons, whereas type IV IRES initiate at non-AUG codons (e.g., GCU). Thereby Type I to III require the initiator tRNA that delivers methionine by the help of eIF2/GTP (eIF2/GTP/Met-tRNA^{iMet}). Activation of eIF2 kinases under stress phosphorylates the alpha subunit of eIF2 which inhibits translation that initiates at AUG. Thereby translation directed by type IV IRESs are not inhibited by eIF2 phosphorylation.

[0210] The term “internal ribosome entry site”, abbreviated “IRES”, relates to an RNA element that recruits ribosomes to the internal region of mRNAs to initiate translation in a cap-independent manner. IRESs are commonly located in the 5′-UTR of RNA viruses. However, mRNAs of viruses from dicistroviridae family possess two open reading frames (ORFs), and translation of each is directed by two distinct IRESs. It has also been suggested that some mammalian cellular mRNAs also have IRESs. These cellular IRES elements are thought to be located in eukaryotic mRNAs encoding genes involved in stress survival, and other processes critical to survival. The location for IRES elements is often in the 5′-UTR, but can also occur elsewhere in mRNAs.

[0211] The term “internal ribosome entry site” includes IRESs that are present in the viruses of the Picornaviridae family such as poliovirus (PV) and encephalomyocarditis virus and pathogenic viruses, including human immunodeficiency virus, hepatitis C virus (HCV), and foot and mouth disease virus. Although these viral IRESs contain diverse sequences, many of them have similar secondary structures and initiate translation through similar mechanisms. In addition, the activities of IRESs often require assistance from other factors known as IRES-transacting factors (ITAFs). Based on the structures and the requirement of translation initiation factors (IFs) and ITAFs, the viral IRESs are classified into four types as described herein. Any

of these IRES types is useful according to the invention, with Type IV IRESs being particularly preferred.

[0212] Two groups of viral IRESs, Type I and Type II, cannot bind to the 40S small ribosomal subunit directly. Instead, they recruit the 40S small ribosomal subunit through different ITAFs and require canonical IFs in the cap-dependent translation (i.e., eIF2, eIF3, eIF4A, eIF48, and eIF4G). The major difference between Type I and Type II IRESs is the requirement of 40S ribosome scanning, with 40S ribosome scanning being unnecessary for Type II IRES.

[0213] Examples of Type I IRESs include IRESs found in poliovirus (PV) and rhinovirus. Examples of Type II IRESs include IRESs found in encephalomyocarditis virus (EMCV), foot-and-mouth disease virus (FMDV) and Theiler's murine encephalomyelitis viruses (TMEV).

[0214] Type III IRESs can directly interact with 40S small ribosomal subunit with specialized RNA structure, but their activities usually require assistance of several IFs including eIF2 and eIF3 and initiator Met-tRNAi. Examples include IRESs found in hepatitis C-virus (HCV), classical swine fever virus (CSFV) and porcine teschovirus (PTV).

[0215] Type IV viral IRESs generally have strong activities and can initiate translation from a non-AUG start codon without additional ITAFs or even eIF2/Met-tRNAi/GTP ternary complex. These IRESs are folded to a compact structure that directly interacts with the 40S small ribosomal subunit. Examples include IRESs found in dicistroviruses such as cricket paralysis virus (CrPV), plautia stall intestine virus (PSIV), and Taura-Syndrom-Virus (TSV).

[0216] The term "internal ribosome entry site" also includes IRESs found in cellular mRNAs, many of which encode proteins required in stress response, e.g. In conditions of apoptosis, mitosis, hypoxia, and nutrient limitation. The cellular IRESs can be roughly classified into two types based on the mechanisms of ribosome recruitment: Type I IRESs interact with ribosomes through ITAFs that bound on the cis-elements, e.g., RNA binding motifs and N-6-methyladenosine (m6A) modification, whereas Type II IRESs contain a short cis-element that pairs with 18S rRNA to recruit ribosomes.

[0217] In an embodiment, the rRNA described herein may have modified nucleotides/nucleosides/backbone modifications.

[0218] The term "RNA modification" as used herein may refer to chemical modifications comprising backbone modifications as well as sugar modifications or base modifications.

[0219] In this context, a modified rRNA molecule as defined herein may contain nucleotide analogues/modifications, e.g., backbone modifications, sugar modifications or base modifications. A backbone modification in connection with the present invention is a modification, in which phosphates of the backbone of the nucleotides contained in an rRNA molecule as defined herein are chemically modified. A sugar modification in connection with the present invention is a chemical modification of the sugar of the nucleotides of the rRNA molecule as defined herein. Furthermore, a base modification in connection with the present invention is a chemical modification of the base moiety of the nucleotides of the rRNA molecule. In this context, nucleotide analogues or modifications are preferably selected from nucleotide analogues, which are applicable for transcription and/or translation.

[0220] Sugar Modifications: The modified nucleosides and nucleotides, which may be incorporated into a modified rRNA molecule as described herein, can be modified in the sugar moiety. For example, the 2' hydroxyl group (OH) can be modified or replaced with a number of different "oxy" or "deoxy" substituents. Examples of "oxy"-2' hydroxyl group modifications include, but are not limited to, alkoxy or aryloxy (---OR , e.g., R=H , alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar); polyethyleneglycols (PEG), $\text{---O(CH}_2\text{CH}_2\text{O)}_n\text{CH}_2\text{CH}_2\text{OR}$; "locked" nucleic acids (LNA) in which the 2' hydroxyl is connected, e.g., by a methylene bridge, to the 4' carbon of the same ribose sugar; and amino groups (---O-amino , wherein the amino group, e.g., NRR, can be alkylamino, dialkylamino, heterocyclyl, arylamino, diarylamino, heteroarylamino, or diheteroaryl amino, ethylene diamine, polyamino) or aminoalkoxy. "Deoxy" modifications include hydrogen, amino (e.g. NH_2 ; alkylamino, dialkylamino, heterocyclyl, arylamino, diaryl amino, heteroaryl amino, diheteroaryl amino, or amino acid); or the amino group can be attached to the sugar through a linker, wherein the linker comprises one or more of the atoms C, N, and O. The sugar group can also contain one or more carbons that possess the opposite stereochemical configuration than that of the corresponding carbon in ribose. Thus, a modified RNA molecule can include nucleotides containing, for instance, arabinose as the sugar.

[0221] Backbone Modifications: The phosphate backbone may further be modified in the modified nucleosides and nucleotides, which may be incorporated into a modified RNA molecule as described herein. The phosphate groups of the backbone can be modified by replacing one or more of the oxygen atoms with a different substituent. Further, the modified nucleosides and nucleotides can include the full replacement of an unmodified phosphate moiety with a modified phosphate as described herein. Examples of modified phosphate groups include, but are not limited to, phosphorothioate, phosphoroselenates, borano phosphates, borano phosphate esters, hydrogen phosphonates, phosphoroamidates, alkyl or aryl phosphonates and phosphotriesters. Phosphorodithioates have both non-linking oxygens replaced by sulfur. The phosphate linker can also be modified by the replacement of a linking oxygen with nitrogen (bridged phosphoroamidates), sulfur (bridged phosphorothioates) and carbon (bridged methylene-phosphonates).

[0222] Base Modifications: The modified nucleosides and nucleotides, which may be incorporated into a modified rRNA molecule as described herein can further be modified in the nucleobase moiety. Examples of nucleobases found in RNA include, but are not limited to, adenine, guanine, cytosine and uracil. For example, the nucleosides and nucleotides described herein can be chemically modified on the major groove face. In some embodiments, the major groove chemical modifications can include an amino group, a thiol group, an alkyl group, or a halo group.

[0223] In particular embodiments of the present invention, the nucleotide analogues/modifications are selected from base modifications, which are preferably selected from 2-amino-6-chloropurineriboside-5'-triphosphate, 2-aminopurine-riboside-5'-triphosphate; 2-aminoadenosine-5'-triphosphate, 2'-amino-2'-deoxy-cytidine-triphosphate, 2-thiocytidine-5'-triphosphate, 2-thiouridine-5'-triphosphate, 2'-fluorothymidine-5'-triphosphate, 2'-0-methyl inosine-5'-triphosphate 4-thio-uridine-5'-triphosphate, 5-aminoallylcytidine-5'-triphosphate, 5-aminoallyluridine-5'-triphosphate, 5-bromocytidine-5'-triphosphate, 5-bromouridine-5'-triphosphate, 5-bromo-2'-deoxycytidine-5'-triphosphate, 5-bromo-2'-deoxyuridine-5'-triphosphate, 5-iodocytidine-5'-triphosphate, 5-iodo-2'-deoxycytidine-5'-triphosphate, 5-iodouridine-5'-triphosphate, 5-iodo-2'-deoxyuridine-5'-triphosphate, 5-methylcytidine-5'-triphosphate, 5-methyluridine-5'-triphosphate, 5-propynyl-2'-deoxycytidine-5'-triphosphate, 5-propynyl-2'-deoxyuridine-5'-triphosphate, 6-azacytidine-5'-triphosphate, 6-azauridine-5'-triphosphate, 6-choropurineriboside-5'-triphosphate, 7-deaza-adenosine-5'-triphosphate, 7-deazaguanosine-5'-triphosphate, 8-azaadenosine-5'-triphosphate, 8-azidoadenosine-5'-triphosphate, benzimidazole-riboside-5'-triphosphate, N1-methyladenosine-5'-triphosphate, N1-methylguanosine-5'-triphosphate, N6-methyladenosine-5'-triphosphate, 06-methylguanosine-5'-triphosphate, N6-methylguanosine-5'-triphosphate, pseudo-uridine-5'-triphosphate, or puromycin-5'-triphosphate, xanthosine-5'-triphosphate. Particular preference may be given to nucleotides for base modifications selected from the group of base-modified nucleotides consisting of 5-methylcytidine-5'-triphosphate, 7-deazaguanosine-5'-triphosphate, 5-bromocytidine-5'-triphosphate, and pseudouridine-5'-biphosphate. In some embodiments, modified nucleosides include pyridin-4-one ribonucleoside, 5-aza-uridine, 2-thio-5-aza-uridine, 2-thiouridine, 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyl-uridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thiouridine, 1-taurinomethyl-4-thio-uridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydro-pseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxy-uridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, and 4-methoxy-2-thio-pseudouridine.

[0224] In some embodiments, modified nucleosides include 5-aza-cytidine, pseudoisocytidine, 3-methyl-cytidine, N4-acetylcytidine, 5-formylcytidine, N4-methylcytidine, 5-hydroxymethylcytidine, 1-methyl-pseudoisocytidine, pyrrolo-cytidine, pyrrolo-pseudoisocytidine, 2-thio-cytidine, 2-thio-5-methyl-cytidine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-

cytidine, 2-methoxy-5-methyl-cytidine, 4-methoxy-pseudoisocytidine, and 4-methoxy-1-methyl-pseudoisocytidine.

[0225] In other embodiments, modified nucleosides include 2-aminopurine, 2,6-diaminopurine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-aminopurine, 7-deaza-8-aza-2-aminopurine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diamino-purine, 1-methyladenosine, N6-methyladenosine, N6-isopentenyladenosine, N6-(cis-hydroxyisopentenyl)adenosine, 2-methylthio-N6-(cis-hydroxyisopentenyl)adenosine, N6-glycylcarbamoyladenosine, N6-threonylcarbamoyladenosine, 2-methyl-thio-N6-threonylcarbamoyladenosine, N6,N6-dimethyladenosine, 7-methyladenine, 2-methylthio-adenine, and 2-methoxy-adenine. In other embodiments, modified nucleosides include inosine, 1-methyl-inosine, wyosine, wybutosine, 7-deaza-guanosine, 7-deaza-8-aza-guanosine, 6-thio-guanosine, 6-thio-7-deaza-guanosine, 6-thio-7-deaza-1-aza-guanosine, 7-methyl-guanosine, 6-thio-7-methyl-guanosine, 7-methylinosine, 6-methoxy-guanosine, 1-methylguanosine, N2-methylguanosine, N2,N2-dimethylguanosine, 8-oxo-guanosine, 7-methyl-8-oxo-guanosine, 1-methyl-6-thio-guanosine, N2-methyl-6-thio-guanosine, and N2,N2-dimethyl-6-thio-guanosine.

[0226] In some embodiments, the nucleotide can be modified on the major groove face and can include replacing hydrogen on C-5 of uracil with a methyl group or a halo group. In specific embodiments, a modified nucleoside is 5'-0-(1-thiophosphate)-adenosine, 5'-0-(1-thiophosphate)-cytidine, 5'-0-(1-thiophosphate)-guanosine, 5'-0-(1-thiophosphate)-uridine or 5'-0-(1-thiophosphate)-pseudouridine.

[0227] In further embodiments, a modified rRNA may comprise nucleoside modifications selected from 6-aza-cytidine, 2-thio-cytidine, a-thio-cytidine, pseudo-iso-cytidine, 5-aminoallyl-uridine, 5-iodo-uridine, N1-methyl-pseudouridine, 5,6-dihydrouridine, a-thio-uridine, 4-thio-uridine, 6-aza-uridine, 5-hydroxy-uridine, deoxy-thymidine, 5-methyl-uridine, pyrrolo-cytidine, inosine, a-thio-guanosine, 6-methyl-guanosine, 5-methyl-cytidine, 8-oxo-guanosine, 7-deaza-guanosine, N1-methyl-adenosine, 2-amino-6-chloro-purine, N6-methyl-2-amino-purine, pseudo-iso-cytidine, 6-chloro-purine, N6-methyl-adenosine, a-thio-adenosine, 8-azido-adenosine, 7-deaza-adenosine.

[0228] In certain preferred embodiments, the rRNA comprises a modified nucleoside in place of at least one (e.g., every) uridine.

[0229] The term “uracil,” as used herein, describes one of the nucleobases that can occur in the nucleic acid of RNA. The structure of uracil is: ##STR00004##

[0230] The term “uridine,” as used herein, describes one of the nucleosides that can occur in RNA. The structure of uridine is: ##STR00005##

[0231] UTP (uridine 5'-triphosphate) has the following structure: ##STR00006##

[0232] Pseudo-UTP (pseudouridine 5'-triphosphate) has the following structure: ##STR00007##

[0233] “Pseudouridine” is one example of a modified nucleoside that is an isomer of uridine, where the uracil is attached to the pentose ring via a carbon-carbon bond instead of a nitrogen-carbon glycosidic bond.

[0234] Another exemplary modified nucleoside is N1-methyl-pseudouridine (m1ψ), which has the structure: ##STR00008##

N1-methyl-pseudo-UTP has the following structure: ##STR00009##

[0235] Another exemplary modified nucleoside is 5-methyl-uridine (m5U), which has the structure: ##STR00010##

[0236] In certain preferred embodiments, one or more uridine in the rRNA described herein is replaced by a modified nucleoside. In some embodiments, the modified nucleoside is a modified uridine.

[0237] In certain preferred embodiments, RNA comprises a modified nucleoside in place of at least one uridine. In some embodiments, RNA comprises a modified nucleoside in place of each uridine.

[0238] In certain preferred embodiments, the modified nucleoside is independently selected from pseudouridine (ψ), N1-methyl-pseudouridine (m1ψ), and 5-methyl-uridine (m5ψ). In some embodiments, the modified nucleoside comprises pseudouridine (ψ). In some embodiments, the modified nucleoside comprises N1-methyl-pseudouridine (m1ψ). In some embodiments, the modified nucleoside comprises 5-methyl-uridine (m5U). In some embodiments, RNA may comprise more than one type of modified nucleoside, and the modified nucleosides are independently selected from pseudouridine (ψ), N1-methyl-pseudouridine (m1ψ), and 5-methyl-uridine (m5U). In some embodiments, the modified nucleosides comprise pseudouridine (ψ) and N1-methyl-pseudouridine (m1ψ). In some embodiments, the modified nucleosides comprise pseudouridine (ψ) and 5-methyl-uridine (m5U). In some embodiments, the modified nucleosides comprise N1-methyl-pseudouridine (m1ψ) and 5-methyl-uridine (m5U). In some embodiments, the modified nucleosides comprise pseudouridine (ψ), N1-methyl-pseudouridine (m1ψ), and 5-methyl-uridine (m5U).

[0239] In certain preferred embodiments, the modified nucleoside replacing one or more, e.g., all, uridine in the rRNA may be any one or more of 3-methyl-uridine (m.sup.3U), 5-methoxy-uridine (mo.sup.5U), 5-aza-uridine, 6-aza-uridine, 2-thio-5-aza-uridine, 2-thio-uridine (s.sup.2U), 4-thio-uridine (s.sup.4U), 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxy-uridine (ho.sup.5U), 5-aminoallyl-uridine, 5-halo-uridine (e.g., 5-iodo-uridine or 5-bromo-uridine), uridine 5-oxyacetic acid (cmo.sup.5U), uridine 5-oxyacetic acid methyl ester (mcmo.sup.5U), 5-carboxymethyl-uridine (cm.sup.5U), 1-carboxymethyl-pseudouridine, 5-carboxyhydroxymethyl-uridine (chm.sup.5U), 5-carboxyhydroxymethyl-uridine methyl ester (mchm.sup.5U), 5-methoxycarbonylmethyl-uridine (mcm.sup.5U), 5-methoxycarbonylmethyl-2-thio-uridine (mcm.sup.5s.sup.2U), 5-aminomethyl-2-thio-uridine (nm.sup.5s.sup.2U), 5-methylaminomethyl-uridine (mnm.sup.5U), 1-ethyl-pseudouridine, 5-methylaminomethyl-2-thio-uridine (mnm.sup.5s.sup.2U), 5-methylaminomethyl-2-seleno-uridine (mnm.sup.5se.sup.2U), 5-carbamoylmethyl-uridine (ncm.sup.5U), 5-carboxymethylaminomethyl-uridine (cmnm.sup.5U), 5-carboxymethylaminomethyl-2-thio-uridine (cmnm.sup.5s.sup.2U), 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyl-uridine (tm.sup.5U), 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine (tm5s2U), 1-taurinomethyl-4-thio-pseudouridine, 5-methyl-2-thio-uridine (m.sup.5s.sup.2U), 1-methyl-4-thio-pseudouridine (m.sup.1s.sup.4ψ), 4-thio-1-methyl-pseudouridine, 3-methyl-pseudouridine (m.sup.3ψ), 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine (D), dihydropseudouridine, 5,6-dihydrouridine, 5-methyl-dihydrouridine (m.sup.5D), 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxy-uridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, N1-methyl-pseudouridine, 3-(3-amino-3-carboxypropyl)uridine (acp.sup.3U), 1-methyl-3-(3-amino-3-carboxypropyl)pseudouridine (acp.sup.3ψ), 5-(isopentenylaminomethyl)uridine (inm.sup.5U), 5-(isopentenylaminomethyl)-2-thio-uridine (inm.sup.5s.sup.2U), α-thio-uridine, 2'-O-methyl-uridine (Um), 5,2'-O-dimethyl-uridine (m.sup.5Um), 2'-O-methyl-pseudouridine (ψm), 2-thio-2'-O-methyl-uridine (s.sup.2Um), 5-methoxycarbonylmethyl-2'-O-methyl-uridine (mcm.sup.5Um), 5-carbamoylmethyl-2'-O-methyl-uridine (ncm.sup.5Um), 5-carboxymethylaminomethyl-2'-O-methyl-uridine (cmnm.sup.5Um), 3,2'-O-dimethyl-uridine (m.sup.3Um), 5-(isopentenylaminomethyl)-2'-O-methyl-uridine (inm.sup.5Um), 1-thio-uridine, deoxythymidine, 2'-F-ara-uridine, 2'-F-uridine, 2'-OH-ara-uridine, 5-(2-carbomethoxyvinyl) uridine, 5-[3-(1-E-propenylamino)uridine, or any other modified uridine known in the art.

[0240] In an embodiment, the rRNA comprises other modified nucleosides or comprises further modified nucleosides, e.g., modified cytidine such as those described above. For example, in one embodiment, in the rRNA 5-methylcytidine is substituted partially or completely, preferably completely, for cytidine. In one embodiment, the rRNA comprises 5-methylcytidine and one or more selected from pseudouridine (ψ), N1-methyl-pseudouridine (m1ψ), and 5-methyl-uridine (m5U). In an embodiment, the rRNA comprises 5-methylcytidine and N1-methyl-pseudouridine (m1ψ). In some embodiments, the rRNA comprises 5-methylcytidine in place of each cytidine and N1-methyl-pseudouridine (m1ψ) in place of each uridine.

Functional Non-Structural Protein

[0241] The term “non-structural protein” relates to a protein encoded by a virus but that is not part of the viral particle. This term typically includes

the various enzymes and factors that control the virus to replicate itself, such as RNA replicase or other template-directed polymerases. The term “non-structural protein” includes each and every co- or post-translationally modified form, including carbohydrate-modified (such as glycosylated) and lipid-modified forms of a non-structural protein and preferably relates to an “alphavirus non-structural protein”.

[0242] In some embodiments, the term “alphavirus non-structural protein” refers to any one or more of individual non-structural proteins of alphavirus origin (nsP1, nsP2, nsP3, nsP4), or to a poly-protein comprising the polypeptide sequence of more than one non-structural protein of alphavirus origin. In some embodiments, “alphavirus non-structural protein” refers to nsP123 and/or to nsP4. In other embodiments, “alphavirus non-structural protein” refers to nsP1234. In one embodiment, the protein of interest encoded by an open reading frame consists of all of nsP1, nsP2, nsP3 and nsP4 as one single, optionally deavable poly-protein: nsP1234. In one embodiment, the protein of interest encoded by an open reading frame consists of nsP1, nsP2 and nsP3 as one single, optionally cleavable polyprotein: nsP123. In that embodiment, nsP4 may be a further protein of interest and may be encoded by a further open reading frame.

[0243] In some embodiments, non-structural protein is capable of forming a complex or association, e.g., in a host cell. In some embodiments, “alphavirus non-structural protein” refers to a complex or association of nsP123 (synonymously P123) and nsP4. In some embodiments, “alphavirus non-structural protein” refers to a complex or association of nsP1, nsP2, and nsP3. In some embodiments, “alphavirus non-structural protein” refers to a complex or association of nsP1, nsP2, nsP3 and nsP4. In some embodiments, “alphavirus non-structural protein” refers to a complex or association of any one or more selected from the group consisting of nsP1, nsP2, nsP3 and nsP4. In some embodiments, the alphavirus non-structural protein comprises at least nsP4.

[0244] The terms “complex” or “association” refer to two or more same or different protein molecules that are in spatial proximity. Proteins of a complex are preferably in direct or indirect physical or physicochemical contact with each other. A complex or association can consist of multiple different proteins (heteromultimer) and/or of multiple copies of one particular protein (homomultimer). In the context of alphavirus non-structural protein, the term “complex or association” describes a multitude of at least two protein molecules, of which at least one is an alphavirus non-structural protein. The complex or association can consist of multiple copies of one particular protein (homomultimer) and/or of multiple different proteins (heteromultimer). In the context of a multimer, “multi” means more than one, such as two, three, four, five, six, seven, eight, nine, ten, or more than ten.

[0245] The term “functional non-structural protein” includes non-structural protein that has replicase function. Thus, “functional non-structural protein” includes alphavirus replicase. “Replicase function” comprises the function of an RNA-dependent RNA polymerase (RdRP), i.e., an enzyme which is capable to catalyze the synthesis of (−) strand RNA based on a (+) strand RNA template, and/or which is capable to catalyze the synthesis of (+) strand RNA based on a (−) strand RNA template. Thus, the term “functional non-structural protein” can refer to a protein or complex that synthesizes (−) stranded RNA, using the (+) stranded (e.g. genomic) RNA as template, to a protein or complex that synthesizes new (+) stranded RNA, using the (−) stranded complement of genomic RNA as template, and/or to a protein or complex that synthesizes a subgenomic transcript, using a fragment of the (−) stranded complement of genomic RNA as template. The functional non-structural protein may additionally have one or more additional functions, such as, e.g., a protease (for auto-cleavage), helicase, terminal adenylyltransferase (for poly(A) tail addition), methyltransferase and guanylyltransferase (for providing a nucleic acid with a 5′-cap), nuclear localization sites, triphosphatase (Gould et al., 2010, Antiviral Res. 87:111-124; Rupp et al., 2015, J. Gen. Virol. 96:2483-500).

[0246] The term “replicase” includes RNA-dependent RNA polymerase. According to the invention, the term “replicase” includes “alphavirus replicase”, including a RNA-dependent RNA polymerase from a naturally occurring alphavirus (alphavirus found in nature) and a RNA-dependent RNA polymerase from a variant or derivative of an alphavirus, such as from an attenuated alphavirus.

[0247] The term “replicase” comprises all variants, in particular post-translationally modified variants, conformations, isoforms and homologs of alphavirus replicase, which are expressed by alphavirus-infected cells or which are expressed by cells that have been transfected with a nucleic acid that codes for alphavirus replicase. Moreover, the term “replicase” comprises all forms of replicase that have been produced and can be produced by recombinant methods. For example, a replicase comprising a tag that facilitates detection and/or purification of the replicase in the laboratory, e.g.; a myc-tag, a HA-tag or an oligohistidine tag (His-tag) may be produced by recombinant methods.

[0248] Optionally, the alphavirus replicase is additionally functionally defined by the capacity of binding to any one or more of alphavirus conserved sequence element 1 (CSE 1) or complementary sequence thereof, conserved sequence element 2 (CSE 2) or complementary sequence thereof, conserved sequence element 3 (CSE 3) or complementary sequence thereof, conserved sequence element 4 (CSE 4) or complementary sequence thereof. Preferably, the replicase is capable of binding to CSE 2 [i.e., to the (+) strand] and/or to CSE 4 [i.e., to the (+) strand], or of binding to the complement of CSE 1 [i.e. to the (−) strand] and/or to the complement of CSE 3 [i.e., to the (−) strand].

[0249] The origin of the alphavirus replicase is not limited to any particular alphavirus. In a preferred embodiment, the alphavirus replicase comprises non-structural protein from Semliki Forest virus, including a naturally occurring Semliki Forest virus and a variant or derivative of Semliki Forest virus, such as an attenuated Semliki Forest virus. In an alternative preferred embodiment, the alphavirus replicase comprises non-structural protein from Sindbis virus, including a naturally occurring Sindbis virus and a variant or derivative of Sindbis virus, such as an attenuated Sindbis virus. In an alternative preferred embodiment, the alphavirus replicase comprises non-structural protein from Venezuelan equine encephalitis virus (VEEV), including a naturally occurring VEEV and a variant or derivative of VEEV, such as an attenuated VEEV. In an alternative preferred embodiment, the alphavirus replicase comprises non-structural protein from chikungunya virus (CHIKV), including a naturally occurring CHIKV and a variant or derivative of CHIKV, such as an attenuated CHIKV.

[0250] A replicase can also comprise non-structural proteins from more than one virus, e.g., from more than one alphavirus. Thus, heterologous complexes or associations comprising alphavirus non-structural protein and having replicase function are equally comprised by the present invention. Merely for illustrative purposes, replicase may comprise one or more non-structural proteins (e.g., nsP1, nsP2) from a first alphavirus, and one or more non-structural proteins (nsP3, nsP4) from a second alphavirus. Non-structural proteins from more than one different alphavirus may be encoded by separate open reading frames, or may be encoded by a single open reading frame as poly-protein, e.g., nsP1234.

[0251] In some embodiments, functional non-structural protein is capable of forming membranous replication complexes and/or vacuoles in cells in which the functional non-structural protein is expressed.

[0252] If functional non-structural protein, i.e., non-structural protein with replicase function, is encoded by a nucleic acid molecule according to the present invention, it is preferable that the subgenomic promoter of the replicon, if present, is compatible with said replicase. Compatible in this context means that the replicase is capable of recognizing the subgenomic promoter, if present. In one embodiment, this is achieved when the subgenomic promoter is native to the virus from which the replicase is derived, i.e. the natural origin of these sequences is the same virus. In an alternative embodiment, the subgenomic promoter is not native to the virus from which the virus replicase is derived, provided that the virus replicase is capable of recognizing the subgenomic promoter. In other words, the replicase is compatible with the subgenomic promoter (cross-virus compatibility). Examples of cross-virus compatibility concerning subgenomic promoter and replicase originating from different alphaviruses are known in the art. Any combination of subgenomic promoter and replicase is possible as long as cross-virus compatibility exists. Cross-virus compatibility can readily be tested by the skilled person working the present invention by incubating a replicase to be tested together with an RNA, wherein the RNA has a subgenomic promoter to be tested, at conditions suitable for RNA synthesis from the a subgenomic promoter. If a subgenomic transcript is prepared, the subgenomic promoter and the replicase are determined to be compatible. Various examples of cross-virus compatibility are known.

[0253] The replicon can preferably be replicated by the functional non-structural protein. In particular, the RNA replicon that encodes functional non-structural protein can be replicated by the functional non-structural protein encoded by the replicon. In a preferred embodiment, the RNA replicon

comprises an open reading frame encoding functional alphavirus non-structural protein. In one embodiment, the replicon comprises a further open reading frame encoding a protein of interest. This embodiment is particularly suitable in some methods for producing a protein of interest according to the present invention. In one embodiment, the further open reading frame encoding a protein of interest is located downstream from the 5' replication recognition sequence and upstream from the IRES (and upstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus) and/or downstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus. The further open reading frame encoding a protein of interest located downstream from the 5' replication recognition sequence and upstream from the IRES (and upstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus) may be expressed as a fusion protein with sequences encoded by the 5' replication recognition sequence. The further open reading frame encoding a protein of interest located downstream from the 5' replication recognition sequence and upstream from the IRES (and upstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus) may be or may not be controlled by a subgenomic promoter. The one or more further open reading frames encoding one or more proteins of interest located downstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus are generally controlled by (a) subgenomic promoter(s).

[0254] It is preferable that the open reading frame encoding functional non-structural protein does not overlap with the 5' replication recognition sequence. In one embodiment, the open reading frame encoding functional non-structural protein does not overlap with the subgenomic promoter, if present. Embodiments thereof are disclosed in WO 2017/162460, herein incorporated by reference.

Uncoupling of Sequence Elements Required for Replication and Protein-Coding Regions

[0255] Versatile alphavirus-derived vectors are difficult to develop because the open reading frame encoding nsP1234 overlaps with the 5' replication recognition sequence of the alphavirus genome (coding sequence for nsP1) and typically also with the subgenomic promoter comprising CSE 3 (coding sequence for nsP4).

[0256] The RNA replicon described herein generally comprises sequence elements required for replication by the replicase, in particular a 5' replication recognition sequence. In an embodiment, the coding sequence for non-structural protein is under the control of an IRES and thus an IRES is located upstream of the coding sequence for non-structural protein. Thus, in one embodiment, the 5' replication recognition sequence which normally overlaps with the coding sequence for the N-terminal fragment of the alphavirus non-structural protein, is located upstream of the IRES and does not overlap with the coding sequence for non-structural protein.

[0257] In an embodiment, coding sequences of the 5' replication recognition sequence such as nsP1 coding sequences are fused in frame to a gene of interest which is located upstream from the IRES.

[0258] In an embodiment, the 5' replication recognition sequence does not encode any protein or fragment thereof, such as an alphavirus non-structural protein or fragment thereof. Thus, in the RNA replicon according to the invention, the sequence elements required for replication by the replicase and protein-coding regions may be uncoupled. The uncoupling may be achieved by the removal of at least one initiation codon in the 5' replication recognition sequence compared to a native virus genomic RNA, e.g., native alphavirus genomic RNA.

[0259] Thus, the rRNA may comprise a 5' replication recognition sequence, wherein the 5' replication recognition sequence is characterized in that it comprises the removal of at least one initiation codon compared to a native virus 5' replication recognition sequence, e.g., native alphavirus 5' replication recognition sequence.

[0260] The 5' replication recognition sequence that is characterized in that it comprises the removal of at least one initiation codon compared to a native virus 5' replication recognition sequence can be referred to herein as “modified 5' replication recognition sequence” or “5' replication recognition sequence according to the invention”. As described herein below, the 5' replication recognition sequence according to the invention may optionally be characterized by the presence of one or more additional nucleotide changes, such as those detected by the methods of the present invention.

[0261] A nucleic acid construct that is capable of being replicated by a replicase, preferably an alphaviral replicase, is termed replicable RNA or replicon. According to the invention, the term “replicon” defines an RNA molecule that can be replicated by RNA-dependent RNA polymerase, yielding—without DNA intermediate—one or multiple identical or essentially identical copies of the RNA replicon. “Without DNA intermediate” means that no deoxyribonucleic acid (DNA) copy or complement of the replicon is formed in the process of forming the copies of the RNA replicon, and/or that no deoxyribonucleic acid (DNA) molecule is used as a template in the process of forming the copies of the RNA replicon, or complement thereof. The replicase function is typically provided by functional non-structural protein, e.g., functional alphavirus non-structural protein.

[0262] According to the invention, the terms “can be replicated” and “capable of being replicated” generally describe that one or more identical or essentially identical copies of a nucleic acid can be prepared. When used together with the term “replicase”, such as in “capable of being replicated by a replicase”, the terms “can be replicated” and “capable of being replicated” describe functional characteristics of a nucleic acid molecule, e.g. a RNA replicon, with respect to a replicase. These functional characteristics comprise at least one of (i) the replicase is capable of recognizing the replicon and (ii) the replicase is capable of acting as RNA-dependent RNA polymerase (RdRP). Preferably, the replicase is capable of both (i) recognizing the replicon and (ii) acting as RNA-dependent RNA polymerase.

[0263] The expression “capable of recognizing” describes that the replicase is capable of physically associating with the replicon, and preferably, that the replicase is capable of binding to the replicon, typically non-covalently. The term “binding” can mean that the replicase has the capacity of binding to any one or more of a conserved sequence element 1 (CSE 1) or complementary sequence thereof (if comprised by the replicon), conserved sequence element 2 (CSE 2) or complementary sequence thereof (if comprised by the replicon), conserved sequence element 3 (CSE 3) or complementary sequence thereof (if comprised by the replicon), conserved sequence element 4 (CSE 4) or complementary sequence thereof (if comprised by the replicon). Preferably, the replicase is capable of binding to CSE 2 [i.e., to the (+) strand] and/or to CSE 4 [i.e., to the (+) strand], or of binding to the complement of CSE 1 [i.e. to the (–) strand] and/or to the complement of CSE 3 [i.e., to the (–) strand].

[0264] In one embodiment, the expression “capable of acting as RdRP” means that the replicase is capable to catalyze the synthesis of the (–) strand complement of viral genomic (+) strand RNA, wherein the (+) strand RNA has template function, and/or that the replicase is capable to catalyze the synthesis of (+) strand viral genomic RNA, wherein the (–) strand RNA has template function. In general, the expression “capable of acting as RdRP” can also include that the replicase is capable to catalyze the synthesis of a (+) strand subgenomic transcript wherein a (–) strand RNA has template function, and wherein synthesis of the (+) strand subgenomic transcript is typically initiated at a subgenomic promoter. In one embodiment, the virus is an alphavirus.

[0265] The expressions “capable of binding” and “capable of acting as RdRP” refer to the capability at normal physiological conditions. In particular, they refer to the conditions inside a cell, which expresses functional non-structural protein or which has been transfected with a nucleic acid that codes for functional non-structural protein. The cell is preferably a eukaryotic cell. The capability of binding and/or the capability of acting as RdRP can be experimentally tested, e.g. In a cell-free in vitro system or in a eukaryotic cell. Optionally, said eukaryotic cell is a cell from a species to which the particular virus that represents the origin of the replicase is infectious. For example, when the virus replicase from a particular virus is used that is infectious to humans, the normal physiological conditions are conditions in a human cell. More preferably, the eukaryotic cell (in one example human cell) is from the same tissue or organ to which the particular virus that represents the origin of the replicase is infectious.

[0266] According to the invention, “compared to a native alphavirus sequence” and similar terms refer to a sequence that is a variant of a native alphavirus sequence. The variant is typically not itself a native alphavirus sequence.

[0267] In one embodiment, the RNA replicon comprises a 3' replication recognition sequence. A 3' replication recognition sequence is a nucleic acid sequence that can be recognized by functional non-structural protein. In other words, functional non-structural protein is capable of recognizing the 3' replication recognition sequence. Preferably, the 3' replication recognition sequence is located at the 3' end of the replicon (if the replicon does not

comprise a poly(A) tail), or immediately upstream of the poly(A) tail (if the replicon comprises a poly(A) tail). In one embodiment, the 3' replication recognition sequence consists of or comprises CSE 4.

[0268] In one embodiment, the 5' replication recognition sequence and the 3' replication recognition sequence are capable of directing replication of the RNA replicon according to the present invention in the presence of functional non-structural protein. Thus, when present alone or preferably together, these recognition sequences direct replication of the RNA replicon in the presence of functional non-structural protein.

[0269] It is preferable that a functional non-structural protein is provided that is capable of recognizing both the 5' replication recognition sequence and the 3' replication recognition sequence of the replicon. In one embodiment, this is achieved when the 3' replication recognition sequence is native to the alphavirus from which the functional alphavirus non-structural protein is derived, and when the 5' replication recognition sequence is native to the alphavirus from which the functional alphavirus non-structural protein is derived or is a variant of the 5' replication recognition sequence that is native to the alphavirus from which the functional alphavirus non-structural protein is derived. Native means that the natural origin of these sequences is the same alphavirus. In an alternative embodiment, the 5' replication recognition sequence and/or the 3' replication recognition sequence are not native to the alphavirus from which the functional alphavirus non-structural protein is derived, provided that the functional alphavirus non-structural protein is capable of recognizing both the 5' replication recognition sequence and the 3' replication recognition sequence of the replicon. In other words, the functional alphavirus non-structural protein is compatible to the 5' replication recognition sequence and the 3' replication recognition sequence. When a non-native functional alphavirus non-structural protein is capable of recognizing a respective sequence or sequence element, the functional alphavirus non-structural protein is said to be compatible (cross-virus compatibility). Any combination of (3'/5') replication recognition sequences and CSEs, respectively, with functional alphavirus non-structural protein is possible as long as cross-virus compatibility exists. Cross-virus compatibility can readily be tested by the skilled person working the present invention by incubating a functional alphavirus non-structural protein to be tested together with an RNA, wherein the RNA has 3'- and 5' replication recognition sequences to be tested, at conditions suitable for RNA replication, e.g. in a suitable host cell. If replication occurs, the (3'/5') replication recognition sequences and the functional alphavirus non-structural protein are determined to be compatible.

[0270] The removal of at least one initiation codon within the 5' replication recognition sequence provides several advantages. Absence of an initiation codon in the nucleic acid sequence encoding nsP1* (N-terminal fragment of nsP1) will typically cause that nsP1* is not translated. Further, since nsP1* is not translated, the open reading frame encoding the protein of interest ("GOI 2") is the most upstream open reading frame accessible to the ribosome; thus, when the replicon is present in a cell, translation is initiated at the first AUG of the open reading frame (RNA) encoding the gene of interest.

[0271] The removal of at least one initiation codon can be achieved by any suitable method known in the art. For example, a suitable DNA molecule encoding the replicon according to the invention, i.e., characterized by the removal of an initiation codon, can be designed in silico and subsequently synthesized in vitro (gene synthesis); alternatively, a suitable DNA molecule may be obtained by site-directed mutagenesis of a DNA sequence encoding a replicon. In any case, the respective DNA molecule may serve as template for in vitro transcription, thereby providing the replicon according to the invention.

[0272] The removal of at least one initiation codon compared to a native 5' replication recognition sequence is not particularly limited and may be selected from any nucleotide modification, including substitution of one or more nucleotides (including, on DNA level, a substitution of A and/or T and/or G of the initiation codon); deletion of one or more nucleotides (including, on DNA level, a deletion of A and/or T and/or G of the initiation codon), and insertion of one or more nucleotides (including, on DNA level, an insertion of one or more nucleotides between A and T and/or between T and G of the initiation codon). Irrespective of whether the nucleotide modification is a substitution, an insertion or a deletion, the nucleotide modification must not result in the formation of a new initiation codon (as an illustrative example: an insertion, at DNA level, must not be an insertion of an ATG).

[0273] The 5' replication recognition sequence of the RNA replicon that is characterized by the removal of at least one initiation codon (i.e. the modified 5' replication recognition sequence according to the present invention) is preferably a variant of a 5' replication recognition sequence of the genome of an alphavirus found in nature. In one embodiment, the modified 5' replication recognition sequence according to the present invention is preferably characterized by a degree of sequence identity of 80% or more, preferably 85% or more, more preferably 90% or more, even more preferably 95% or more, to the 5' replication recognition sequence of the genome of at least one alphavirus found in nature.

[0274] In one embodiment, the 5' replication recognition sequence of the RNA replicon that may be characterized by the removal of at least one initiation codon comprises a sequence homologous to about 250 nucleotides at the 5' end of an alphavirus, i.e. at the 5' end of the alphaviral genome. In a preferred embodiment, it comprises a sequence homologous to about 250 to 500, preferably about 300 to 500 nucleotides at the 5' end of an alphavirus, i.e., at the 5' end of the alphaviral genome. "At the 5' end of the alphaviral genome" means a nucleic acid sequence beginning at, and including, the most upstream nucleotide of the alphaviral genome. In other words, the most upstream nucleotide of the alphaviral genome is designated nucleotide no. 1, and, e.g., "250 nucleotides at the 5' end of the alphaviral genome" means nucleotides 1 to 250 of the alphaviral genome. In one embodiment, the 5' replication recognition sequence of the RNA replicon is characterized by a degree of sequence identity of 80% or more, preferably 85% or more, more preferably 90% or more, even more preferably 95% or more, to at least 250 nucleotides at the 5' end of the genome of at least one alphavirus found in nature. At least 250 nucleotides includes, e.g., 250 nucleotides, 300 nucleotides, 400 nucleotides, 500 nucleotides.

[0275] The 5' replication recognition sequence of an alphavirus found in nature is typically characterized by at least one initiation codon and/or by conserved secondary structural motifs. For example, the native 5' replication recognition sequence of Semliki Forest virus (SFV) comprises five specific AUG base triplets. According to Frolov et al., 2001, RNA 7:1638-1651, analysis by MFOLD revealed that the native 5' replication recognition sequence of Semliki Forest virus is predicted to form four stem loops (SL), termed stem loops 1 to 4 (SL1, SL2, SL3, SL4). According to Frolov et al., analysis by MFOLD revealed that also the native 5' replication recognition sequence of a different alphavirus, Sindbis virus, is predicted to form four stem loops: SL1, SL2, SL3, SL4.

[0276] It is known that the 5' end of the alphaviral genome comprises sequence elements that enable replication of the alphaviral genome by functional alphavirus non-structural protein. In one embodiment of the present invention, the 5' replication recognition sequence of the RNA replicon comprises a sequence homologous to conserved sequence element 1 (CSE 1) and/or a sequence homologous to conserved sequence element 2 (CSE 2) of an alphavirus.

[0277] Conserved sequence element 2 (CSE 2) of alphavirus genomic RNA typically is represented by SL3 and SL4 which is preceded by SL2 comprising at least the native initiation codon that encodes the first amino acid residue of alphavirus non-structural protein nsP1. In this description, however, in some embodiments, the conserved sequence element 2 (CSE 2) of alphavirus genomic RNA refers to a region spanning from SL2 to SL4 and comprising the native initiation codon that encodes the first amino acid residue of alphavirus non-structural protein nsP1. In a preferred embodiment, the RNA replicon comprises CSE 2 or a sequence homologous to CSE 2. In one embodiment, the RNA replicon comprises a sequence homologous to CSE 2 that is preferably characterized by a degree of sequence identity of 80% or more, preferably 85% or more, more preferably 90% or more, even more preferably 95% or more, to the sequence of CSE 2 of at least one alphavirus found in nature.

[0278] In an embodiment, the 5' replication recognition sequence comprises a sequence that is homologous to CSE 2 of an alphavirus. The CSE 2 of an alphavirus may comprise a fragment of an open reading frame of a non-structural protein from an alphavirus.

[0279] Thus, in an embodiment, the RNA replicon is characterized in that it comprises a sequence homologous to an open reading frame of a non-structural protein or a fragment thereof from an alphavirus. The sequence homologous to an open reading frame of a non-structural protein or a fragment thereof is typically a variant of an open reading frame of a non-structural protein or a fragment thereof of an alphavirus found in nature. In one embodiment, the sequence homologous to an open reading frame of a non-structural protein or a fragment thereof is preferably characterized by

a degree of sequence identity of 80% or more, preferably 85% or more, more preferably 90% or more, even more preferably 95% or more, to an open reading frame of a non-structural protein or a fragment thereof of at least one alphavirus found in nature.

[0280] In an embodiment, the sequence homologous to an open reading frame of a non-structural protein that is comprised by the replicon of the present invention does not comprise the native initiation codon of a non-structural protein, and more preferably does not comprise any initiation codon of a non-structural protein. In an embodiment, the sequence homologous to CSE 2 is characterized by the removal of all initiation codons compared to a native alphavirus CSE 2 sequence. Thus, the sequence homologous to CSE 2 does preferably not comprise any initiation codon.

[0281] When the sequence homologous to an open reading frame does not comprise any initiation codon, the sequence homologous to an open reading frame is not itself an open reading frame since it does not serve as a template for translation.

[0282] In one embodiment, the 5' replication recognition sequence comprises a sequence homologous to an open reading frame of a non-structural protein or a fragment thereof from an alphavirus, wherein the sequence homologous to an open reading frame of a non-structural protein or a fragment thereof from an alphavirus is characterized in that it comprises the removal of at least one initiation codon compared to the native alphavirus sequence.

[0283] In an embodiment, the sequence homologous to an open reading frame of a non-structural protein or a fragment thereof from an alphavirus is characterized in that it comprises the removal of at least the native start codon of the open reading frame of a non-structural protein. Preferably, it is characterized in that it comprises the removal of at least the native start codon of the open reading frame encoding nsP1.

[0284] The native start codon is the AUG base triplet at which translation on ribosomes in a host cell begins when an RNA is present in a host cell. In other words, the native start codon is the first base triplet that is translated during ribosomal protein synthesis, e.g., in a host cell that has been inoculated with RNA comprising the native start codon. In one embodiment, the host cell is a cell from a eukaryotic species that is a natural host of the specific alphavirus that comprises the native alphavirus 5' replication recognition sequence. In an embodiment, the host cell is a BHK21 cell from the cell line "BHK21 [C13](ATCC® CCL10™)", available from American Type Culture Collection, Manassas, Virginia, USA.

[0285] The genomes of many alphaviruses have been fully sequenced and are publicly accessible, and the sequences of non-structural proteins encoded by these genomes are publicly accessible as well. Such sequence information allows to determine the native start codon in silico.

[0286] In an embodiment, the sequence homologous to an open reading frame of a non-structural protein or a fragment thereof from an alphavirus is characterized in that it comprises the removal of one or more initiation codons other than the native start codon of the open reading frame of a non-structural protein. In an embodiment, said nucleic acid sequence is additionally characterized by the removal of the native start codon. For example, in addition to the removal of the native start codon, any one or two or three or four or more than four (e.g., five) initiation codons may be removed.

[0287] If the replicon is characterized by the removal of the native start codon, and optionally by the removal of one or more initiation codons other than the native start codon, of the open reading frame of a non-structural protein, the sequence homologous to an open reading frame is not itself an open reading frame since it does not serve as a template for translation.

[0288] The one or more initiation codon other than the native start codon that is removed, preferably in addition to removal of the native start codon, is preferably selected from an AUG base triplet that has the potential to initiate translation. An AUG base triplet that has the potential to initiate translation may be referred to as "potential initiation codon". Whether a given AUG base triplet has the potential to initiate translation can be determined in silico or in a cell-based in vitro assay.

[0289] In one embodiment, it is determined in silico whether a given AUG base triplet has the potential to initiate translation: in that embodiment, the nucleotide sequence is examined, and an AUG base triplet is determined to have the potential to initiate translation if it is part of an AUGG sequence, preferably part of a Kozak sequence.

[0290] In one embodiment, it is determined in a cell-based in vitro assay whether a given AUG base triplet has the potential to initiate translation: an RNA replicon characterized by the removal of the native start codon and comprising the given AUG base triplet downstream of the position of the removal of the native start codon is introduced into a host cell. In one embodiment, the host cell is a cell from a eukaryotic species that is a natural host of the specific alphavirus that comprises the native alphavirus 5' replication recognition sequence. In a preferred embodiment, the host cell is a BHK21 cell from the cell line "BHK21 [C13](ATCC® CCL10™)", available from American Type Culture Collection, Manassas, Virginia, USA. It is preferable that no further AUG base triplet is present between the position of the removal of the native start codon and the given AUG base triplet. If, following transfer of the RNA replicon—characterized by the removal of the native start codon and comprising the given AUG base triplet—into the host cell, translation is initiated at the given AUG base triplet, the given AUG base triplet is determined to have the potential to initiate translation. Whether translation is initiated can be determined by any suitable method known in the art. For example, the replicon may encode, downstream of the given AUG base triplet and in-frame with the given AUG base triplet, a tag that facilitates detection of the translation product (if any), e.g. a myc-tag or a HA-tag; whether or not an expression product having the encoded tag is present may be determined e.g. by Western Blot. In this embodiment, it is preferable that no further AUG base triplet is present between the given AUG base triplet and the nucleic acid sequence encoding the tag. The cell-based in vitro assay can be performed individually for more than one given AUG base triplet: in each case, it is preferable that no further AUG base triplet is present between the position of the removal of the native start codon and the given AUG base triplet. This can be achieved by removing all AUG base triplets (if any) between the position of the removal of the native start codon and the given AUG base triplet. Thereby, the given AUG base triplet is the first AUG base triplet downstream of the position of the removal of the native start codon.

[0291] Preferably, the 5' replication recognition sequence of the RNA replicon according to the present invention is characterized by the removal of all potential initiation codons. Thus, according to the invention, the 5' replication recognition sequence preferably does not comprise an open reading frame that can be translated to protein.

[0292] In an embodiment, the 5' replication recognition sequence of the RNA replicon according to the invention is characterized by a secondary structure that is equivalent to the (predicted) secondary structure of the 5' replication recognition sequence of viral genomic RNA. To this end, the RNA replicon may comprise one or more nucleotide changes compensating for nucleotide pairing disruptions within one or more stem loops introduced by the removal of at least one initiation codon.

[0293] In an embodiment, the 5' replication recognition sequence of the RNA replicon according to the invention is characterized by a secondary structure that is equivalent to the secondary structure of the 5' replication recognition sequence of alphaviral genomic RNA. In a preferred embodiment, the 5' replication recognition sequence of the RNA replicon according to the invention is characterized by a predicted secondary structure that is equivalent to the predicted secondary structure of the 5' replication recognition sequence of alphaviral genomic RNA. According to the present invention, the secondary structure of an RNA molecule is preferably predicted by the web server for RNA secondary structure prediction <http://ma.urmc.rochester.edu/RNAstructureWeb/Servers/Predicti/Predicti.html>.

[0294] By comparing the secondary structure or predicted secondary structure of a 5' replication recognition sequence of an RNA replicon characterized by the removal of at least one initiation codon compared to the native alphavirus 5' replication recognition sequence, the presence or absence of a nucleotide pairing disruption can be identified. For example, at least one base pair may be absent at a given position, compared to a native alphavirus 5' replication recognition sequence, e.g. a base pair within a stem loop, in particular the stem of the stem loop.

[0295] In an embodiment, one or more stem loops of the 5' replication recognition sequence are not deleted or disrupted. More preferably, stem loops 3 and 4 are not deleted or disrupted. Preferably, none of the stem loops of the 5' replication recognition sequence is deleted or disrupted.

[0296] In one embodiment, the removal of at least one initiation codon does not disrupt the secondary structure of the 5' replication recognition sequence. In an alternative embodiment, the removal of at least one initiation codon does disrupt the secondary structure of the 5' replication recognition sequence. In this embodiment, the removal of at least one initiation codon may be causative for the absence of at least one base pair at a given position, e.g. a base pair within a stem loop, compared to a native 5' replication recognition sequence. If a base pair is absent within a stem

loop, compared to a native 5' replication recognition sequence, the removal of at least one initiation codon is determined to introduce a nucleotide pairing disruption within the stem loop. A base pair within a stem loop is typically a base pair in the stem of the stem loop.

[0297] In an embodiment, the RNA replicon comprises one or more nucleotide changes compensating for nucleotide pairing disruptions within one or more stem loops introduced by the removal of at least one initiation codon.

[0298] If the removal of at least one initiation codon introduces a nucleotide pairing disruption within a stem loop, compared to a native 5' replication recognition sequence, one or more nucleotide changes may be introduced which are expected to compensate for the nucleotide pairing disruption, and the secondary structure or predicted secondary structure obtained thereby may be compared to a native 5' replication recognition sequence.

[0299] Based on the common general knowledge and on the disclosure herein, certain nucleotide changes can be expected by the skilled person to compensate for nucleotide pairing disruptions. For example, if a base pair is disrupted at a given position of the secondary structure or predicted secondary structure of a given 5' replication recognition sequence of an RNA replicon characterized by the removal of at least one initiation codon, compared to the native 5' replication recognition sequence, a nucleotide change that restores a base pair at that position, preferably without re-introducing an initiation codon, is expected to compensate for the nucleotide pairing disruption.

[0300] In an embodiment, the 5' replication recognition sequence of the replicon does not overlap with, or does not comprise, a translatable nucleic acid sequence, i.e. translatable into a peptide or protein, in particular a nsP, in particular nsP1, or a fragment of any thereof. For a nucleotide sequence to be "translatable", it requires the presence of an initiation codon; the initiation codon encodes the most N-terminal amino acid residue of the peptide or protein. In one embodiment, the 5' replication recognition sequence of the replicon does not overlap with, or does not comprise, a translatable nucleic acid sequence encoding an N-terminal fragment of nsP1.

[0301] In some scenarios, the RNA replicon comprises at least one subgenomic promoter. In a preferred embodiment, the subgenomic promoter of the replicon does not overlap with, or does not comprise, a translatable nucleic acid sequence, i.e. translatable into a peptide or protein, in particular a nsP, in particular nsP4, or a fragment of any thereof. In one embodiment, the subgenomic promoter of the replicon does not overlap with, or does not comprise, a translatable nucleic acid sequence that encodes a C-terminal fragment of nsP4. A RNA replicon having a subgenomic promoter that does not overlap with, or does not comprise, a translatable nucleic acid sequence, e.g. translatable into the C-terminal fragment of nsP4, may be generated by deleting part of the coding sequence for nsP4 (typically the part encoding the N-terminal part of nsP4), and/or by removing AUG base triplets in the part of the coding sequence for nsP4 that has not been deleted. If AUG base triplets in the coding sequence for nsP4 or a part thereof are removed, the AUG base triplets that are removed are preferably potential initiation codons. Alternatively, if the subgenomic promoter does not overlap with a nucleic acid sequence that encodes nsP4, the entire nucleic acid sequence encoding nsP4 may be deleted.

[0302] In an embodiment, the RNA replicon does not comprise an open reading frame encoding a truncated non-structural protein, e.g., a truncated alphavirus non-structural protein. In the context of this embodiment, it is particularly preferable that the RNA replicon does not comprise an open reading frame encoding the N-terminal fragment of nsP1, and optionally does not comprise an open reading frame encoding the C-terminal fragment of nsP4. The N-terminal fragment of nsP1 is a truncated alphavirus protein; the C-terminal fragment of nsP4 is also a truncated alphavirus protein.

[0303] In some embodiments, the replicon according to the present invention does not comprise stem loop 2 (SL2) of the 5' terminus of the genome of an alphavirus. According to Frolov et al., supra, stem loop 2 is a conserved secondary structure found at the 5' terminus of the genome of an alphavirus, upstream of CSE 2, but is dispensable for replication.

[0304] The RNA replicon according to the present invention is preferably a single stranded RNA molecule. The RNA replicon according to the present invention is typically a (+) stranded RNA molecule. In one embodiment, the RNA replicon of the present invention is an isolated nucleic acid molecule. The RNA replicon according to the present invention comprises at least one modified nucleotide, and preferably comprises one or more sequence changes, in particular those detected by the methods disclosed herein for identifying sequence changes that restore or improve the function of an rRNA comprising at least one modified nucleotide.

[0305] At least one open reading frame encoding at least one gene product of interest

[0306] In one embodiment, the RNA replicon according to the present invention comprises at least one open reading frame encoding a gene product of interest, such as a peptide of interest or a protein of interest. Preferably, the protein of interest is encoded by a heterologous nucleic acid sequence. The gene encoding the peptide or protein of interest is synonymously termed "gene of interest" or "transgene". In various embodiments, the peptide or protein of interest is encoded by a heterologous nucleic acid sequence. According to the present invention, the term "heterologous" refers to the fact that a nucleic acid sequence is not naturally functionally or structurally linked to a virus nucleic acid sequence, e.g., an alphavirus nucleic acid sequence.

[0307] The replicon according to the present invention may encode a single polypeptide or multiple polypeptides. Multiple polypeptides can be encoded as a single polypeptide (fusion polypeptide) or as separate polypeptides. In some embodiments, the replicon according to the present invention may comprise more than one open reading frames, each of which may independently be selected to be under the control of a subgenomic promoter or not. Alternatively, a poly-protein or fusion polypeptide comprises individual polypeptides separated by a 2A self-cleaving peptides (e.g. from foot-and-mouth disease virus 2A protein), or protease cleavage site or an intein.

[0308] Proteins of interest may, e.g., be selected from the group consisting of reporter proteins, pharmaceutically active peptides or proteins, inhibitors of intracellular interferon (IFN) signaling. According to the invention, a protein of interest preferably does not include functional non-structural protein from a self-replicating virus, e.g., functional alphavirus non-structural protein.

Reporter Protein

[0309] In one embodiment, an open reading frame encodes a reporter protein, e.g., a cell-surface expressed protein such as CD90. In that embodiment, the open reading frame comprises a reporter gene. Certain genes may be chosen as reporters because the characteristics they confer on cells or organisms expressing them may be readily identified and measured, or because they are selectable markers. Reporter genes are often used as an indication of whether a certain gene has been taken up by or expressed in the cell or organism population. Preferably, the expression product of the reporter gene is visually detectable. Common visually detectable reporter proteins typically possess fluorescent or luminescent proteins. Examples of specific reporter genes include the gene that encodes jellyfish green fluorescent protein (GFP), which causes cells that express it to glow green under blue light, the enzyme luciferase (Luc), which catalyzes a reaction with luciferin to produce light, and the red fluorescent protein (RFP). Variants of any of these specific reporter genes are possible, as long as the variants possess visually detectable properties. For example, eGFP is a point mutant variant of GFP. The reporter protein embodiment is particularly suitable for testing expression.

[0310] Pharmaceutically Active Gene Product, Such as a Peptide or Protein or Nucleic Acid

[0311] According to the invention, in one embodiment, rRNA comprises or consists of pharmaceutically active rRNA. A "pharmaceutically active RNA" may be RNA that encodes a pharmaceutically active peptide or protein. Preferably, the RNA replicon according to the present invention encodes a pharmaceutically active peptide or protein or other gene product. Preferably, an open reading frame encodes a pharmaceutically active peptide or protein. Preferably, the RNA replicon comprises an open reading frame that encodes a pharmaceutically active peptide or protein, optionally under control of the subgenomic promoter.

[0312] A "pharmaceutically active peptide or protein" has a positive or advantageous effect on the condition or disease state of a subject when administered to the subject in a therapeutically effective amount. Preferably, a pharmaceutically active peptide or protein has curative or palliative properties and may be administered to ameliorate, relieve, alleviate, reverse, delay onset of or lessen the severity of one or more symptoms of a disease or disorder. A pharmaceutically active peptide or protein may have prophylactic properties and may be used to delay the onset of a disease or to lessen the severity of such disease or pathological condition. The term "pharmaceutically active peptide or protein" includes entire proteins or polypeptides, and can also refer to pharmaceutically active fragments thereof. It can also include pharmaceutically active analogs of a peptide or

protein. The term “pharmaceutically active peptide or protein” includes peptides and proteins that are antigens, i.e., the peptide or protein elicits an immune response in a subject which may be therapeutic or partially or fully protective.

[0313] In one embodiment, the pharmaceutically active peptide or protein is or comprises an immunologically active compound or an antigen or an epitope.

[0314] According to the invention, the term “immunologically active compound” relates to any compound altering an immune response, preferably by inducing and/or suppressing maturation of immune cells, inducing and/or suppressing cytokine biosynthesis, and/or altering humoral immunity by stimulating antibody production by B cells. In one embodiment, the immune response involves stimulation of an antibody response (usually including immunoglobulin G (IgG)). Immunologically active compounds possess potent immunostimulating activity including, but not limited to, antiviral and antitumor activity, and can also down-regulate other aspects of the immune response, for example shifting the immune response away from a Th2 immune response, which is useful for treating a wide range of Th2 mediated diseases.

[0315] According to the invention, the term “antigen” or “immunogen” covers any substance that will elicit an immune response. In particular, an “antigen” relates to any substance that reacts specifically with antibodies or T-lymphocytes (T-cells). According to the present invention, the term “antigen” comprises any molecule which comprises at least one epitope. Preferably, an antigen in the context of the present invention is a molecule which, optionally after processing, induces an immune reaction, which is preferably specific for the antigen. According to the present invention, any suitable antigen may be used, which is a candidate for an immune reaction, wherein the immune reaction may be both a humoral as well as a cellular immune reaction. In the context of the embodiments of the present invention, the antigen is preferably presented by a cell, preferably by an antigen presenting cell, in the context of MHC molecules, which results in an immune reaction against the antigen. An antigen is preferably a product which corresponds to or is derived from a naturally occurring antigen. Such naturally occurring antigens may include or may be derived from allergens, viruses, bacteria, fungi, parasites and other infectious agents and pathogens or an antigen may also be a tumor antigen. According to the present invention, an antigen may correspond to a naturally occurring product, for example, a viral protein, or a part thereof. In preferred embodiments, the antigen is a surface polypeptide, i.e. a polypeptide naturally displayed on the surface of a cell, a pathogen, a bacterium, a virus, a fungus, a parasite, an allergen, or a tumor. The antigen may elicit an immune response against a cell, a pathogen, a bacterium, a virus, a fungus, a parasite, an allergen, or a tumor.

[0316] The term “pathogen” refers to pathogenic biological material capable of causing disease in an organism, preferably a vertebrate organism. Pathogens include microorganisms such as bacteria, unicellular eukaryotic organisms (protozoa), fungi, as well as viruses.

[0317] The terms “epitope”, “antigen peptide”, “antigen epitope”, “immunogenic peptide” and “MHC binding peptide” are used interchangeably herein and refer to an antigenic determinant in a molecule such as an antigen, i.e., to a part in or fragment of an immunologically active compound that is recognized by the immune system, for example, that is recognized by a T cell, in particular when presented in the context of MHC molecules. An epitope of a protein preferably comprises a continuous or discontinuous portion of said protein and is preferably between 5 and 100, preferably between 5 and 50, more preferably between 8 and 30, most preferably between 10 and 25 amino acids in length, for example, the epitope may be preferably 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 amino acids in length. According to the invention an epitope may bind to MHC molecules such as MHC molecules on the surface of a cell and thus, may be a “MHC binding peptide” or “antigen peptide”. The term “major histocompatibility complex” and the abbreviation “MHC” include MHC class I and MHC class II molecules and relate to a complex of genes which is present in all vertebrates. MHC proteins or molecules are important for signaling between lymphocytes and antigen presenting cells or diseased cells in immune reactions, wherein the MHC proteins or molecules bind peptides and present them for recognition by T cell receptors. The proteins encoded by the MHC are expressed on the surface of cells, and display both self-antigens (peptide fragments from the cell itself) and non-self-antigens (e.g., fragments of invading microorganisms) to a T cell. Preferred such immunogenic portions bind to an MHC class I or class II molecule. As used herein, an immunogenic portion is said to “bind to” an MHC class I or class II molecule if such binding is detectable using any assay known in the art. The term “MHC binding peptide” relates to a peptide which binds to an MHC class I and/or an MHC class II molecule. In the case of class I MHC/peptide complexes, the binding peptides are typically 8-10 amino acids long although longer or shorter peptides may be effective. In the case of class II MHC/peptide complexes, the binding peptides are typically 10-25 amino acids long and are in particular 13-18 amino acids long, whereas longer and shorter peptides may be effective.

[0318] In an embodiment, the protein of interest according to the present invention comprises an epitope suitable for vaccination of a target organism. A person skilled in the art will know that one of the principles of immunobiology and vaccination is based on the fact that an immunoprotective reaction to a disease is produced by immunizing an organism with an antigen, which is immunologically relevant with respect to the disease to be treated. An antigen is selected from the group comprising a self-antigen and non-self-antigen. A non-self-antigen is preferably a bacterial antigen, a virus antigen, a fungus antigen, an allergen or a parasite antigen. It is preferred that the antigen comprises an epitope that is capable of eliciting an immune response in a target organism. For example, the epitope may elicit an immune response against a bacterium, a virus, a fungus, a parasite, an allergen, or a tumor, such as a cytotoxic T cell response.

[0319] In some embodiments the non-self-antigen is a bacterial antigen. In some embodiments, the antigen elicits an immune response against a bacterium which infects animals, including birds, fish and mammals, including domesticated animals. Preferably, the bacterium against which the immune response is elicited is a pathogenic bacterium.

[0320] In some embodiments the non-self-antigen is a virus antigen. A virus antigen may for example be a peptide from a virus surface protein, e.g. a capsid polypeptide or a spike polypeptide, such as from Coronavirus. In some embodiments, the antigen elicits an immune response against a virus which infects animals, including birds, fish and mammals, including domesticated animals. Preferably, the virus against which the immune response is elicited is a pathogenic virus, such as Ebola virus.

[0321] In some embodiments the non-self-antigen is a polypeptide or a protein from a fungus. In some embodiments, the antigen elicits an immune response against a fungus which infects animals, including birds, fish and mammals, including domesticated animals. Preferably, the fungus against which the immune response is elicited is a pathogenic fungus.

[0322] In some embodiments the non-self-antigen is a polypeptide or protein from a unicellular eukaryotic parasite. In some embodiments, the antigen elicits an immune response against a unicellular eukaryotic parasite, preferably a pathogenic unicellular eukaryotic parasite. Pathogenic unicellular eukaryotic parasites may be e.g. from the genus *Plasmodium*, e.g. *P. falciparum*, *P. vivax*, *P. malanriae* or *P. ovale*, from the genus *Leishmania*, or from the genus *Trypanosoma*, e.g. *T. cruzi* or *T. brucei*.

[0323] In some embodiments, it is not required that the pharmaceutically active peptide or protein is an antigen that elicits an immune response. Suitable pharmaceutically active proteins or peptides may be selected from the group consisting of cytokines and immune system proteins such as immunologically active compounds (e.g., interleukins, colony stimulating factor (CSF), granulocyte colony stimulating factor (G-CSF), granulocyte-macrophage colony stimulating factor (GM-CSF), erythropoietin, tumor necrosis factor (TNF), interferons, integrins, addressins, selectins, homing receptors, T cell receptors, chimeric antigen receptors (CARs), immunoglobulins), hormones (insulin, thyroid hormone, catecholamines, gonadotrophins, trophic hormones, prolactin, oxytocin, dopamine, bovine somatotropin, leptins and the like), growth hormones (e.g., human growth hormone), growth factors (e.g., epidermal growth factor, nerve growth factor, insulin-like growth factor and the like), growth factor receptors, enzymes (tissue plasminogen activator, streptokinase, cholesterol biosynthetic or degradative, steroidogenic enzymes, kinases, phosphodiesterases, methylases, de-methylases, dehydrogenases, cellulases, proteases, lipases, phospholipases, aromatases, cytochromes, adenylate or guanylate cyclases, neuramidases and the like), receptors (steroid hormone receptors, peptide receptors), binding proteins (growth hormone or growth factor binding proteins and the like), transcription and translation factors, tumor growth suppressing proteins (e.g., proteins which inhibit angiogenesis), structural proteins (such as collagen, fibroin, fibrinogen, elastin, tubulin, actin, and myosin), blood proteins (thrombin, serum albumin, Factor VII,

Factor VIII, insulin, Factor IX, Factor X, Factor XIII, tissue plasminogen activator, protein C, von Willebrand factor, antithrombin III, glucocorticoids, erythropoietin granulocyte colony stimulating factor (G-CSF) or modified Factor VIII, anticoagulants and the like. In one embodiment, the pharmaceutically active protein according to the invention is a cytokine which is involved in regulating lymphoid homeostasis, preferably a cytokine which is involved in and preferably induces or enhances development, priming, expansion, differentiation and/or survival of T cells. In one embodiment, the cytokine is an interleukin, e.g. IL-2, IL-7, IL-12, IL-15, or IL-21.

Position of the at Least One Open Reading Frame in the rRNA Molecule

[0324] The rRNA replicon is suitable for expression of one or more genes encoding a peptide of interest or a protein of interest, optionally under control of a subgenomic promoter. Various embodiments are possible. One or more open reading frames, each encoding a peptide of interest or a protein of interest, can be present on the RNA replicon. The most upstream open reading frame of the RNA replicon is referred to as “first open reading frame”. In one embodiment, the first open reading frame encoding a protein of interest is located downstream from the 5' replication recognition sequence and upstream from the IRES (and the open reading frame encoding a functional non-structural protein from a self-replicating virus). In some embodiments, the “first open reading frame” is the only open reading frame of the RNA replicon. Optionally, one or more further open reading frames can be present downstream of the first open reading frame. One or more further open reading frames downstream of the first open reading frame may be referred to as “second open reading frame”, “third open reading frame” and so on, in the order (5' to 3') in which they are present downstream of the first open reading frame. In one embodiment, one or more further open reading frames encoding one or more proteins of interest are located downstream from the open reading frame encoding a functional non-structural protein from a self-replicating virus and are preferably controlled by subgenomic promoters. Preferably, each open reading frame comprises a start codon (base triplet), typically AUG (in the RNA molecule), corresponding to ATG (in a respective DNA molecule).

[0325] If the replicon comprises a 3' replication recognition sequence, it is preferred that all open reading frames are localized upstream of the 3' replication recognition sequence.

[0326] In some embodiments, at least one open reading frame of the replicon is under the control of a subgenomic promoter, preferably an alphavirus subgenomic promoter. The alphavirus subgenomic promoter is very efficient, and is therefore suitable for heterologous gene expression at high levels. Preferably, the subgenomic promoter is a promoter for a subgenomic transcript in an alphavirus. This means that the subgenomic promoter is one which is native to an alphavirus and which preferably controls transcription of the open reading frame encoding one or more structural proteins in said alphavirus. Alternatively, the subgenomic promoter is a variant of a subgenomic promoter of an alphavirus; any variant which functions as promoter for subgenomic RNA transcription in a host cell is suitable. If the replicon comprises a subgenomic promoter, it is preferred that the replicon comprises a conserved sequence element 3 (CSE 3) or a variant thereof.

[0327] Preferably, the at least one open reading frame under control of a subgenomic promoter is localized downstream of the subgenomic promoter. Preferably, the subgenomic promoter controls production of subgenomic RNA comprising a transcript of the open reading frame.

[0328] In some embodiments the first open reading frame is under control of a subgenomic promoter. In one embodiment, when the first open reading frame is under control of the subgenomic promoter, the gene encoded by the first open reading frame can be expressed both from the replicon as well as from a subgenomic transcript thereof (the latter in the presence of functional alphavirus non-structural protein). One or more further open reading frames, each under control of a subgenomic promoter, may be present downstream of the first open reading frame that may be under control of a subgenomic promoter. The genes encoded by the one or more further open reading frames, e.g. by the second open reading frame, may be translated from one or more subgenomic transcripts, each under control of a subgenomic promoter. For example, the RNA replicon may comprise a subgenomic promoter controlling production of a transcript that encodes a second protein of interest.

[0329] In other embodiments the first open reading frame is not under control of a subgenomic promoter. In one embodiment, when the first open reading frame is not under control of a subgenomic promoter, the gene encoded by the first open reading frame can be expressed from the replicon. One or more further open reading frames, each under control of a subgenomic promoter, may be present downstream of the first open reading frame. The genes encoded by the one or more further open reading frames may be expressed from subgenomic transcripts.

[0330] In a cell which comprises the replicon according to the present invention, the replicon may be amplified by functional non-structural protein. Additionally, if the replicon comprises one or more open reading frames under control of a subgenomic promoter, one or more subgenomic transcripts are expected to be prepared by functional non-structural protein.

[0331] If a replicon comprises more than one open reading frame encoding a protein of interest, it is preferable that each open reading frame encodes a different protein. For example, the protein encoded by the second open reading frame is different from the protein encoded by the first open reading frame.

Other Features of Replicable RNA Molecules According to the Invention

[0332] RNA molecules according to the invention may optionally be characterized by further features, e.g. by a 5'-cap, a 5'-UTR, a 3'-UTR, a poly(A) sequence, and/or adaptation of the codon usage for optimized translation and/or stabilization of the RNA molecule, as detailed below.

Cap

[0333] In some embodiments, the replicon according to the present invention comprises a 5'-cap.

[0334] The terms “5'-cap”, “cap”, “5'-cap structure”, “cap structure” are used synonymously to refer to a dinucleotide that is found on the 5' end of some eukaryotic primary transcripts such as precursor messenger RNA. A 5'-cap is a structure wherein a (optionally modified) guanosine is bonded to the first nucleotide of an mRNA molecule via a 5' to 5' triphosphate linkage (or modified triphosphate linkage in the case of certain cap analogs). The terms can refer to a conventional cap or to a cap analog.

[0335] “RNA which comprises a 5'-cap” or “RNA which is provided with a 5'-cap” or “RNA which is modified with a 5'-cap” or “capped RNA” refers to RNA which comprises a 5'-cap. For example, providing an RNA with a 5'-cap may be achieved by in vitro transcription of a DNA template in presence of said 5'-cap, wherein said 5'-cap is co-transcriptionally incorporated into the generated RNA strand, or the RNA may be generated, for example, by in vitro transcription, and the 5'-cap may be attached to the RNA post-transcriptionally using capping enzymes, for example, capping enzymes of vaccinia virus. In capped RNA, the 3' position of the first base of a (capped) RNA molecule is linked to the 5' position of the subsequent base of the RNA molecule (“second base”) via a phosphodiester bond.

[0336] In one embodiment, the RNA replicon comprises a 5'-cap. In one embodiment, the RNA replicon does not comprise a 5'-cap.

[0337] The term “conventional 5'-cap” refers to a naturally occurring 5'-cap, preferably to the 7-methylguanosine cap. In the 7-methylguanosine cap, the guanosine of the cap is a modified guanosine wherein the modification consists of a methylation at the 7-position.

[0338] In the context of the present invention, the term “5'-cap analog” refers to a molecular structure that resembles a conventional 5'-cap, but is modified to possess the ability to stabilize RNA if attached thereto, preferably in vivo and/or in a cell. A cap analog is not a conventional 5'-cap.

[0339] For the case of eukaryotic mRNA, the 5'-cap has been generally described to be involved in efficient translation of mRNA: in general, in eukaryotes, translation is initiated only at the 5' end of a messenger RNA (mRNA) molecule, unless an internal ribosomal entry site (IRES) is present. Eukaryotic cells are capable of providing an RNA with a 5'-cap during transcription in the nucleus: newly synthesized mRNAs are usually modified with a 5'-cap structure, e.g.; when the transcript reaches a length of 20 to 30 nucleotides. First, the 5' terminal nucleotide pppN (ppp representing triphosphate; N representing any nucleoside) is converted in the cell to 5' GpppN by a capping enzyme having RNA 5'-triphosphatase and guanylyltransferase activities. The GpppN may subsequently be methylated in the cell by a second enzyme with (guanine-7)-methyltransferase activity to form the mono-methylated m.sup.7GpppN cap. In one embodiment, the 5'-cap used in the present invention is a natural 5'-cap.

[0340] In the present invention, a natural 5'-cap dinucleotide is typically selected from the group consisting of a non-methylated cap dinucleotide (G(5')ppp(5')N; also termed GpppN) and a methylated cap dinucleotide ((m.sup.7G(5')ppp(5')N; also termed m.sup.7GpppN). m.sup.7GpppN

(wherein N is G) is represented by the following formula:

##STR00011##

[0341] Capped RNA of the present invention can be prepared in vitro, and therefore, does not depend on a capping machinery in a host cell. The most frequently used method to make capped RNAs in vitro is to transcribe a DNA template with either a bacterial or bacteriophage RNA polymerase in the presence of all four ribonucleoside triphosphates and a cap dinucleotide such as m.sup.7G(5')ppp(5')G (also called m.sup.7GpppG). The RNA polymerase initiates transcription with a nucleophilic attack by the 3'-OH of the guanosine moiety of m.sup.7GpppG on the α -phosphate of the next templated nucleoside triphosphate (pppN), resulting in the intermediate m.sup.7GpppGpN (wherein N is the second base of the RNA molecule). The formation of the competing GTP-initiated product pppGpN is suppressed by setting the molar ratio of cap to GTP between 5 and 10 during in vitro transcription.

[0342] In preferred embodiments of the present invention, the 5'-cap (if present) is a 5'-cap analog. These embodiments are particularly suitable if the RNA is obtained by in vitro transcription, e.g. Is an in vitro transcribed RNA (IVT-RNA). Cap analogs have been initially described to facilitate large scale synthesis of RNA transcripts by means of in vitro transcription.

[0343] For messenger RNA, some cap analogs (synthetic caps) have been generally described to date, and they can all be used in the context of the present invention. Ideally, a cap analog is selected that is associated with higher translation efficiency and/or increased resistance to in vitro degradation and/or increased resistance to in vitro degradation.

[0344] Preferably, a cap analog is used that can only be incorporated into an RNA chain in one orientation. Pasquinelli et al., 1995, RNA J. 1:957-967) demonstrated that during in vitro transcription, bacteriophage RNA polymerases use the 7-methylguanosine unit for initiation of transcription, whereby around 40-50% of the transcripts with cap possess the cap dinucleotide in a reverse orientation (i.e., the initial reaction product is Gpppm.sup.7GpN). Compared to the RNAs with a correct cap, RNAs with a reverse cap are not functional with respect to translation of a nucleic acid sequence into protein. Thus, it is desirable to incorporate the cap in the correct orientation, i.e., resulting in an RNA with a structure essentially corresponding to m.sup.7GpppGpN etc. It has been shown that the reverse integration of the cap-dinucleotide is inhibited by the substitution of either the 2'- or the 3'-OH group of the methylated guanosine unit (Stepinski et al., 2001, RNA J. 7:1486-1495; Peng et al., 2002, Org. Lett. 24:161-164). RNAs which are synthesized in presence of such "anti reverse cap analogs" are translated more efficiently than RNAs which are in vitro transcribed in presence of the conventional 5'-cap m.sup.7GpppG. To that end, one cap analog in which the 3'-OH group of the methylated guanosine unit is replaced by OCH.sub.b is described, e.g., by Holtkamp et al., 2006, Blood 108:4009-4017 (7-methyl(3'-O-methyl)GpppG; anti-reverse cap analog (ARCA)). ARCA is a suitable cap dinucleotide according to the present invention.

##STR00012##

[0345] In an embodiment, the RNA of the present invention is essentially not susceptible to decapping. This is important because, in general, the amount of protein produced from synthetic mRNAs introduced into cultured mammalian cells is limited by the natural degradation of mRNA. One in vivo pathway for mRNA degradation begins with the removal of the mRNA cap. This removal is catalyzed by a heterodimeric pyrophosphatase, which contains a regulatory subunit (Dcp1) and a catalytic subunit (Dcp2). The catalytic subunit cleaves between the α and β phosphate groups of the triphosphate bridge. In the present invention, a cap analog may be selected or present that is not susceptible, or less susceptible, to that type of cleavage. A suitable cap analog for this purpose may be selected from a cap dinucleotide according to formula (I):

##STR00013## [0346] wherein R.sup.1 is selected from the group consisting of optionally substituted alkyl, optionally substituted alkenyl, optionally substituted alkynyl, optionally substituted cycloalkyl, optionally substituted heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, [0347] R.sup.2 and R.sup.3 are independently selected from the group consisting of H, halo, OH, and optionally substituted alkoxy, or R.sup.2 and R.sup.3 together form O—X—O, wherein X is selected from the group consisting of optionally substituted CH.sub.2, CH.sub.2CH.sub.2, CH.sub.2CH.sub.2CH.sub.2, CH.sub.2CH(CH.sub.3), and [0348] C(CH.sub.3).sub.2, or R.sup.2 is combined with the hydrogen atom at position 4' of the ring to which R.sup.2 is attached to form —O—CH.sub.2— or —CH.sub.2—O—, [0349] R.sup.5 is selected from the group consisting of S, Se, and BH.sub.3, [0350] R.sup.4 and R.sup.6 are independently selected from the group consisting of O, S, Se, and BH.sub.3. [0351] n is 1, 2, or 3.

[0352] Preferred embodiments for R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6 are disclosed in WO 2011/015347 A1 and may be selected accordingly in the present invention.

[0353] For example, in an embodiment, the RNA of the present invention comprises a phosphorothioate-cap-analog. Phosphorothioate-cap-analogs are specific cap analogs in which one of the three non-bridging O atoms in the triphosphate chain is replaced with an S atom, i.e., one of R.sup.4, R.sup.5 or R.sup.6 in Formula (I) is S. Phosphorothioate-cap-analogs have been described by Kowalska et al., 2008, RNA, 14:1119-1131, as a solution to the undesired decapping process, and thus to increase the stability of RNA in vivo in particular, the substitution of an oxygen atom for a sulphur atom at the beta-phosphate group of the 5'-cap results in stabilization against Dcp2. In that embodiment, which is preferred in the present invention, R.sup.5 in Formula (I) is S; and R.sup.4 and R.sup.6 are O.

[0354] In a further embodiment, the RNA of the present invention comprises a phosphorothioate-cap-analog wherein the phosphorothioate modification of the RNA 5'-cap is combined with an "anti-reverse cap analog" (ARCA) modification. Respective ARCA-phosphorothioate-cap-analogs are described in WO 2008/157688 A2, and they can all be used in the RNA of the present invention. In that embodiment, at least one of R.sup.2 or R.sup.3 in Formula (I) is not OH, preferably one among R.sup.2 and R.sup.3 is methoxy (OCH.sub.3), and the other one among R.sup.2 and R.sup.3 is preferably OH. In a preferred embodiment, an oxygen atom is substituted for a sulphur atom at the beta-phosphate group (so that R.sup.5 in Formula (I) is S; and R.sup.4 and R.sup.6 are O). It is believed that the phosphorothioate modification of the ARCA ensures that the α , β , and γ phosphorothioate groups are precisely positioned within the active sites of cap-binding proteins in both the translational and decapping machinery. At least some of these analogs are essentially resistant to pyrophosphatase Dcp1/Dcp2. Phosphorothioate-modified ARCAs were described to have a much higher affinity for eIF4E than the corresponding ARCAs lacking a phosphorothioate group.

[0355] A respective cap analog that is particularly preferred in the present invention, i.e., m.sub.2.sup.7,2'-OGpp.sub.spG, is termed beta-S-ARCA (WO 2008/157688 A2; Kuhn et al., 2010, Gene Ther. 17:961-971). Thus, in one embodiment of the present invention, the RNA of the present invention is modified with beta-S-ARCA. beta-S-ARCA is represented by the following structure:

##STR00014##

[0356] In general, the replacement of an oxygen atom for a sulphur atom at a bridging phosphate results in phosphorothioate diastereomers which are designated D1 and D2, based on their elution pattern in HPLC. Briefly, the D1 diastereomer of beta-S-ARCA" or "beta-S-ARCA(D1)" is the diastereomer of beta-S-ARCA which elutes first on an HPLC column compared to the D2 diastereomer of beta-S-ARCA (beta-S-ARCA(D2)) and thus exhibits a shorter retention time. Determination of the stereochemical configuration by HPLC is described in WO 2011/015347 A1.

[0357] In a first particularly preferred embodiment of the present invention, RNA of the present invention is modified with the beta-S-ARCA(D2) diastereomer. The two diastereomers of beta-S-ARCA differ in sensitivity against nucleases. It has been shown that RNA carrying the D2 diastereomer of beta-S-ARCA is almost fully resistant against Dcp2 cleavage (only 6% cleavage compared to RNA which has been synthesized in presence of the unmodified ARCA 5'-cap), whereas RNA with the beta-S-ARCA(D1) 5'-cap exhibits an intermediary sensitivity to Dcp2 cleavage (71% cleavage). It has further been shown that the increased stability against Dcp2 cleavage correlates with increased protein expression in mammalian cells. In particular, it has been shown that RNAs carrying the beta-S-ARCA(D2) cap are more efficiently translated in mammalian cells than RNAs carrying the beta-S-ARCA(D1) cap. Therefore, in one embodiment of the present invention, RNA of the present invention is modified with a cap analog according to Formula (I), characterized by a stereochemical configuration at the P atom comprising the substituent R.sup.5 in Formula (I) that corresponds to that at the P.sub. β atom of the D2 diastereomer of beta-S-ARCA. In that embodiment, R.sup.5 in Formula (I) is S;

R.sup.4 and R.sup.6 are O. Additionally, at least one of R.sup.2 or R.sup.3 in Formula (I) is preferably not OH, preferably one among R.sup.2 and R.sup.3 is methoxy (OCH₃), and the other one among R.sup.2 and R.sup.3 is preferably OH.

[0358] In a second particularly preferred embodiment, RNA of the present invention is modified with the beta-S-ARCA(D1) diastereomer. This embodiment is particularly suitable for transfer of capped RNA into immature antigen presenting cells, such as for vaccination purposes. It has been demonstrated that the beta-S-ARCA(D1) diastereomer, upon transfer of respectively capped RNA into immature antigen presenting cells, is particularly suitable for increasing the stability of the RNA, increasing translation efficiency of the RNA, prolonging translation of the RNA, increasing total protein expression of the RNA, and/or increasing the immune response against an antigen or antigen peptide encoded by said RNA (Kuhn et al., 2010, Gene Ther. 17:961-971). Therefore, in an alternative embodiment of the present invention, RNA of the present invention is modified with a cap analog according to Formula (I), characterized by a stereochemical configuration at the P atom comprising the substituent R.sup.5 in Formula (I) that corresponds to that at the P.sub.β atom of the D1 diastereomer of beta-S-ARCA. Respective cap analogs and embodiments thereof are described in WO 2011/015347 A1 and Kuhn et al., 2010, Gene Ther. 17:961-971. Any cap analog described in WO 2011/015347 A1, wherein the stereochemical configuration at the P atom comprising the substituent R.sup.5 corresponds to that at the P.sub.β atom of the D1 diastereomer of beta-S-ARCA, may be used in the present invention. Preferably, R.sup.5 in Formula (I) is S; and R.sup.4 and R.sup.6 are O. Additionally, at least one of R.sup.2 or R.sup.3 in Formula (I) is preferably not OH, preferably one among R.sup.2 and R.sup.3 is methoxy (OCH₃), and the other one among R.sup.2 and R.sup.3 is preferably OH.

[0359] In one embodiment, RNA of the present invention is modified with a 5'-cap structure according to Formula (I) wherein any one phosphate group is replaced by a boranophosphate group or a phosphoroselenoate group. Such caps have increased stability both in vitro and in vivo. Optionally, the respective compound has a 2'-O- or 3'-O-alkyl group (wherein alkyl is preferably methyl); respective cap analogs are termed BH.sub.3-ARCA's or Se-ARCA's. Compounds that are particularly suitable for capping of mRNA include the β-BH.sub.3-ARCA's and β-Se-ARCA's, as described in WO 2009/149253 A2. For these compounds, a stereochemical configuration at the P atom comprising the substituent R.sup.5 in Formula (I) that corresponds to that at the P.sub.β atom of the D1 diastereomer of beta-S-ARCA is preferred.

[0360] In one embodiment, the 5' cap can be a CleanCap supplied by Trilink Biotechnologies, San Diego, CA having the following structure:
##STR00015##

[0361] The term "untranslated region" or "UTR" relates to a region in a DNA molecule which is transcribed but is not translated into an amino acid sequence, or to the corresponding region in an RNA molecule, such as an mRNA molecule. An untranslated region (UTR) can be present 5' (upstream) of an open reading frame (5'-UTR) and/or 3' (downstream) of an open reading frame (3'-UTR).

[0362] A 3'-UTR, if present, is located at the 3' end of a gene, downstream of the termination codon of a protein-encoding region, but the term "3'-UTR" does preferably not include the poly(A) tail. Thus, the 3'-UTR is upstream of the poly(A) tail (if present), e.g. directly adjacent to the poly(A) tail.

[0363] A 5'-UTR, if present, is located at the 5' end of a gene, upstream of the start codon of a protein-encoding region. A 5'-UTR is downstream of the 5'-cap (if present), e.g. directly adjacent to the 5'-cap.

[0364] 5'- and/or 3'-untranslated regions may, according to the invention, be functionally linked to an open reading frame, so as for these regions to be associated with the open reading frame in such a way that the stability and/or translation efficiency of the RNA comprising said open reading frame are increased.

[0365] In some embodiments, the RNA replicon according to the present invention comprises a 5'-UTR and/or a 3'-UTR.

[0366] UTRs are implicated in stability and translation efficiency of RNA. Both can be improved, besides structural modifications concerning the 5'-cap and/or the 3' poly(A)-tail as described herein, by selecting specific 5' and/or 3' untranslated regions (UTRs). Sequence elements within the UTRs are generally understood to influence translational efficiency (mainly 5'-UTR) and RNA stability (mainly 3'-UTR). It is preferable that a 5'-UTR is present that is active in order to increase the translation efficiency and/or stability of the RNA replicon. Independently or additionally, it is preferable that a 3'-UTR is present that is active in order to increase the translation efficiency and/or stability of the RNA replicon.

[0367] The terms "active in order to increase the translation efficiency" and/or "active in order to increase the stability", with reference to a first nucleic acid sequence (e.g. a UTR), means that the first nucleic acid sequence is capable of modifying, in a common transcript with a second nucleic acid sequence, the translation efficiency and/or stability of said second nucleic acid sequence in such a way that said translation efficiency and/or stability is increased in comparison with the translation efficiency and/or stability of said second nucleic acid sequence in the absence of said first nucleic acid sequence.

[0368] In one embodiment, the RNA replicon according to the present invention comprises a 5'-UTR and/or a 3'-UTR which is heterologous or non-native to the alphavirus from which the functional alphavirus non-structural protein is derived. This allows the untranslated regions to be designed according to the desired translation efficiency and RNA stability. Thus, heterologous or non-native UTRs allow for a high degree of flexibility, and this flexibility is advantageous compared to native alphaviral UTRs.

[0369] Preferably, the RNA replicon according to the present invention comprises a 5'-UTR and/or a 3'-UTR that is not of virus origin; particularly not of alphavirus origin. In one embodiment, the RNA replicon comprises a 5'-UTR derived from a eukaryotic 5'-UTR and/or a 3'-UTR derived from a eukaryotic 3'-UTR.

[0370] A 5'-UTR according to the present invention can comprise any combination of more than one nucleic acid sequence, optionally separated by a linker. A 3'-UTR according to the present invention can comprise any combination of more than one nucleic acid sequence, optionally separated by a linker.

[0371] The term "linker" according to the invention relates to a nucleic acid sequence added between two nucleic acid sequences to connect said two nucleic acid sequences. There is no particular limitation regarding the linker sequence.

[0372] A 3'-UTR typically has a length of 200 to 2000 nucleotides, e.g. 500 to 1500 nucleotides. The 3'-untranslated regions of immunoglobulin mRNAs are relatively short (fewer than about 300 nucleotides), while the 3'-untranslated regions of other genes are relatively long. For example, the 3'-untranslated region of tPA is about 800 nucleotides in length, that of factor VIII is about 1800 nucleotides in length and that of erythropoietin is about 560 nucleotides in length. The 3-untranslated regions of mammalian mRNA typically have a homology region known as the AAUAAA hexanucleotide sequence. This sequence is presumably the poly(A) attachment signal and is frequently located from 10 to 30 bases upstream of the poly(A) attachment site. 3'-untranslated regions may contain one or more inverted repeats which can fold to give stem-loop structures which act as barriers for exonucleases or interact with proteins known to increase RNA stability (e.g. RNA-binding proteins).

[0373] The human beta-globin 3'-UTR, particularly two consecutive identical copies of the human beta-globin 3'-UTR, contributes to high transcript stability and translational efficiency (Holtkamp et al., 2006, Blood 108:4009-4017). Thus, in one embodiment, the RNA replicon according to the present invention comprises two consecutive identical copies of the human beta-globin 3'-UTR. Thus, it comprises in the 5' → 3' direction: (a) optionally a 5'-UTR; (b) an open reading frame; (c) a 3'-UTR; said 3'-UTR comprising two consecutive identical copies of the human beta-globin 3'-UTR, a fragment thereof, or a variant of the human beta-globin 3'-UTR or fragment thereof.

[0374] In an embodiment, the RNA replicon according to the present invention comprises a 3'-UTR which is active in order to increase translation efficiency and/or stability, but which is not the human beta-globin 3'-UTR, a fragment thereof, or a variant of the human beta-globin 3'-UTR or fragment thereof.

[0375] In an embodiment, the RNA replicon according to the present invention comprises a 5'-UTR which is active in order to increase translation efficiency and/or stability.

Poly(A) Sequence

[0376] In some embodiments, the replicon comprising or essentially consists of or consists of a 3'-poly(A) sequence. If the replicon comprises conserved sequence element 4 (CSE 4), the 3'-poly(A) sequence of the replicon is preferably present downstream of CSE 4, most preferably directly adjacent to CSE 4.

[0377] According to the invention, in one embodiment, a poly(A) sequence comprises or essentially consists of or consists of at least 20, preferably at least 26, preferably at least 40, preferably at least 80, preferably at least 100 and preferably up to 500, preferably up to 400, preferably up to 300, preferably up to 200, and in particular up to 150, A nucleotides, and in particular about 120 A nucleotides. In this context "essentially consists of" means that most nucleotides in the poly(A) sequence, typically at least 50%, and preferably at least 75% by number of nucleotides in the "poly(A) sequence", are A nucleotides (adenylate), but permits that remaining nucleotides are nucleotides other than A nucleotides, such as U nucleotides (uridylate), G nucleotides (guanylate), C nucleotides (cytidylate). In this context "consists of" means that all nucleotides in the poly(A) sequence, i.e. 100% by number of nucleotides in the poly(A) sequence, are A nucleotides. The term "A nucleotide" or "A" refers to adenylate.

[0378] Indeed, it has been demonstrated that a 3' poly(A) sequence of about 120 A nucleotides has a beneficial influence on the levels of RNA in transfected eukaryotic cells, as well as on the levels of protein that is translated from an open reading frame that is present upstream (5') of the 3' poly(A) sequence (Holtkamp et al., 2006, Blood, vol. 108, pp. 4009-4017).

[0379] In alphaviruses, a 3' poly(A) sequence of at least 11 consecutive adenylate residues, or at least 25 consecutive adenylate residues, is thought to be important for efficient synthesis of the minus strand. In particular, in alphaviruses, a 3' poly(A) sequence of at least 25 consecutive adenylate residues is understood to function together with conserved sequence element 4 (CSE 4) to promote synthesis of the (-) strand (Hardy & Rice, 2005, J. Virol. 79:4630-4639).

[0380] The present invention provides for a 3' poly(A) sequence to be attached during RNA transcription, i.e. during preparation of in vitro transcribed RNA, based on a DNA template comprising repeated dT nucleotides (deoxythymidylate) in the strand complementary to the coding strand. The DNA sequence encoding a poly(A) sequence (coding strand) is referred to as poly(A) cassette.

[0381] In a preferred embodiment of the present invention, the 3' poly(A) cassette present in the coding strand of DNA essentially consists of dA nucleotides, but is interrupted by a random sequence having an equal distribution of the four nucleotides (dA, dC, dG, dT). Such random sequence may be 5 to 50, preferably 10 to 30, more preferably 10 to 20 nucleotides in length. Such a cassette is disclosed in WO 2016/005004 A1. Any poly(A) cassette disclosed in WO 2016/005004 A1 may be used in the present invention. A poly(A) cassette that essentially consists of dA nucleotides, but is interrupted by a random sequence having an equal distribution of the four nucleotides (dA, dC, dG, dT) and having a length of, e.g., 5 to 50 nucleotides shows, on DNA level, constant propagation of plasmid DNA in *E. coli* and is still associated, on RNA level, with the beneficial properties with respect to supporting RNA stability and translational efficiency.

[0382] Consequently, in a preferred embodiment of the present invention, the 3' poly(A) sequence contained in an RNA molecule described herein essentially consists of A nucleotides, but is interrupted by a random sequence having an equal distribution of the four nucleotides (A, C, G, U). Such random sequence may be 5 to 50, preferably 10 to 30, more preferably 10 to 20 nucleotides in length.

Codon Usage

[0383] In general, the degeneracy of the genetic code will allow the substitution of certain codons (base triplets coding for an amino acid) that are present in an RNA sequence by other codons (base triplets), while maintaining the same coding capacity (so that the replacing codon encodes the same amino acid as the replaced codon). In some embodiments of the present invention, at least one codon of an open reading frame comprised by an RNA (rRNA) molecule differs from the respective codon in the respective open reading frame in the species from which the open reading frame originates. In that embodiment, the coding sequence of the open reading frame is said to be "adapted" or "modified". The coding sequence of an open reading frame comprised by the replicon may be adapted.

[0384] For example, when the coding sequence of an open reading frame is adapted, frequently used codons may be selected: WO 2009/024567 A1 describes the adaptation of a coding sequence of a nucleic acid molecule, involving the substitution of rare codons by more frequently used codons. Since the frequency of codon usage depends on the host cell or host organism, that type of adaptation is suitable to fit a nucleic acid sequence to expression in a particular host cell or host organism. Generally, speaking, more frequently used codons are typically translated more efficiently in a host cell or host organism, although adaptation of all codons of an open reading frame is not always required.

[0385] For example, when the coding sequence of an open reading frame is adapted, the content of G (guanylate) residues and C (cytidylate) residues may be altered by selecting codons with the highest GC-rich content for each amino acid. RNA molecules with GC-rich open reading frames were reported to have the potential to reduce immune activation and to improve translation and half-life of RNA (Thess et al., 2015, Mol. Ther. 23:1457-1465).

[0386] In particular, the coding sequence for non-structural protein can be adapted as desired. This freedom is possible because the open reading frame encoding non-structural protein does not overlap with the 5' replication recognition sequence of the replicon.

Safety Features of Embodiments of the Present Invention

[0387] The following features are preferred in the present invention, alone or in any suitable combination:

[0388] The replicons of the present invention are not particle-forming. This means that, following inoculation of a host cell by a replicon of the present invention, the host cell does not produce virus particles, such as next generation virus particles. In one embodiment, the RNA replicon according to the invention is completely free of genetic information encoding any virus structural protein, e.g., alphavirus structural protein, such as core nucleocapsid protein C, envelope protein P62, and/or envelope protein E1. Preferably, the replicon according to the present invention does not comprise a virus packaging signal, e.g., an alphavirus packaging signal. For example, the alphavirus packaging signal comprised in the coding region of nsP2 of SFV (White et al., 1998, J. Virol. 72:4320-4326) may be removed, e.g. by deletion or mutation. A suitable way of removing the alphavirus packaging signal includes adaptation of the codon usage of the coding region of nsP2. The degeneration of the genetic code may allow to delete the function of the packaging signal without affecting the amino acid sequence of the encoded nsP2.

DNA

[0389] The present invention also provides a DNA comprising a nucleic acid sequence encoding the RNA replicon according to the present invention.

[0390] Preferably, the DNA is double-stranded.

[0391] In a preferred embodiment, the DNA is a plasmid. The term "plasmid", as used herein, generally relates to a construct of extrachromosomal genetic material, usually a circular DNA duplex, which can replicate independently of chromosomal DNA.

[0392] The DNA of the present invention may comprise a promoter that can be recognized by a DNA-dependent RNA-polymerase. This allows for transcription of the encoded RNA in vivo or in vitro, e.g. of the RNA of the present invention. IVT vectors may be used in a standardized manner as template for in vitro transcription. Examples of promoters preferred according to the invention are promoters for SP6, T3 or T7 polymerase.

[0393] In one embodiment, the DNA of the present invention is an isolated nucleic acid molecule.

Methods of Preparing RNA

[0394] The RNA molecule according to the present invention may be obtainable by in vitro transcription. In vitro-transcribed RNA (IVT-RNA) is of particular interest in the present invention. IVT-RNA is obtainable by transcription from a nucleic acid molecule (particularly a DNA molecule). The DNA molecule(s) of the present invention are suitable for such purposes, particularly if comprising a promoter that can be recognized by a DNA-dependent RNA-polymerase.

[0395] rRNA according to the present invention can be synthesized in vitro. This allows to add cap-analogs to the in vitro transcription reaction. Typically, the poly(A) tail is encoded by a poly-(dT) sequence on the DNA template. Alternatively, capping and poly(A) tail addition can be achieved enzymatically after transcription.

[0396] The *in vitro* transcription methodology is known to the skilled person. For example, as mentioned in WO 2011/015347 A1, a variety of *in vitro* transcription kits is commercially available.

Kit

[0397] The present invention also provides a kit comprising the at least two RNA replicons according to the invention.

[0398] In one embodiment, the constituents of the kit are present as separate entities. For example, one constituent of the kit may be present in one entity, and another constituent of the kit may be present in a separate entity. For example, an open or closed container is a suitable entity. A closed container is preferred. The container used should preferably be RNase-free or essentially RNase-free.

[0399] In one embodiment, the kit of the present invention comprises RNA for inoculation with a cell and/or for administration to a human or animal subject.

[0400] The kit according to the present invention optionally comprises a label or other form of information element, e.g. an electronic data carrier.

The label or information element preferably comprises instructions, e.g. printed written instructions or instructions in electronic form that are optionally printable. The instructions may refer to at least one suitable possible use of the kit.

Pharmaceutical Composition

[0401] The RNA replicon composition described herein may be present in the form of a pharmaceutical composition. A pharmaceutical composition according to the invention may comprise at least one nucleic acid molecule according to the present invention. A pharmaceutical composition according to the invention comprises a pharmaceutically acceptable diluent and/or a pharmaceutically acceptable excipient and/or a pharmaceutically acceptable carrier and/or a pharmaceutically acceptable vehicle. The choice of pharmaceutically acceptable carrier, vehicle, excipient or diluent is not particularly limited. Any suitable pharmaceutically acceptable carrier, vehicle, excipient or diluent known in the art may be used.

[0402] In one embodiment of the present invention, a pharmaceutical composition can further comprise a solvent such as an aqueous solvent or any solvent that makes it possible to preserve the integrity of the rRNA. In a preferred embodiment, the pharmaceutical composition is an aqueous solution comprising RNA. The aqueous solution may optionally comprise solutes, e.g. salts.

[0403] In one embodiment of the present invention, the pharmaceutical composition is in the form of a freeze-dried composition. A freeze-dried composition is obtainable by freeze-drying a respective aqueous composition.

[0404] In one embodiment, the pharmaceutical composition comprises at least one cationic entity. In general, cationic lipids, cationic polymers and other substances with positive charges may form complexes with negatively charged nucleic acids. It is possible to stabilize the RNA according to the invention by complexation with cationic compounds, preferably polycationic compounds such as for example a cationic or polycationic peptide or protein. In one embodiment, the pharmaceutical composition according to the present invention comprises at least one cationic molecule selected from the group consisting of protamine, polyethylene imine, a poly-L-lysine, a poly-L-arginine, a histone or a cationic lipid.

[0405] According to the present invention, a cationic lipid is a cationic amphiphilic molecule, e.g., a molecule which comprises at least one hydrophilic and lipophilic moiety. The cationic lipid can be monocationic or polycationic. Cationic lipids typically have a lipophilic moiety, such as a sterol, an acyl or diacyl chain, and have an overall net positive charge. The head group of the lipid typically carries the positive charge. The cationic lipid preferably has a positive charge of 1 to 10 valences, more preferably a positive charge of 1 to 3 valences, and more preferably a positive charge of 1 valence. Examples of cationic lipids include, but are not limited to, 1,2-d-O-octadecenyl-3-trimethylammonium propane (DOTMA); dimethyldioctadecylammonium (DDAB); 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP); 1,2-dioleoyl-3-dimethylaminopropane (DODMA); 1,2-dioleoyl-3-dimethylammonium-propane (DODAP); 1,2-diacyloxy-3-dimethylammonium propanes; 1,2-dialkyl-3-dimethylammonium propanes; dioctadecyldimethyl ammonium chloride (DODAC), 1,2-dimyristoyloxypropyl-1,3-dimethylhydroxyethyl ammonium (DMRIE), and 2,3-dioleoyl-N-[2(spermine carboxamide)ethyl]-N,N-dimethyl-1-propanamium trifluoroacetate (DOSPA). Cationic lipids also include lipids with a tertiary amine group, including 1,2-dilinoleoyl-N,N-dimethyl-3-aminopropane (DLinDMA). Cationic lipids are suitable for formulating RNA in lipid formulations as described herein, such as liposomes, emulsions and lipoplexes. Typically, positive charges are contributed by at least one cationic lipid and negative charges are contributed by the RNA. In one embodiment, the pharmaceutical composition comprises at least one helper lipid, in addition to a cationic lipid. The helper lipid may be a neutral or an anionic lipid. The helper lipid may be a natural lipid, such as a phospholipid, or an analogue of a natural lipid, or a fully synthetic lipid, or lipid-like molecule, with no similarities with natural lipids. In the case where a pharmaceutical composition includes both a cationic lipid and a helper lipid, the molar ratio of the cationic lipid to the neutral lipid can be appropriately determined in view of stability of the formulation and the like.

[0406] In one embodiment, the pharmaceutical composition according to the present invention comprises protamine. According to the invention, protamine is useful as cationic carrier agent. The term "protamine" refers to any of various strongly basic proteins of relatively low molecular weight that are rich in arginine and are found associated especially with DNA in place of somatic histories in the sperm cells of animals such as fish. In particular, the term "protamine" refers to proteins found in fish sperm that are strongly basic, are soluble in water, are not coagulated by heat, and comprise multiple arginine monomers. According to the invention, the term "protamine" as used herein is meant to comprise any protamine amino acid sequence obtained or derived from native or biological sources including fragments thereof and multimeric forms of said amino acid sequence or fragment thereof. Furthermore, the term encompasses (synthesized) polypeptides which are artificial and specifically designed for specific purposes and cannot be isolated from native or biological sources.

[0407] In some embodiments, the compositions of the invention may comprise one or more adjuvants. Adjuvants may be added to vaccines to stimulate the immune system's response; adjuvants do not typically provide immunity themselves. Exemplary adjuvants include without limitation the following: Inorganic compounds (e.g. alum, aluminum hydroxide, aluminum phosphate, calcium phosphate hydroxide); mineral oil (e.g. paraffin oil), cytokines (e.g. IL-1, IL-2, IL-12); immunostimulatory polynucleotide (such as RNA or DNA; e.g., CpG-containing oligonucleotides); saponins (e.g. plant saponins from Quillaja, Soybean, Polygala senega); oil emulsions or liposomes; polyoxy ethylene ether and poly oxy ethylene ester formulations; polyphosphazene (PCPP); muramyl peptides; imidazoquinolone compounds; thiosemicarbazone compounds; the Flt3 ligand (WO 2010/066418 A1); or any other adjuvant that is known by a person skilled in the art. A preferred adjuvant for administration of RNA according to the present invention is the Flt3 ligand (WO 2010/066418 A1). When Flt3 ligand is administered together with RNA that codes for an antigen, a strong increase in antigen-specific CD8⁺ T cells may be observed.

[0408] The pharmaceutical composition according to the invention can be buffered, (e.g., with an acetate buffer, a citrate buffer, a succinate buffer, a Tris buffer, a phosphate buffer).

RNA-Containing Particles

[0409] In some embodiments, owing to the instability of non-protected RNA, it is advantageous to provide the RNA molecules of the present invention in complexed or encapsulated form. Respective pharmaceutical compositions are provided in the present invention. In particular, in some embodiments, the pharmaceutical composition of the present invention comprises nucleic acid-containing particles, preferably RNA-containing particles. Respective pharmaceutical compositions are referred to as particulate formulations. In particulate formulations according to the present invention, a particle comprises nucleic acid according to the invention and a pharmaceutically acceptable carrier or a pharmaceutically acceptable vehicle that is suitable for delivery of the nucleic acid. The nucleic acid-containing particles may be, for example, in the form of proteinaceous particles or in the form of lipid-containing particles. Suitable proteins or lipids are referred to as particle forming agents. Proteinaceous particles and lipid-containing particles have been described previously to be suitable for delivery of alphaviral RNA in particulate form (e.g. Strauss & Strauss, 1994, Microbiol. Rev. 58:491-562). In particular, alphavirus structural proteins (provided e.g. by a helper virus) are a suitable carrier for delivery of RNA in the form of proteinaceous particles.

[0410] In one embodiment, the particulate formulation of the present invention is a nanoparticulate formulation. In that embodiment, the composition

according to the present invention comprises nucleic acid according to the invention in the form of nanoparticles. Nanoparticulate formulations can be obtained by various protocols and with various complexing compounds. Lipids, polymers, oligomers, or amphiphiles are typical constituents of nanoparticulate formulations.

[0411] As used herein, the term “nanoparticle” refers to any particle having a diameter making the particle suitable for systemic, in particular parenteral, administration, of, in particular, nucleic acids, typically a diameter of 1000 nanometers (nm) or less. In one embodiment, the nanoparticles have an average diameter in the range of from about 50 nm to about 1000 nm, preferably from about 50 nm to about 400 nm, preferably about 100 nm to about 300 nm such as about 150 nm to about 200 nm. In one embodiment, the nanoparticles have a diameter in the range of about 200 to about 700 nm, about 200 to about 600 nm, preferably about 250 to about 550 nm, in particular about 300 to about 500 nm or about 200 to about 400 nm. In one embodiment, the average diameter is between about 50 to 150 nm, preferably, about 60 to 120 nm. In one embodiment, the average diameter is less than 50 nm,

[0412] In one embodiment, the polydispersity index (PI) of the nanoparticles described herein, as measured by dynamic light scattering, is 0.5 or less, preferably 0.4 or less or even more preferably 0.3 or less. The “polydispersity index” (PI) is a measurement of homogeneous or heterogeneous size distribution of the individual particles (such as liposomes) in a particle mixture and indicates the breadth of the particle distribution in a mixture. The PI can be determined, for example, as described in WO 2013/143555 A1.

[0413] As used herein, the term “nanoparticulate formulation” or similar terms refer to any particulate formulation that contains at least one nanoparticle. In some embodiments, a nanoparticulate composition is a uniform collection of nanoparticles. In some embodiments, a nanoparticulate composition is a lipid-containing pharmaceutical formulation, such as a liposome formulation or an emulsion.

Lipid-Containing Pharmaceutical Compositions

[0414] In one embodiment, the pharmaceutical composition of the present invention comprises at least one lipid. Preferably, at least one lipid is a cationic lipid. Said lipid-containing pharmaceutical composition comprises nucleic acid according to the present invention. In one embodiment, the pharmaceutical composition according to the invention comprises RNA encapsulated in a vesicle, e.g. in a liposome. In one embodiment, the pharmaceutical composition according to the invention comprises RNA in the form of an emulsion. In one embodiment, the pharmaceutical composition according to the invention comprises rRNA in a complex with a cationic compound, thereby forming e.g. so-called lipoplexes or polyplexes. Encapsulation of RNA within vesicles such as liposomes is distinct from, for instance, lipid/RNA complexes. Lipid/RNA complexes are obtainable e.g. when RNA is e.g. mixed with pre-formed liposomes.

[0415] In one embodiment, the pharmaceutical composition according to the invention comprises rRNA encapsulated in a vesicle. Such formulation is a particular particulate formulation according to the invention. A vesicle is a lipid bilayer rolled up into a spherical shell, enclosing a small space and separating that space from the space outside the vesicle. Typically, the space inside the vesicle is an aqueous space, i.e. comprises water. Typically, the space outside the vesicle is an aqueous space, i.e. comprises water. The lipid bilayer is formed by one or more lipids (vesicle-forming lipids). The membrane enclosing the vesicle is a lamellar phase, similar to that of the plasma membrane. The vesicle according to the present invention may be a multilamellar vesicle, a unilamellar vesicle, or a mixture thereof. When encapsulated in a vesicle, the rRNA is typically separated from any external medium. Thus, it is present in protected form, functionally equivalent to the protected form in a natural alphavirus. Suitable vesicles are particles, particularly nanoparticles, as described herein.

[0416] For example, RNA (rRNA) may be encapsulated in a liposome. In that embodiment, the pharmaceutical composition is or comprises a liposome formulation. Encapsulation within a liposome will typically protect RNA from RNase digestion. It is possible that the liposomes include some external RNA (e.g. on their surface), but at least half of the RNA (and ideally all of it) is encapsulated within the core of the liposome.

[0417] Liposomes are microscopic lipidic vesicles often having one or more bilayers of a vesicle-forming lipid, such as a phospholipid, and are capable of encapsulating a drug, e.g. RNA. Different types of liposomes may be employed in the context of the present invention, including, without being limited thereto, multilamellar vesicles (MLV), small unilamellar vesicles (SUV), large unilamellar vesicles (LUV), sterically stabilized liposomes (SSL), multivesicular vesicles (MV), and large multivesicular vesicles (LMV) as well as other bilayered forms known in the art. The size and lamellarity of the liposome will depend on the manner of preparation. There are several other forms of supramolecular organization in which lipids may be present in an aqueous medium, comprising lamellar phases, hexagonal and inverse hexagonal phases, cubic phases, micelles, reverse micelles composed of monolayers. These phases may also be obtained in the combination with DNA or RNA, and the interaction with RNA and DNA may substantially affect the phase state. Such phases may be present in nanoparticulate RNA formulations of the present invention.

[0418] Liposomes may be formed using standard methods known to the skilled person. Respective methods include the reverse evaporation method, the ethanol injection method, the dehydration-rehydration method, sonication or other suitable methods. Following liposome formation, the liposomes can be sized to obtain a population of liposomes having a substantially homogeneous size range.

[0419] In a preferred embodiment of the present invention, the rRNA is present in a liposome which includes at least one cationic lipid. Respective liposomes can be formed from a single lipid or from a mixture of lipids, provided that at least one cationic lipid is used. Preferred cationic lipids have a nitrogen atom which is capable of being protonated; preferably, such cationic lipids are lipids with a tertiary amine group. A particularly suitable lipid with a tertiary amine group is 1,2-dilinoleyloxy-N,N-dimethyl-3-aminopropane (DLinDMA). In one embodiment, the RNA according to the present invention is present in a liposome formulation as described in WO 2012/006378 A1: a liposome having a lipid bilayer encapsulating an aqueous core including RNA, wherein the lipid bilayer comprises a lipid with a pKa in the range of 5.0 to 7.6, which preferably has a tertiary amine group. Preferred cationic lipids with a tertiary amine group include DLinDMA (pKa 5.8) and are generally described in WO 2012/031046 A2. According to WO 2012/031046 A2, liposomes comprising a respective compound are particularly suitable for encapsulation of RNA and thus liposomal delivery of RNA. In one embodiment, the RNA according to the present invention is present in a liposome formulation, wherein the liposome includes at least one cationic lipid whose head group includes at least one nitrogen atom (N) which is capable of being protonated, wherein the liposome and the RNA have a N:P ratio of between 1:1 and 20:1. According to the present invention, “N:P ratio” refers to the molar ratio of nitrogen atoms (N) in the cationic lipid to phosphate atoms (P) in the RNA comprised in a lipid containing particle (e.g. liposome), as described in WO 2013/006825 A1. The N:P ratio of between 1:1 and 20:1 is implicated in the net charge of the liposome and in efficiency of delivery of RNA to a vertebrate cell.

[0420] In one embodiment, the rRNA according to the present invention is present in a liposome formulation that comprises at least one lipid which includes a polyethylene glycol (PEG) moiety, wherein RNA is encapsulated within a PEGylated liposome such that the PEG moiety is present on the liposome's exterior, as described in WO 2012/031043 A1 and WO 2013/033563 A1.

[0421] In one embodiment, the rRNA according to the present invention is present in a liposome formulation, wherein the liposome has a diameter in the range of 60-180 nm, as described in WO 2012/030901 A1.

[0422] In one embodiment, the rRNA according to the present invention is present in a liposome formulation, wherein the rRNA-containing liposomes have a net charge close to zero or negative, as disclosed in WO 2013/143555 A1.

[0423] In other embodiments, the rRNA according to the present invention is present in the form of an emulsion. Emulsions have been previously described to be used for delivery of nucleic acid molecules, such as rRNA molecules, to cells. Preferred herein are oil-in-water emulsions. The respective emulsion particles comprise an oil core and a cationic lipid. More preferred are cationic oil-in-water emulsions in which the RNA according to the present invention is complexed to the emulsion particles. The emulsion particles comprise an oil core and a cationic lipid. The cationic lipid can interact with the negatively charged rRNA, thereby anchoring the rRNA to the emulsion particles. In an oil-in-water emulsion, emulsion particles are dispersed in an aqueous continuous phase. For example, the average diameter of the emulsion particles may typically be from about 80 nm to 180 nm. In one embodiment, the pharmaceutical composition of the present invention is a cationic oil-in-water emulsion, wherein the

emulsion particles comprise an oil core and a cationic lipid as described in WO 2012/006380 A2. The rRNA according to the present invention may be present in the form of an emulsion comprising a cationic lipid wherein the N:P ratio of the emulsion is at least 4:1, as described in WO 2013/006834 A1. The rRNA according to the present invention may be present in the form of a cationic lipid emulsion, as described in WO 2013/006837 A1. In particular, the composition may comprise rRNA complexed with a particle of a cationic oil-in-water emulsion, wherein the ratio of oil/lipid is at least about 8:1 (mole:mole).

[0424] In other embodiments, the pharmaceutical composition according to the invention comprises RNA in the format of a lipoplex. The term, “lipoplex” or “RNA lipoplex” refers to a complex of lipids and nucleic acids such as RNA. Lipoplexes can be formed of cationic (positively charged) liposomes and the anionic (negatively charged) nucleic acid. The cationic liposomes can also include a neutral “helper” lipid. In the simplest case, the lipoplexes form spontaneously by mixing the nucleic acid with the liposomes with a certain mixing protocol, however various other protocols may be applied. It is understood that electrostatic interactions between positively charged liposomes and negatively charged nucleic acid are the driving force for the lipoplex formation (WO 2013/143555 A1). In one embodiment of the present invention, the net charge of the RNA lipoplex particles is close to zero or negative. It is known that electro-neutral or negatively charged lipoplexes of RNA and liposomes lead to substantial RNA expression in spleen dendritic cells (DCs) after systemic administration and are not associated with the elevated toxicity that has been reported for positively charged liposomes and lipoplexes (cf. WO 2013/143555 A1). Therefore, in one embodiment of the present invention, the pharmaceutical composition according to the invention comprises RNA in the format of nanoparticles, preferably lipoplex nanoparticles, in which (i) the number of positive charges in the nanoparticles does not exceed the number of negative charges in the nanoparticles and/or (ii) the nanoparticles have a neutral or net negative charge and/or (iii) the charge ratio of positive charges to negative charges in the nanoparticles is 1.4:1 or less and/or (iv) the zeta potential of the nanoparticles is 0 or less. As described in WO 2013/143555 A1, zeta potential is a scientific term for electrokinetic potential in colloidal systems. In the present invention, (a) the zeta potential and (b) the charge ratio of the cationic lipid to the RNA in the nanoparticles can both be calculated as disclosed in WO 2013/143555 A1. In summary, pharmaceutical compositions which are nanoparticulate lipoplex formulations with a defined particle size, wherein the net charge of the particles is close to zero or negative, as disclosed in WO 2013/143555 A1, are preferred pharmaceutical compositions in the context of the present invention.

[0425] In one embodiment, nucleic acid such as the rRNA described herein is administered in the form of lipid nanoparticles (LNPs). The LNP may comprise any lipid capable of forming a particle to which the one or more nucleic acid molecules are attached, or in which the one or more nucleic acid molecules are encapsulated.

[0426] In one embodiment, the LNP comprises one or more cationic lipids, and one or more stabilizing lipids. Stabilizing lipids include neutral lipids and pegylated lipids.

[0427] In one embodiment, the LNP comprises a cationic lipid, a neutral lipid, a steroid, a polymer conjugated lipid; and the RNA, encapsulated within or associated with the lipid nanoparticle.

[0428] In one embodiment, the LNP comprises from 40 to 55 mol percent, from 40 to 50 mol percent, from 41 to 49 mol percent, from 41 to 48 mol percent, from 42 to 48 mol percent, from 43 to 48 mol percent, from 44 to 48 mol percent, from 45 to 48 mol percent, from 46 to 48 mol percent, from 47 to 48 mol percent, or from 47.2 to 47.8 mol percent of the cationic lipid. In one embodiment, the LNP comprises about 47.0, 47.1, 47.2, 47.3, 47.4, 47.5, 47.6, 47.7, 47.8, 47.9 or 48.0 mol percent of the cationic lipid.

[0429] In one embodiment, the neutral lipid is present in a concentration ranging from 5 to 15 mol percent, from 7 to 13 mol percent, or from 9 to 11 mol percent. In one embodiment, the neutral lipid is present in a concentration of about 9.5, 10 or 10.5 mol percent.

[0430] In one embodiment, the steroid is present in a concentration ranging from 30 to 50 mol percent, from 35 to 45 mol percent or from 38 to 43 mol percent. In one embodiment, the steroid is present in a concentration of about 40, 41, 42, 43, 44, 45 or 46 mol percent.

[0431] In one embodiment, the LNP comprises from 1 to 10 mol percent, from 1 to 5 mol percent, or from 1 to 2.5 mol percent of the polymer conjugated lipid.

[0432] In one embodiment, the LNP comprises from 40 to 50 mol percent a cationic lipid; from 5 to 15 mol percent of a neutral lipid; from 35 to 45 mol percent of a steroid; from 1 to 10 mol percent of a polymer conjugated lipid; and the RNA, encapsulated within or associated with the lipid nanoparticle.

[0433] In one embodiment, the mol percent is determined based on total mol of lipid present in the lipid nanoparticle.

[0434] In one embodiment, the neutral lipid is selected from the group consisting of DSPC, DPPC, DMPC, DOPC, POPC, DOPE, DOPG, DPPG, POPE, DPPE, DMPE, DSPE, and SM. In one embodiment, the neutral lipid is selected from the group consisting of DSPC, DPPC, DMPC, DOPC, POPC, DOPE and SM. In one embodiment, the neutral lipid is DSPC.

[0435] In one embodiment, the steroid is cholesterol.

[0436] In one embodiment, the polymer conjugated lipid is a pegylated lipid. In one embodiment, the pegylated lipid has the following structure.

##STR00016## [0437] or a pharmaceutically acceptable salt, tautomer or stereoisomer thereof, wherein: [0438] R_{sup.12} and R_{sup.13} are each independently a straight or branched, saturated or unsaturated alkyl chain containing from 10 to 30 carbon atoms, wherein the alkyl chain is optionally interrupted by one or more ester bonds; and w has a mean value ranging from 30 to 60. In one embodiment, R_{sup.12} and R_{sup.13} are each independently straight, saturated alkyl chains containing from 12 to 16 carbon atoms. In one embodiment, w has a mean value ranging from 40 to 55. In one embodiment, the average w is about 45. In one embodiment, R_{sup.12} and R_{sup.13} are each independently a straight, saturated alkyl chain containing about 14 carbon atoms, and w has a mean value of about 45.

[0439] In one embodiment, the pegylated lipid is DMG-PEG 2000, e.g., having the following structure:

##STR00017##

[0440] In some embodiments, the cationic lipid component of the LNPs has the structure of Formula (III):

##STR00018## [0441] or a pharmaceutically acceptable salt, tautomer, prodrug or stereoisomer thereof, wherein: [0442] one of L_{sup.1} or L_{sup.2} is —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O).sub.x—, —S—S—, —C(=O)S—, SC(=O)—, —NR_{sup.a}C(=O)—, —C(=O)NR_{sup.a}—, NR_{sup.a}C(=O)NR_{sup.a}—, —OC(=O)NR_{sup.a}— or —NR_{sup.a}C(=O)O—, and the other of L_{sup.1} or L_{sup.2} is —O(C=O)—, —(C=O)O—, —C(=O)—, —O—, —S(O).sub.x—, —S—S—, —C(=O)S—, SC(=O)—, —NR_{sup.a}C(=O)—, —C(=O)NR_{sup.a}—, NR_{sup.a}C(=O)NR_{sup.a}—, —OC(=O)NR_{sup.a}— or —NR_{sup.a}C(=O)O— or a direct bond; [0443] G_{sup.1} and G_{sup.2} are each independently unsubstituted C_{sub.1}-C_{sub.12} alkylene or C_{sub.1}-C_{sub.12} alkenylene; [0444] G_{sup.3} is C_{sub.1}-C_{sub.24} alkylene, C_{sub.1}-C_{sub.24} alkenylene, C_{sub.3}-C_{sub.8} cycloalkylene, C_{sub.3}-C_{sub.8} cycloalkenylene; [0445] R_{sup.a} is H or C_{sub.1}-C_{sub.12} alkyl; [0446] R_{sup.1} and R_{sup.2} are each independently C_{sub.6}-C_{sub.24} alkyl or C_{sub.6}-C_{sub.24} alkenyl; [0447] R_{sup.3} is H, OR_{sup.5}, CN, —C(=O)OR_{sup.4}, —OC(=O)R_{sup.4} or —NR_{sup.5}C(=O)R_{sup.4}; [0448] R_{sup.4} is C_{sub.1}-C_{sub.12} alkyl; [0449] R_{sup.5} is H or C_{sub.1}-C_{sub.6} alkyl; and [0450] x is 0, 1 or 2.

[0451] In some of the foregoing embodiments of Formula (III), the lipid has one of the following structures (IIIA) or (IIIB):

##STR00019## [0452] wherein: [0453] A is a 3 to 8-membered cycloalkyl or cycloalkylene ring; [0454] R_{sup.6} is, at each occurrence, independently H, OH or C_{sub.1}-C_{sub.24} alkyl; [0455] n is an integer ranging from 1 to 15.

[0456] In some of the foregoing embodiments of Formula (III), the lipid has structure (IIIA), and in other embodiments, the lipid has structure (IIIB).

[0457] In other embodiments of Formula (III), the lipid has one of the following structures (IIIC) or (IIID):

##STR00020## [0458] wherein y and z are each independently integers ranging from 1 to 12.

[0459] In any of the foregoing embodiments of Formula (III), one of L_{sup.1} or L_{sup.2} is —O(C=O)—. For example, in some embodiments each of L_{sup.1} and L_{sup.2} are —O(C=O)—. In some different embodiments of any of the foregoing, L_{sup.1} and L_{sup.2} are each independently —(C=O)O— or —O(C=O)—. For example, in some embodiments each of L_{sup.1} and L_{sup.2} is —(C=O)O—.

[0460] In some different embodiments of Formula (III), the lipid has one of the following structures (IIIE) or (IIIF):

##STR00021##

[0461] In some of the foregoing embodiments of Formula (III), the lipid has one of the following structures (IIIG), (IIIH), (IIII), or (IIIJ):

##STR00022##

[0462] In some of the foregoing embodiments of Formula (III), n is an integer ranging from 2 to 12, for example from 2 to 8 or from 2 to 4. For example, in some embodiments, n is 3, 4, 5 or 6. In some embodiments, n is 3. In some embodiments, n is 4. In some embodiments, n is 5. In some embodiments, n is 6.

[0463] In some other of the foregoing embodiments of Formula (III), y and z are each independently an integer ranging from 2 to 10. For example, in some embodiments, y and z are each independently an integer ranging from 4 to 9 or from 4 to 6.

[0464] In some of the foregoing embodiments of Formula (III), R_{sup.6} is H. In other of the foregoing embodiments, R_{sup.6} is C_{sub.1}-C_{sub.2} alkyl. In other embodiments, R_{sup.6} is OH.

[0465] In some embodiments of Formula (III), G_{sup.3} is unsubstituted. In other embodiments, G₃ is substituted. In various different embodiments, G_{sup.3} is linear C_{sub.1}-C_{sub.24} alkylene or linear C_{sub.1}-C_{sub.24} alkenylene.

[0466] In some other foregoing embodiments of Formula (III), R_{sup.1} or R_{sup.2}, or both, is C_{sub.6}-C_{sub.2} alkenyl. For example, in some embodiments, R_{sup.1} and R_{sup.2} each, independently have the following structure:

##STR00023## [0467] wherein: [0468] R_{sup.7a} and R_{sup.7b} are, at each occurrence, independently H or C_{sub.1}-C_{sub.12} alkyl; and [0469] a is an integer from 2 to 12, [0470] wherein R_{sup.7a}, R_{sup.7b} and a are each selected such that R_{sup.1} and R_{sup.2} each independently comprise from 6 to 20 carbon atoms. For example, in some embodiments a is an integer ranging from 5 to 9 or from 8 to 12.

[0471] In some of the foregoing embodiments of Formula (III), at least one occurrence of R_{sup.7a} is H. For example, in some embodiments, R_{sup.7b} is H at each occurrence. In other different embodiments of the foregoing, at least one occurrence of R_{sup.7b} is C_{sub.1}-C_{sub.8} alkyl. For example, in some embodiments, C_{sub.1}-C_{sub.8} alkyl is methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, tert-butyl, n-hexyl or n-octyl.

[0472] In different embodiments of Formula (III), R_{sup.1} or R_{sup.2}, or both, has one of the following structures:

##STR00024##

[0473] In some of the foregoing embodiments of Formula (III), R_{sup.3} is OH, CN, —C(=O)OR_{sup.4}, —OC(=O)R_{sup.4} or —NHC(=O)R_{sup.4}. In some embodiments, R_{sup.4} is methyl or ethyl.

[0474] In various different embodiments, the cationic lipid of Formula (III) has one of the structures set forth in the table below.

[0475] Representative Compounds of Formula (III).

TABLE-US-00002 No. Structure III-1 [00025]  III-2 [00026]  III-3 [00027]  III-4 [00028]  III-5 [00029]  III-6 [00030]  III-7 [00031]  III-8 [00032]  III-9 [00033]  III-10 [00034]  III-11 [00035]  III-12 [00036]  III-13 [00037]  III-14 [00038]  III-15 [00039]  III-16 [00040]  III-17 [00041]  III-18 [00042]  III-19 [00043]  III-20 [00044]  III-21 [00045]  III-22 [00046]  III-23 [00047]  III-24 [00048]  III-25 [00049]  III-26 [00050]  III-27 [00051]  III-28 [00052] III-29 [00053] III-30 [00054] III-31 [00055] III-32 [00056] III-33 [00057] III-34 [00058] III-35 [00059] III-36 [00060]

[0476] In some embodiments, the LNP comprises a lipid of Formula (III), RNA, a neutral lipid, a steroid and a pegylated lipid. In some embodiments, the lipid of Formula (III) is compound III-3. In some embodiments, the neutral lipid is DSPC. In some embodiments, the steroid is cholesterol. In some embodiments, the pegylated lipid is ALC-0159.

[0477] In some embodiments, the cationic lipid is present in the LNP in an amount from about 40 to about 50 mole percent. In one embodiment, the neutral lipid is present in the LNP in an amount from about 5 to about 15 mole percent. In one embodiment, the steroid is present in the LNP in an amount from about 35 to about 45 mole percent. In one embodiment, the pegylated lipid is present in the LNP in an amount from about 1 to about 10 mole percent.

[0478] In some embodiments, the LNP comprises compound III-3 in an amount from about 40 to about 50 mole percent, DSPC in an amount from about 5 to about 15 mole percent, cholesterol in an amount from about 35 to about 45 mole percent, and ALC-0159 in an amount from about 1 to about 10 mole percent.

[0479] In some embodiments, the LNP comprises compound III-3 in an amount of about 47.5 mole percent, DSPC in an amount of about 10 mole percent, cholesterol in an amount of about 40.7 mole percent, and ALC-0159 in an amount of about 1.8 mole percent.

[0480] In various different embodiments, the cationic lipid has one of the structures set forth in the table below.

TABLE-US-00003 No. Structure A [00061]  B [00062]  C [00063]  D [00064]  E [00065]  F [00066] 

[0481] In some embodiments, the LNP comprises a cationic lipid shown in the above table, e.g., a cationic lipid of Formula (B) or Formula (D), in particular a cationic lipid of Formula (D), RNA, a neutral lipid, a steroid and a pegylated lipid. In some embodiments, the neutral lipid is DSPC. In some embodiments, the steroid is cholesterol. In some embodiments, the pegylated lipid is DMG-PEG 2000.

[0482] In one embodiment, the LNP comprises a cationic lipid that is an ionizable lipid-like material (lipidoid). In one embodiment, the cationic lipid has the following structure:

##STR00067##

[0483] The N/P value is preferably at least about 4. In some embodiments, the N/P value ranges from 4 to 20, 4 to 12, 4 to 10, 4 to 8, or 5 to 7. In one embodiment, the N/P value is about 6.

[0484] LNP described herein may have an average diameter that in one embodiment ranges from about 30 nm to about 200 nm, or from about 60 nm to about 120 nm.

RNA Targeting

[0485] Some aspects of the disclosure involve the targeted delivery of the rRNA disclosed herein (e.g., RNA encoding vaccine antigens and/or immunostimulants).

[0486] In one embodiment, the disclosure involves targeting lung. Targeting lung is in particular preferred if the RNA administered is RNA encoding vaccine antigen. RNA may be delivered to lung, for example, by administering the RNA which may be formulated as particles as described herein, e.g., lipid particles, by inhalation.

[0487] In one embodiment, the disclosure involves targeting the lymphatic system, in particular secondary lymphoid organs, more specifically spleen. Targeting the lymphatic system, in particular secondary lymphoid organs, more specifically spleen is in particular preferred if the RNA administered is RNA encoding vaccine antigen.

[0488] In one embodiment, the target cell is a spleen cell. In one embodiment, the target cell is an antigen presenting cell such as a professional antigen presenting cell in the spleen. In one embodiment, the target cell is a dendritic cell in the spleen.

[0489] The "lymphatic system" is part of the circulatory system and an important part of the immune system, comprising a network of lymphatic vessels that carry lymph. The lymphatic system consists of lymphatic organs, a conducting network of lymphatic vessels, and the circulating lymph.

the primary or central lymphoid organs generate lymphocytes from immature progenitor cells. The thymus and the bone marrow constitute the primary lymphoid organs. Secondary or peripheral lymphoid organs, which include lymph nodes and the spleen, maintain mature naïve lymphocytes and initiate an adaptive immune response.

[0490] RNA may be delivered to spleen by so-called lipoplex formulations, in which the RNA is bound to liposomes comprising a cationic lipid and optionally an additional or helper lipid to form injectable nanoparticle formulations. The liposomes may be obtained by injecting a solution of the lipids in ethanol into water or a suitable aqueous phase. RNA lipoplex particles may be prepared by mixing the liposomes with RNA. Spleen targeting RNA lipoplex particles are described in WO 2013/143683, herein incorporated by reference. It has been found that RNA lipoplex particles having a net negative charge may be used to preferentially target spleen tissue or spleen cells such as antigen-presenting cells, in particular dendritic cells. Accordingly, following administration of the RNA lipoplex particles, RNA accumulation and/or RNA expression in the spleen occurs. Thus, RNA lipoplex particles of the disclosure may be used for expressing RNA in the spleen. In an embodiment, after administration of the RNA lipoplex particles, no or essentially no RNA accumulation and/or RNA expression in the lung and/or liver occurs. In one embodiment, after administration of the RNA lipoplex particles, RNA accumulation and/or RNA expression in antigen presenting cells, such as professional antigen presenting cells in the spleen occurs. Thus, RNA lipoplex particles of the disclosure may be used for expressing RNA in such antigen presenting cells. In one embodiment, the antigen presenting cells are dendritic cells and/or macrophages.

[0491] The electric charge of the RNA lipoplex particles of the present disclosure is the sum of the electric charges present in the at least one cationic lipid and the electric charges present in the RNA. The charge ratio is the ratio of the positive charges present in the at least one cationic lipid to the negative charges present in the RNA. The charge ratio of the positive charges present in the at least one cationic lipid to the negative charges present in the RNA is calculated by the following equation: $\text{charge ratio} = [(\text{cationic lipid concentration (mol)}) * (\text{the total number of positive charges in the cationic lipid})] / [(\text{RNA concentration (mol)}) * (\text{the total number of negative charges in RNA})]$.

[0492] The spleen targeting RNA lipoplex particles described herein at physiological pH preferably have a net negative charge such as a charge ratio of positive charges to negative charges from about 1.9:2 to about 1:2, or about 1.6:2 to about 1:2, or about 1.6:2 to about 1.1:2. In specific embodiments, the charge ratio of positive charges to negative charges in the RNA lipoplex particles at physiological pH is about 1.9:2.0, about 1.8:2.0, about 1.7:2.0, about 1.6:2.0, about 1.5:2.0, about 1.4:2.0, about 1.3:2.0, about 1.2:2.0, about 1.1:2.0, or about 1:2.0.

[0493] Immunostimulants may be provided to a subject by administering to the subject RNA encoding an immunostimulant in a formulation for preferential delivery of RNA to liver or liver tissue. The delivery of RNA to such target organ or tissue is preferred, in particular, if it is desired to express large amounts of the immunostimulant and/or if systemic presence of the immunostimulant, in particular in significant amounts, is desired or required.

[0494] RNA delivery systems have an inherent preference to the liver. This pertains to lipid-based particles, cationic and neutral nanoparticles, in particular lipid nanoparticles such as liposomes, nanomicelles and lipophilic ligands in bioconjugates. Liver accumulation is caused by the discontinuous nature of the hepatic vasculature or the lipid metabolism (liposomes and lipid or cholesterol conjugates).

[0495] For in vivo delivery of RNA to the liver, a drug delivery system may be used to transport the RNA into the liver by preventing its degradation. For example, polyplex nanomicelles consisting of a poly(ethylene glycol) (PEG)-coated surface and an mRNA-containing core is a useful system because the nanomicelles provide excellent in vivo stability of the RNA, under physiological conditions. Furthermore, the stealth property provided by the polyplex nanomicelle surface, composed of dense PEG palisades, effectively evades host immune defenses.

[0496] Examples of suitable immunostimulants for targeting liver are cytokines involved in T cell proliferation and/or maintenance. Examples of suitable cytokines include IL2 or IL7, fragments and variants thereof, and fusion proteins of these cytokines, fragments and variants, such as extended-PK cytokines.

[0497] In another embodiment, RNA encoding an immunostimulant may be administered in a formulation for preferential delivery of RNA to the lymphatic system, in particular secondary lymphoid organs, more specifically spleen. The delivery of an immunostimulant to such target tissue is preferred, in particular, if presence of the immunostimulant in this organ or tissue is desired (e.g., for inducing an immune response, in particular in case immunostimulants such as cytokines are required during T-cell priming or for activation of resident immune cells), while it is not desired that the immunostimulant is present systemically, in particular in significant amounts (e.g., because the immunostimulant has systemic toxicity).

[0498] Examples of suitable immunostimulants are cytokines involved in T cell priming. Examples of suitable cytokines include IL12, IL15, IFN- α , or IFN- β , fragments and variants thereof, and fusion proteins of these cytokines, fragments and variants, such as extended-PK cytokines.

Methods for Producing a Protein

[0499] The present invention also provides a method for producing a protein of interest in a cell comprising the steps of: [0500] (a) obtaining the RNA replicon composition according to the invention, which comprises an open reading frame encoding the protein of interest, and [0501] (b) inoculating the RNA replicon into the cell.

[0502] In various embodiments of the method, the RNA replicon is as defined above for the RNA replicon of the invention, as long as the RNA replicon comprises an open reading frame encoding the protein of interest, optionally an open reading frame encoding functional non-structural protein, and can be replicated by the functional non-structural protein, wherein the rRNA may comprising at least one modified nucleotide and one or more point mutations in a regulatory sequence that restores or improves a function of the modified rRNA.

[0503] The cell into which one or more nucleic molecule can be inoculated can be referred to as “host cell”. According to the invention, the term “host cell” refers to any cell which can be transformed or transfected with an exogenous nucleic acid molecule. The term “cell” preferably is an intact cell, i.e. a cell with an intact membrane that has not released its normal intracellular components such as enzymes, organelles, or genetic material. An intact cell preferably is a viable cell, i.e. a living cell capable of carrying out its normal metabolic functions. The term “host cell” comprises, according to the invention, prokaryotic (e.g. *E. coli*) or eukaryotic cells (e.g. human and animal cells, plant cells, yeast cells and insect cells). Particular preference is given to mammalian cells such as cells from humans, mice, hamsters, pigs, domesticated animals including horses, cows, sheep and goats, as well as primates. The cells may be derived from a multiplicity of tissue types and comprise primary cells and cell lines. Specific examples include keratinocytes, peripheral blood leukocytes, bone marrow stem cells and embryonic stem cells. In other embodiments, the host cell is an antigen-presenting cell, in particular a dendritic cell, a monocyte or a macrophage. A nucleic acid may be present in the host cell in a single or in several copies and, in one embodiment is expressed in the host cell.

[0504] The cell may be a prokaryotic cell or a eukaryotic cell. Prokaryotic cells are suitable herein e.g. for propagation of DNA according to the invention, and eukaryotic cells are suitable herein e.g. for expression of the open reading frame of the replicon.

[0505] In the method of the present invention, any of the RNA replicon compositions according to the invention, or the kit according to the invention, or the pharmaceutical composition according to the invention, can be used. RNA can be used in the form of a pharmaceutical composition, or as naked RNA e.g. for electroporation.

[0506] In the method for producing a protein in a cell according to the present invention, the cell may be an antigen presenting cell, and the method may be used for expressing the RNA encoding the antigen. To this end, the invention may involve introduction of RNA encoding antigen into antigen presenting cells such as dendritic cells. For transfection of antigen presenting cells such as dendritic cells a pharmaceutical composition comprising RNA encoding the antigen may be used.

[0507] In one embodiment, a method for producing a protein in a cell is an in vitro method. In one embodiment, a method for production of a protein in a cell does not comprise the removal of a cell from a human or animal subject by surgery or therapy.

[0508] In this embodiment, the cell inoculated according to the invention may be administered to a subject so as to produce the protein in the subject and to provide the subject with the protein. The cell may be autologous, syngeneic, allogeneic or heterologous with respect to the subject.

[0509] In other embodiments, the cell in a method for producing a protein in a cell may be present in a subject, such as a patient. In these embodiments, the method for producing a protein in a cell is an in vivo method which comprises administration of RNA molecules to the subject.

[0510] In this respect, the invention also provides a method for producing a protein of interest in a subject comprising the steps of: [0511] (a) obtaining the RNA replicon according to the invention, which comprises an open reading frame encoding the protein of interest, and [0512] (b) administering the RNA replicon to the subject.

[0513] In various embodiments of the method, the RNA replicon is as defined above for the RNA replicon of the invention, as long as the RNA replicon comprises an open reading frame encoding the protein of interest, optionally an open reading frame encoding functional non-structural protein, and can be replicated by the functional non-structural protein, wherein the rRNA may comprise at least one modified nucleotide and one or more point mutations in a regulatory sequence that restores or improves a function of the modified rRNA.

[0514] Any of the RNA replicon according to the invention, or the kit according to the invention, or the pharmaceutical composition according to the invention can be used in the method for producing a protein in a subject according to the invention. For example, in the method of the invention, RNA can be used in the format of a pharmaceutical composition, e.g. as described herein, or as naked RNA.

[0515] In view of the capacity to be administered to a subject, each of the RNA replicon according to the invention, or the kit according to the invention, or the pharmaceutical composition according to the invention, may be referred to as “medicament”, or the like. The present invention foresees that the RNA replicon, the kit, the pharmaceutical composition of the present invention is provided for use as a medicament. The medicament can be used to treat a subject. By “treat” is meant to administer a compound or composition or other entity as described herein to a subject. The term includes methods for treatment of the human or animal body by therapy.

[0516] The above described medicament does typically not comprise a DNA, and is thus associated with additional safety features compared to DNA vaccines described in the prior art (e.g. WO 2008/119827 A1).

[0517] An alternative medical use according to the present invention comprises a method for producing a protein in a cell according to the present invention, wherein the cell may be an antigen presenting cell such as a dendritic cell, followed by the introduction of said cell to a subject. For example, RNA encoding a pharmaceutically active protein, such as an antigen, may be introduced (transfected) into antigen-presenting cells ex vivo, e.g. antigen-presenting cells taken from a subject, and the antigen-presenting cells, optionally clonally propagated ex vivo, may be reintroduced into the same or a different subject. Transfected cells may be reintroduced into the subject using any means known in the art.

[0518] The medicament according to the present invention may be administered to a subject in need thereof. The medicament of the present invention can be used in prophylactic as well as in therapeutic methods of treatment of a subject.

[0519] The medicament according to the invention is administered in an effective amount. An “effective amount” concerns an amount that is sufficient, alone or together with other doses, to cause a reaction or a desired effect. In the case of treatment of a certain disease or a certain condition in a subject, the desired effect is the inhibition of disease progression. This includes the deceleration of disease progression, in particular the interruption of disease progression. The desired effect in the treatment of a disease or a condition can also be a delay of disease outbreak or the inhibition of disease outbreak.

[0520] The effective amount will depend on the condition being treated, the severity of the disease, the individual parameters of the patient, including age, physiological condition, size and weight, duration of the treatment, type of accompanying therapy (if any), the specific mode of administration and other factors.

Vaccination

[0521] The term “immunization” or “vaccination” generally refers to a process of treating a subject for therapeutic or prophylactic reasons. A treatment, particularly a prophylactic treatment, is or comprises preferably a treatment aiming to induce or enhance an immune response of a subject, e.g. against one or more antigens. If, according to the present invention, it is desired to induce or enhance an immune response by using rRNA as described herein, the immune response may be triggered or enhanced by the rRNA. In one embodiment, the invention provides a prophylactic treatment which is or comprises preferably the vaccination of a subject. An embodiment of the present invention wherein the replicon encodes, as a protein of interest, a pharmaceutically active peptide or protein which is an immunologically active compound or an antigen is particularly useful for vaccination.

[0522] RNA has been previously described for vaccination against foreign agents including pathogens or cancer (reviewed recently by Ulmer et al., 2012, Vaccine 30:4414-4418). In contrast to common approaches in the prior art, the replicon according to the present invention is a particularly suitable element for efficient vaccination because of the ability to be replicated by functional alphavirus non-structural protein as described herein. The vaccination according to the present invention can be used for example for induction of an immune response to weakly immunogenic proteins. In the case of the RNA vaccines according to the invention, the protein antigen is never exposed to serum antibodies, but is produced by transfected cells themselves after translation of the RNA. Therefore, anaphylaxis should not be a problem. The invention therefore permits the repeated immunization of a patient without risk of allergic reactions.

[0523] In methods involving vaccination according to the present invention, the medicament of the present invention is administered to a subject, in particular if treating a subject having a disease involving the antigen or at risk of falling ill with the disease involving the antigen is desired.

[0524] In methods involving vaccination according to the present invention, the protein of interest encoded by the replicon according to the present invention codes for example for a bacterial antigen, against which an immune response is to be directed, or for a viral antigen, against which an immune response is to be directed, or for a cancer antigen, against which an immune response is to be directed, or for an antigen of a unicellular organism, against which an immune response is to be directed. The efficacy of vaccination can be assessed by known standard methods such as by measurement of antigen-specific IgG antibodies from the organism. In methods involving allergen-specific immunotherapy according to the present invention, the protein of interest encoded by the replicon according to the present invention codes for an antigen relevant to an allergy. Allergen-specific immunotherapy (also known as hypo-sensitization) is defined as the administration of preferably increasing doses of an allergen vaccine to an organism with one or more allergies, in order to achieve a state in which the symptoms that are associated with a subsequent exposure to the causative allergen are alleviated. The efficacy of an allergen-specific immunotherapy can be assessed by known standard methods such as by measurement of allergen-specific IgG and IgE antibodies from the organism.

[0525] The medicament of the present invention can be administered to a subject, e.g. for treatment of the subject, including vaccination of the subject.

[0526] The term “subject” relates to vertebrates, particularly mammals. For example, mammals in the context of the present invention are humans, non-human primates, domesticated mammals such as dogs, cats, sheep, cattle, goats, pigs, horses etc., laboratory animals such as mice, rats, rabbits, guinea pigs, etc. as well as animals in captivity such as animals of zoos. The term “subject” also relates to non-mammalian vertebrates such as birds (particularly domesticated birds such as chicken, ducks, geese, turkeys) and to fish (particularly farmed fish, e.g., salmon or catfish). The term “animal” as used herein also includes humans.

[0527] The administration to domesticated animals such as dogs, cats, rabbits, guinea pigs, hamsters, sheep, cattle, goats, pigs, horses, chicken, ducks, geese, turkeys, or wild animals, e.g., foxes, is preferred in some embodiments. For example, a prophylactic vaccination according to the present invention may be suitable to vaccinate an animal population, e.g. in the farming industry, or a wild animal population. Other animal populations in captivity, such as pets, or animals of zoos, may be vaccinated.

[0528] In an embodiment, the medicament can be administered more than once. Multiple doses can be administered such that individual doses can be administered at different intervals. For example, a dose can be administered 14 to 35 days after the previous dose has been administered. In an embodiment, a dose is administered 21 days after the previous dose. In an embodiment, a dose is administered 35 days after the previous dose.

[0529] In an embodiment, when administered to a subject, the replicon used as a medication does not comprise sequences from a type of virus, e.g., alphavirus, that is infectious to the species or genus to which the treated subject belongs. Preferably, in that case, the replicon does not comprise any nucleotide sequence from an alphavirus that can infect the respective species or genus. This embodiment bears the advantage that no recombination with infectious (e.g. fully functional or wild-type) alphavirus is possible, even if the subject to which the RNA is administered is (e.g. accidentally) affected by infectious alphavirus. As an illustrative example, for treatment of pigs, the replicon used does not comprise any nucleotide sequence from an alphavirus that can infect pigs.

Mode of Administration

[0530] The medicament according to the present invention can be applied to a subject in any suitable route.

[0531] For example, the medicament may be administered systemically, for example intravenously (i.v.), intramuscularly (i.m.), subcutaneously (s.c.), intradermally (i.d.) or by inhalation.

[0532] In one embodiment, the medicament according to the present invention is administered to muscle tissue, such as skeletal muscle, or skin, e.g. subcutaneously. It is generally understood that transfer of RNA into the skin or muscles leads to high and sustained local expression, paralleled by a strong induction of humoral and cellular immune responses (Johansson et al., 2012, PLoS. One. 7:e29732; Geall et al., 2012, Proc. Natl. Acad. Sci. U.S.A 109:14604-14609).

[0533] Alternatives to administration to muscle tissue or skin include, but are not limited to: intradermal, intranasal, intraocular, intraperitoneal, intravenous, interstitial, buccal, transdermal, or sublingual administration. Intradermal and intramuscular administration are two preferred routes.

[0534] Administration can be achieved in various ways. In one embodiment, the medicament according to the present invention is administered by injection. In a preferred embodiment, injection is via a needle. Needle-free injection may be used as an alternative.

[0535] The present invention is described in detail and is illustrated by the figures and examples, which are used only for illustration purposes and are not meant to be limiting. Owing to the description and the examples, further embodiments which are likewise included in the invention are accessible to the skilled worker.

Description

DESCRIPTION OF THE FIGURES

[0536] FIG. 1: C57Bl/6 IFNAR^{-/-} mice were injected intramuscularly with saRNA coding for Ebola virus GP or NP formulated by polyplexes (PLX) or buffer control on days 0 and 21 or on day 0 and 35. Serum samples were collected before immunization (day -1) before the boost immunization (d20 or d34) and at the indicated times. 1A) Schematic overview of the experimental set-up for testing different prime-boost intervals. 1B) Average PLX size as well as size distribution was determined by dynamic light scattering characterization on a DynaPro PlateReader II, using dynamic light scattering (DLS) for calculating the hydrodynamic size of nanoparticles on a Wyatt device. LNP samples were diluted in PBS and measured in duplicates. Ten data points are recorded per well, each lasting 10 seconds. Average size (Z-average in nm) and polydispersity (polydispersity index, PDI) were analyzed with Dynamics v.7.8.1 (Wyatt Technology). 1C) Seroconversion of the groups vaccinated at a prime boost interval of 21 days, measured by ELISA. Recombinant Ebola virus GP-Biotin or NP-Biotin fusion-protein was coated onto Streptavidin-plates, incubated with 1:100 diluted sera and an HRP-coupled secondary antibody. Adsorption at 450 nm and 620 nm was measured and the ΔOD was calculated. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Top panel shows GP, bottom panel shows NP. 1D) Seroconversion of the groups vaccinated at a prime boost interval of 35 days, measured as explained in C. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Top panel shows GP, bottom panel shows NP.

[0537] FIG. 2:

[0538] C57Bl/6 IFNAR^{sup.-/-} mice were injected intramuscularly with saRNA Ebola virus vaccine candidates or buffer control on days 0 and 35. Serum samples were collected before immunization (day -1) and on days 20 and 34 after prime immunization and after boost immunization (d48 and d70) with saRNA coding for GP and NP formulated by polyplexes (PLX) or lipid nanoparticles (LNPs) or with GP alone together with an irrelevant “filler” RNA (GP+filler) formulated by LNPs. Splenocytes were isolated on day 48 or day 70 after prime immunization. 2A) Schematic overview of the experimental set-up for testing i.m. application of PLX and LNP. 2B) Size measurement of PLX was performed as described for FIG. 1B. Average LNP size as well as size distribution was determined by dynamic light scattering characterization on a DynaPro PlateReader II, using dynamic light scattering (DLS) for calculating the hydrodynamic size of nanoparticles on a Wyatt device. LNP samples were diluted in PBS and measured in duplicates. Ten data points are recorded per well, each lasting 10 seconds. Average size (Z-average in nm) and polydispersity (polydispersity index, PDI) were analyzed with Dynamics v.7.8.1 (Wyatt Technology). 2C) Seroconversion per group over time. ELISA was performed as described for FIG. 1C. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Dotted lines indicate the lower assay-mediated limit of detection (background level). Left panel shows GP, right panel shows NP. 2D) IgG concentration determination on day 70 using four-parameter logistic (4-PL) fit in GraphPad Prism has been performed against an IgG standard curve with known concentrations. Individual IgG concentrations are shown by dots; group mean values are indicated by lines ($\pm$ SEM). 2E) Neutralizing titers against authentic EBOV are shown by dots; group mean values are indicated by horizontal bars. Dotted line indicates the lower assay-mediated limit of detection. 2F) ELISpot assay was performed using splenocytes isolated on day 48 or day 70 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific GP- or NP-specific peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots (mean values of triplicate measurements have been calculated); group mean values are indicated by horizontal lines. Top panels show GP, bottom panels show NP.

[0539] FIG. 3:

[0540] C57Bl/6 IFNAR^{sup.-/-} mice were injected intramuscular or intradermal with saRNA Ebola virus vaccine candidates or buffer control on days 0 and 35. Combination of saRNA encoding GP and saRNA encoding NP was used in a GP:NP ratio of 2:1. The total RNA amount delivered to the intramuscular route was either 7.5 μ g (high dose) or 1.5 μ g (low dose). RNA amount delivered by the intradermal route was 7.5 μ g only (high dose). Serum samples were collected before immunization (day -1) and on days 21 and 34 after prime immunization and after boost immunization (d50). Splenocytes were isolated on day 50 after prime immunization. 3A) Schematic overview of the experimental set-up for testing i.m. versus i.d. application. 3B) Size measurement of nanoparticles was performed as described for FIG. 2B. 3C) Seroconversion per group over time. ELISA was performed as described for FIG. 1C. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Left panel shows GP, right panel shows NP. Asterisks indicate statistical significance as detailed by bars between relevant groups. * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p < 0.0001$. 3D) IgG concentration determination on day 50 was performed as described for FIG. 2D. Individual IgG concentrations are shown by dots; group mean values are indicated by lines ($\pm$ SEM). Asterisks indicate statistical significance as detailed by bars between relevant groups or compared to buffer control if no bars are marked. * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p < 0.0001$. 3E) Neutralizing titers against authentic EBOV are shown by dots; group mean values are indicated by horizontal lines. Dotted line indicates the lower assay-mediated limit of detection. 3F) ELISpot assay was performed using splenocytes isolated on day 50 after prime immunization. Method was performed as described for FIG. 2F. Group mean values are indicated by horizontal lines. Left panel shows GP, right panel shows NP.

[0541] FIG. 4:

[0542] C57Bl/6 IFNAR^{sup.-/-} mice were injected intramuscular with saRNA Ebola virus vaccine candidates on days 0 and 35. Combination of saRNA encoding GP and saRNA encoding NP was used in a GP:NP ratio of 2:1. The total RNA amount delivered to the intramuscular route was 7.5 μ g. Injections using GP or NP alone have been using equal RNA amount of single saRNAs compared to the combination and filled up with saRNA

encoding the replicase only (filler). saRNA encoding the replicase only was used in the full dose (empty). Serum samples were collected before immunization (day 0) and on days 21 and 34 after prime immunization and after boost immunization (d50) followed by the infection with EBOV on d56 and further serum sample collection on days 5 and 14 after EBOV challenge. Splenocytes were isolated on day 14 after EBOV challenge. **4A)** Schematic overview of the experimental set-up for challenge infection after prime boost immunization **4B)** Size measurement of nanoparticles was performed as described for FIG. 2B. **4C)** Seroconversion per group over time. ELISA was performed as described for FIG. 1C. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars lines ($\pm SEM$). Top panels show GP, bottom panels show NP. Asterisks indicate statistical significance as detailed by bars between relevant groups or compared to buffer control if no bars are marked. ns not significant; $**p \leq 0.01$; $****p < 0.0001$. **4D)** IgG concentration determination was performed as described for FIG. 2D. Individual IgG concentrations are shown by dots; group mean values are indicated by horizontal lines. **4E)** Neutralizing titers against authentic EBOV are shown by dots; group mean values are indicated by unfilled bars ($\pm SEM$). Dotted line indicates the lower assay-mediated limit of detection. Asterisks indicate statistical significance compared to buffer control. $*p \leq 0.05$. **4F)** Body weight curves for the different groups after challenge infection (left graph). Mice that developed severe clinical signs of infection (middle graph) and/or exceeded 15% of body weight loss were euthanized. Survival over time is shown in the graph on the right. **4G)** Infectious EBOV in blood. Serum samples were taken on day 5 and on day 14 after infection and the amount of infectious virus was determined by plaque titration. Group mean values are indicated by unfilled bars ($\pm SEM$). Dotted lines indicate the lower assay-mediated limit of detection. **4H)** EBOV RNA in organs. Detection of viral genome copies by EBOV GP-specific qRT-PCR in liver and spleen samples of the mice obtained at day 14 post infection. Group mean values are indicated by horizontal lines. Asterisks indicate statistical significance as detailed by bars between relevant groups. $****p < 0.0001$.

[0543] FIG. 5:

[0544] C57Bl/6 IFNAR.sup.-/- mice were injected intramuscular with saRNA Ebola virus vaccine candidates or buffer control on day 0 only. Serum samples were collected before immunization (day -1) and on day 15 after prime immunization. Challenge infection was performed on day 21. Blood and organs were isolated on day 14 after EBOV challenge. **5A)** Schematic overview of the experimental set-up for challenge infection after prime only immunization. **5B)** Size measurement of nanoparticles was performed as described for FIG. 2B. **5C)** Seroconversion per group over time. ELISA was performed as described for FIG. 1C. Individual ΔOD values are shown by dots, group mean values are indicated by horizontal bars ($\pm SEM$). Left panel shows GP, right panel shows NP. **5D)** Neutralizing titers against authentic EBOV are shown by dots; group mean values are indicated by horizontal lines. Dotted lines indicate the lower assay-mediated limit of detection. **5E)** Body weight curves for the different groups after challenge infection (left graph). Mice that developed severe clinical signs of infection and/or exceeded 15% of body weight loss were euthanized. Survival over time is shown in the graph on the right. **5F)** Infectious EBOV in blood. Serum samples were taken on day 5 and on day 14 after infection and the amount of infectious virus was determined by plaque titration. Group mean values are indicated by unfilled bars ($\pm SEM$). Dotted lines indicate the lower assay-mediated limit of detection. **5G)** EBOV RNA in organs. Detection of viral genome copies by EBOV GP-specific qRT-PCR in liver and spleen samples of the mice obtained at day 14 post infection. Group mean values are indicated by horizontal lines; dotted lines indicate the lower assay-mediated limit of detection. Asterisks indicate statistical significance as detailed by bars between relevant groups. $****p < 0.0001$.

[0545] FIG. 6:

[0546] Immune response induced in C57Bl/6 IFNAR.sup.-/- mice after prime-boost vaccination of LNP-formulated saRNA combinations encoding CCHFV proteins. C57Bl/6 IFNAR.sup.-/- mice were injected intramuscular with saRNA vaccine candidates (CCHFV GP and CCHFV NP) or buffer control (PBS) on days 0 and 28. Serum samples were collected directly before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (d49). Splenocytes were isolated on day 49 after prime immunization. **6A)** Schematic overview of the experimental setting for testing prime/boost vaccination. **6B)** Seroconversion per group over time. Recombinant CCHFV Gc- or NP-protein was coated onto maxisorp-plates, incubated with 1:100 diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the ΔOD was calculated. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Top panels show GC, bottom panels show NP. Asterisks indicate statistical significance compared to buffer control. $**p \leq 0.01$, $****p < 0.0001$. **6C)** ELISpot assay was performed using splenocytes isolated on day 49 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific CCHFV Gc- or NP-peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm SEM$). Top panels show GC, bottom panels show NP. Asterisks indicate statistical significance compared to buffer control. ns: not significant; $* \leq 0.05$; $**p \leq 0.01$; $*** \leq 0.001$; $****p < 0.0001$.

[0547] FIG. 7:

[0548] Immune response induced in C57Bl/6 IFNAR.sup.-/- mice after prime-boost vaccination of LNP-formulated saRNA combinations encoding CCHFV proteins. C57Bl/6 IFNAR.sup.-/- mice were injected intramuscular with saRNA vaccine candidates (CCHFV GP and CCHFV NP) or buffer control (PBS) on days 0 and 28. Serum samples were collected directly before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (d54). Splenocytes were isolated on day 54 after prime immunization. **7A)** Schematic overview of the experimental setting for testing different Gc+TM:NP ratios. **7B)** Seroconversion per group over time. ELISA was performed as described for FIG. 2B. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. Top panels show GC, bottom panels show NP. Asterisks indicate statistical significance compared to buffer control. $*p \leq 0.05$; $**p \leq 0.01$, $***p \leq 0.001$; $****p < 0.0001$. **7C)** ELISpot assay was performed using splenocytes isolated on day 54 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific CCHFV Gc- or NP-peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm SEM$). Top panels show Gc+TM, bottom panels show NP.

[0549] FIG. 8:

[0550] CCHFV GP and CCHFV NP encoding transreplicons (TR) and replicase (rep) mRNA were separately formulated within LNPs and mixed at indicated molar ratios prior to application. BALB/c mice (n=5 per group) were intramuscularly injected with formulated RNAs on days 0 and 28. Replication-deficient replicase (def. rep) with both TR served as negative control. Serum samples were collected before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (day 49). **8A)** Seroconversion per group over time. Recombinant CCHFV Gc- or NP-protein was coated onto maxisorp-plates, incubated with 1:100 diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the ΔOD was calculated. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. **8B)** ELISpot assay was performed using splenocytes isolated on day 49 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific CCHFV Gc- or NP-peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm SEM$). Asterisks indicate statistical significance compared to single immunization with only one TR. $***p < 0.001$, $*p < 0.05$.

[0551] FIG. 9:

[0552] MERS-CoV S and MERS-CoV NP encoding transreplicons (TR) and replicase (rep) mRNA were separately formulated within LNPs and mixed at indicated molar ratios prior to application. BALB/c mice (n=5 per group) were intramuscularly injected with formulated RNAs or buffer control on days 0 and 28. Serum samples were collected before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (day 49). **9A)** seroconversion per group over time. Recombinant MERS-CoV S1- or NP-protein was coated onto maxisorp-plates, incubated with diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the ΔOD was calculated. Individual ΔOD values are shown by dots; group mean values are indicated by horizontal bars. **9B)** ELISpot assay was performed using splenocytes isolated on day 49 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific MERS-CoV S- or NP-

peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm$ SEM).

[0553] FIG. 10:

[0554] EBOV GP and EBOV NP encoding transreplicons (TR) and replicase (rep) mRNA were separately formulated within LNPs and mixed at indicated molar ratios prior to application. BALB/c mice (n=5 per group) were intramuscularly injected with formulated RNAs on days 0 and 28. Replication-deficient replicase (def. rep) with both TR served as negative control. Serum samples were collected before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (day 49). Seroconversion per group over time. Recombinant Ebola virus GP-Biotin or NP-Biotin fusion-protein was coated onto Streptavidin-plates, incubated with diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the Δ OD was calculated. Individual Δ OD values are shown by dots, group mean values are indicated by horizontal bars.

[0555] FIG. 11

[0556] Transreplicons (TR) encoding for CCHFV GP, CCHFV NP, MERS-CoV S and MERS-CoV NP and replicase (rep) mRNA were separately formulated within LNPs and mixed at indicated molar ratios prior to application. BALB/c mice (n=5 per group) were intramuscularly injected with formulated RNAs or buffer control on days 0 and 28. Serum samples were collected before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (day 49). 11A) seroconversion per group over time. Recombinant CCHFV Gc- or NP-protein or recombinant MERS-CoV S1- or NP-protein was coated onto maxisorp-plates, incubated with diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the Δ OD was calculated. Individual Δ OD values are shown by dots; group mean values are indicated by horizontal bars. 11B) ELISpot assay was performed using splenocytes isolated on day 49 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific CCHFV Gc- or NP- or MERS-CoV S- or NP-peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm$ SEM). Asterisks indicate statistical significance compared to immunization with MERS-CoV S and NP TR and replicase mRNA. ****p<0.0001, ***p<0.001, **p<0.01, *p<0.05.

[0557] FIG. 12

[0558] Transreplicons (TR) encoding for CCHFV GP, CCHFV NP, MERS-CoV S and MERS-CoV NP and replicase (rep) mRNA were separately formulated within LNPs and mixed at indicated molar ratios prior to application. B6.129S2-Ifnar1tm1Agt/Mmjax mice (n=5 per group) were intramuscularly injected with formulated RNAs on days 0 and 28. Replication-deficient replicase (def. rep) with CCHFV GP and CCHFV NP TRs served as negative control. Serum samples were collected before immunization (day 0) and on days 14 and 28 after prime immunization and after boost immunization (day 49). A) Seroconversion per group over time. Recombinant CCHFV Gc- or NP-protein or recombinant MERS-CoV S1- or NP-protein was coated onto maxisorp-plates, incubated with diluted sera and an HRP-coupled secondary antibody. Adsorption at 460 nm and 620 nm was measured and the Δ OD was calculated. Individual Δ OD values are shown by dots; group mean values are indicated by horizontal bars. B) ELISpot assay was performed using splenocytes isolated on day 49 after prime immunization. Splenocytes were stimulated with MHC I and MHC II-specific CCHFV Gc- or NP-peptide pools and IFN- γ secretion was measured to assess T-cell responses. Individual spot counts are shown by dots; group mean values are indicated by unfilled bars ($\pm$ SEM).

EXAMPLES

[0559] In the following study, mice (C57BL/6 IFNAR.sup.-/-) have been immunized with saRNAs encoding EBOV GP or NP or with a combination of both. In addition another viral target, namely crimean congo hemorrhagic fever virus (CCHFV) has been used for the further evaluation of the combination of saRNAs. In detail the c-terminal part of the CCHFV GP in addition to the transmembrane domain (TM) was used together with the NP of CCHFV. For immunization, the intramuscular (i.m.) route is generally preferred and was investigated predominantly. Nevertheless, intradermal (i.d.) application of the vaccine was tested in direct comparison. At different time points after immunization, serum samples have been analyzed to provide information about antibody levels and for their neutralizing activity (only available for the EBOV studies) in virus neutralization assays using life Ebola virus. In addition, T cell immune responses have been analyzed using IFN γ ELISpot assays. The prime-boost regimen was compared to prime-only immunization in challenge infection studies, and the serological responses and the outcome of the infection were compared.

[0560] Besides saRNA, GP and NP of different viruses (EBOV, CCHFV, MERS-CoV) were also combined using the trans-replication system, where the alphaviral replicase is encoded on a non-replicable mRNA and the viral antigens like GP and NP are encoded on replicable trans-replicons. In the following studies, mice (BALB/c) have been immunized i.m. with LNP-formulated mRNA encoding the replicase of VEEV and trans-replicons (TR) encoding GP or NP of EBOV, CCHFV or MERS-CoV or a combination of up to four TR. At different time points after immunization, serum samples have been analyzed to provide information about antibody levels. In addition, T cell immune responses have been analyzed using IFN γ ELISpot assays.

Materials and Methods

Constructs:

[0561] For the Ebola-specific experiments described in FIGS. 1 to 5 the following constructs have been used: saRNA encoding the Semliki forest virus replicase followed by the SFV-derived subgenomic promotor and the antigens of interest, namely Ebola virus Zaire GP and NP.

[0562] For the Crimean Congo hemorrhagic fever virus specific experiments described in FIGS. 6 to 7 the following constructs have been used: saRNA encoding the Venezuelan equine encephalitis virus (VEEV) derived replicase followed by the VEEV-derived subgenomic promotor and the antigens of interest, namely CCHFV Gc+TM and NP.

[0563] For the transamplifying RNA experiments using CCHFV, MERS and EBOV antigens VEEV derived constructs have been used.

[0564] The sequences of all constructs are depicted in TABLE 1 and the sequences of the antigens are depicted in Table 2.

TABLE-US-00004 TABLE 1 DESCRIPTION OF THE VECTOR SEQUENCES SEQ ID NO: Description SEQUENCE Subgenomic promotor (-19/+5; with respect to the start position of subgenomic RNA) 73 24 nt CTCTCTACGGCTAACCTGAATGGA

5' UTR pST1-OPAL + 51 nt-CSE 74 187 nt

ATGGGGGCGCATGAGAGAAGCCAGACCAATTACCTACCCAAAATGGAGAAAGTTCACGTTGACATCGAGGAAGACAGCCATTCTCAGAC

TTTGCAGCGGAGCTTCCCGCAGTTTGAGGTAGAAGCCAAGCAGGTCACTGATAATGACCATGCTAATGCCAGAGCGTTTTCGCATCTGGCTTC

5' UTR pVEE + 51 nt-CSE 75 187 nt

ATGGGCGGCGCATGAGAGAAGCCAGACCAATTACCTACCCAAAATGGAGAAAGTTCACGTTGACATCGAGGAAGACAGCCATTCTCAGAC

TTTGCAGCGGAGCTTCCCGCAGTTTGAGGTAGAAGCCAAGCAGGTCACTGATAATGACCATGCTAATGCCAGAGCGTTTTCGCATCTGGCTTC

3' UTR pST1-OPAL 76 63 nt GGAGCTTAATTCGACGAATAATTGGATTTTATTTTATTTTGAATTGGTTTTTAATATTTCC 3'

UTR pVEE 77 121 nt

TGAATACAGCAGCAATTGGCAAGCTTGCTTACATAGAAGCTCGCGGCGATTGGCATGCCGCCTTAAAATTTTATTTTATTTTCTTTCTTTTCCG

AATCGGATTTTGTGTTTTAATATTTTC polyA tail (A30LinkerA70) 78 110 nt

AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAGCATATGACTAAA

AAAAAAAAAAAAAAAA pST1-OPAL-ZEBOV-GP-A82V-A30LA70 79 nucleotide

AUGGCGAUGUGUGACAUACACGACGCCAAAAGAUUUUGUCCAGCUCCUGCCACCUCGCUACGCGAGAGAUUAACCACCCACGAUGG sequence

CAAAGUGCAUGUUGAUUUGAGGCUGACAGCCAUUCAUCAAGUCUUUGCAGAAGGCAUUUCCGUCGUUCGAGGUGGAGUCAUUGCAGG

CACCAAAUGACCAUGCAAAUGCCAGAGCAUUUUCGACCCUGGCUACCAAAUUGAUCGAGCAGGAGACUGACAAAGACACACUCAUCUUG

CGGAGUGCGCCUUCAGGAGGAGAAUGAUGUCUACGCAACAAUACACUACUGCGUAUUGCCUUAUGCGCAGCGCAGAGAAGACCCCGAAAGGCCUUCG
CUACGCCAAAGAAACUGGCAGCGGCCUCCGGGAAGGUGUCUGGAUAGAGAGAUUCGCAGGAAAAAUACCCGACCUGCAGACCCGUCAUUGGCUAU
AGACGCGUAAUUCUCUACCUUUUGCCUGCAUACAGACGUCACGUGUCGUACGGCAGCCGAAGUGGCCGUAUACCAGGACGUGUAUUGCUGU
UGCACCAACUACGCGUACCAUCACAGGCGAUGAAAGGUGUCAGAACGGCGUAUUGGAUUGGGUUUGACACCACCCCGUUUAUGUUUUGACG
UAGCAGGCGCGUAUCCAACUACGCCACAACACUGGGCCGACGAGCGAGGUAUACAGGCCAGGAACAUAGGACUGUGCAGUAUCCUUA
AGGGAAGACUCGGCAAAACUGUCCAUUCUCCGCAAGAAGCAAUUGAAACCUUUGCGACACAGUCAUGUUCUCGGUAGGUAUCUAUUGUAC
AGAGCAGAAAGCUACUGAGGAGCUGGCACUUACCCUCCGUAUUCCACCUGAAAGGUAACAACAUCCUUUACCUGUAGGUGCGAUACCAUCU
CAUGUGAAGGGUACGUAGUUAAGAAAAUCACUAUGUGCCCCGGCCUGUACGGUAAAACGGUAGGGUACGCCGUGACGUUACACGCGGAG
UUCUAGUGUGCAAGACCACAGACACUGUCAAAAGGAGAAAGAGUCUCAUUCCCUGUAUGCACCUACGUCCCCUCAACCAUCUGUGAUCAA
CUGGCAUACUAGCGACCGACGUCACACCGGAGGACGCACAGAAGUUGUUAGUGGGAUUGAAUCAGAGGAUAGUUGUGAACGGAAGAACA
GAAACACUAACACGAUGAAGAACUAUCUGCUUCCGAUUGUGGCCGUCGCAUUUAGCAAGUGGGCGAGGGAAUACAAGGCAGACCUUGAU
AAAAACCUCUGGGUGUCCGAGAGAGGUCACUUAUUGCUGCUGCUGUUGGGCAUUUAAAACGAGGAAGAUGCACACCAUGUACAAGA
ACACCCAGCAAAUGAGAAGGUGCCUUCAGAGUUUAACUCUGUUCGUAUCCCGACCCUAGUGGUCUACAGGCCUGCAGUCCCGACAGCAG
GCAUUAGAUGCUUUUGGCCAAGAAGACCAAGCGAGAGUUAUACCUGUUCUGCAGCGCGUCGACGCCAGGGAUGUGAACAAAGAGGAG
AGAGGUUGGAGGCCGAGCUGACUAGAGAAGCCUUAACACCCUCGUCCCCAUCGCGCCGGCGGAGACGGGAGUCGUCGACGUCGACGUU
AACUAGAGUAUACACGCAGGUGCAGGGGUCGUGGAAACACCUCGCAGCGCGUUGAAAGUCACCCGCACAGCCGAACGACGUACUACUAGGA
ACGUAGUUCUGUCCCCGCAGACCGUGCUCAAGAGCUCCAAGUUGGCCCCCGUGCACCUCUAGCAGAGCAGGUGAAAAUAAUACACAU
GGAGGGCCGGCCGUUACCAGGUCGACGGAUUGACGGCAGGGUCCUACUACCAUGUGGAUCGGCCAUUCCGGUCCUGAGUUUCAGGC
AGCGAGAGCGCCACUAUGGUGUACAACGAAAGGGAGUUCGUCAACAGGAAACUAUACCAUAUUGCCGUUCACGGACCCUCGUGAACAC
GAGGAGAACUACGAGAAAGUCAGAGCUGAAGAACUGACGCCGAGUACGUGUCCGACGUAGAUAAAAAUUGCUGCGUAAGAGAGAGGA
UCGGGUUUGGUGUUGGUGGAGCUAACCAACCCCGUUCUUGCAUUAUUGCCUACGAAAGGCGUGAAGAUCAGCGCCGUGCCGACCAU
ACUACAGUAGUAGGAGUCUUUGGGUUCGGGAUCAGGCUAGUCUGCUUAUUAAUAGAGCCUCUGACCAACACGAUCUGGUCACCGA
CGGAGCAAAGUGGUGUUUUGCGGAGACCCCAAGCAAUGCGGAUUCUUAUUAUGAUGCAGCUUAAAGGUGAACUUAACCACAACUUG
GAAGUAUGUCAUAAAAGUAUAUCCAGACGUUGCACGCGUCCAGUCACGGCCAUCGUGUCUACGUUGCACUACGGAGGCAAGAUGCAC
AACCCGUGCAACAACCCCAUAAUCAUAGACACCACAGGACAGACCAAGCCCAAGCCAGGAGACAUCGUGUUAACAUGCUCUCCGAGGCUG
AGCAGCUGCAGUUGGACUACCGUGGACACGAAGUCAUGACAGCAGCAGCAUCUCAGGGCCUCACCCGCAAAGGGGUUAUACGCCGUAAGG
AGGUGAAUGAAAAUCCCUUGUAUGCCCCUGCGUCGGAGCAGUGAAUGUACUGCUGACGCGCACUGAGGAUAGGCUGGUGUGGAAAACG
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A82V-A30LA70 87 Nucleotide

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KGKNRLLDGERLADRKIAELFSTKTHIG 94 CCHIFV GC + TM
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CCHFV NP
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Middle East Respiratory Syndrome-related coronavirus (MERS-COV) Spike protein 96 CCHFV NP
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(amino acid

sequence)

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RNA Synthesis by *in Vitro* Transcription

[0565] T7 in vitro transcription was based on protocols provided by MEGAscript® T7 Transcription Kit (Thermo Fisher, formerly Ambion). The general procedure starting with linear DNA template containing the T7 promoter, and particularly with respect to co-transcriptional capping with the synthetic cap analogue beta-S-ARCA(D1) (used in 4:1 ratio regarding GTP), was carried out similarly to as described before (Kuhn et al., 2010, Gene Ther. 17:961-71). High-yielding processes have been modified and optimized with respect to long saRNA with up to 10000 nucleotides (Pokrovskaya & Gurevich, 1994, Anal. Biochem. 220:420-423).

Preparation of Polyplexes (PLX)

[0566] Linear polyethylenimine of 20 to 25 kDa molecular weight was used (in vivo-jetPEI). For the calculation of the N/P ratio of the polyplex formulations the positive charges of nitrogen atoms of amines (N) in PEI and anionic charges (phosphates) of the RNA (P) are taken into consideration. The formulation consists of self-amplifying RNA (saRNA), in vivo-jetPEI and formulation buffer, which consist of 10 mM MES and 5% w/v D-Glucose at pH 6.1. Polyplexes were formulated for a final RNA concentration of 0.1 mg/ml and N/P of 12. The N/P ratio here was calculated as the bulk stoichiometric ratio between the total amount of amines, which are introduced with the in vivo-jetPEI, and phosphates from the saRNA in the bulk solution. Shortly, the method for manufacturing the polyplexes is based on an equivoluminar mixing of an in vivo-jetPEI containing solution and a saRNA containing solution. The saRNA solution is prepared by mixing concentrated formulation Buffer (2×) and saRNA from stock solution. This saRNA solution contains exactly double the concentration required at the final conditions and upon equivoluminar mixing with the PEI solution, the saRNA concentration will be diluted by half, i.e., to the final saRNA concentration. In case it is required, the final concentration of formulation buffer was adjusted through the PEI solution. On a similar fashion, the PEI Solution was prepared with Bio-Grade Sterile ddH.sub.2O and the required volume of in vivo-jetPEI. The concentration of in vivo-jetPEI corresponds to the double amine concentration that is required for the final N/P, i.e., by equivoluminar mixing will be reduced by half and therefore to the final required concentration. The equivoluminar mixing was performed by aspiration of the required volume from the PEI solution and vigorous injection into the saRNA solution. The mixing requires immediate vortexing, so that the injection of PEI solution into the vial containing saRNA solution takes places over the vortexer. After mixing, the formulation is incubated for 15 min at RT and final quality control takes places.

LNP Production

[U567] Lipid nanoparticles (LNPs) were manufactured by controlled mixing of drug substance dissolved in aqueous buffer with ethanolic solution of lipids using a NanoAssemblr (Precision Nanosystems). The resulting aqueous-organic dispersion of LNPs was subjected to dialysis for removal of ethanol. Composition of LNP-C12 for i.m. application (DODMA:Chol:DOPE:PEGcerC16; 40:48:10:2; N/P ratio 4) and composition of LNP-C09 for i.d. application (DODMA:Chol:DSPC:PEGcerC16; 40:48:10:2; N/P ratio 2.67) have been used.

Size and PDI Measurement

[0568] Average LNP and PLX sizes, as well as size distribution, was determined by dynamic light scattering characterization on a DynaPro PlateReader II, using dynamic light scattering (DLS) for calculating the hydrodynamic size of nanoparticles on a Wyatt device. LNP samples were diluted in PBS and measured in duplicates. Ten data points are recorded per well, each lasting 10 seconds. Average size (Z average in nm) and polydispersity (polydispersity index, PDI) were analyzed with Dynamics v.7.8.1 (Wyatt Technology).

Animal Care

[0569] Mice were either delivered at the age of at least six weeks or bred at BioNTech SE's animal facility. Delivered mice were used for experiments after approximately one week of acclimatization. All experiments and protocols were approved by the local authorities (local welfare committee), conducted according to the FELASA recommendations and in compliance with the German animal welfare act and Directive 2010/63/EU. Only animals with an unobjectionable health status were selected for testing procedures and housed under SPF conditions in individually ventilated cages (Sealsafe GM500 IVC Green Line, TECNIPLAST, Hohenpeißenberg, Germany; 500 cm.sup.2) with a maximum of five

animals per cage. The temperature and relative humidity in the cages and animal unit was kept at 20 to 24° C. and 45 to 55%, respectively, and the air change (AC) rate in the cages at 75 AC/hour. The cages with dust-free bedding made of debarked chopped aspen wood (Abedd LAB & VET Service GmbH, Vienna, Austria, product code: LTE E-001) and additional nesting material were changed weekly. Autoclaved sniff M-Z food (sniff Spezialdiäten GmbH, Soest, Germany; product code: V1124) and autoclaved water (tap water) were provided ad libitum and changed at least once weekly. All materials were autoclaved prior to use.

[0570] C57Bl/6 IFNAR^{-/-} mice for immunization with consecutive EBOV challenge infection were bred at the animal facility of the Philipps University Marburg under SPF conditions according to FELASA recommendations. All experiments and protocols were approved by the local authorities (Regierungspräsidium Gießen AZ V54-19 c 20 15 h 01 MR 20/7 Nr. G 47/2018) and performed according to the German animal welfare act and Directive 2010/63/EU. The mouse stem was chosen since wildtype mice are not susceptible to non-adapted EBOV infection. For challenge experiments, mice were kept in groups of max. 5 in negative pressure isocages (IsoCage N, Tecniplast).

Virus

[0571] EBOV (Ebola virus strain Zaire, Mayinga, GenBank: NC_002549) was used for challenge infections of C57Bl/6 IFNAR^{sup.-/-} mice. All experiments with EBOV were carried out under highest safety containment according to national and international regulations in the BSL-4 laboratory of the Philipps University of Marburg, Hans-Meerwein-Str. 2, 35043 Marburg, Germany.

Infection of IFNAR^{SUP.-/-} Knock-Out Mice

[0572] At the BSL4 animal facility, groups of C57Bl/6 IFNAR^{sup.-/-} mice (n=5-6) were infected via the intranasal (i.n.) route according to an adapted protocol (Oestereich et al., 2014, Antiviral Res. 105:17-21 on day 21 (prime-only) or on day 56 (prime-boost) after the first vaccination with 30 µl of DMEM containing 1000 plaque forming units (PRJ) of EBOV under short isoflurane anesthesia. The mice were monitored daily for weight loss and clinical scoring, comprising spontaneous behavior and general condition. Blood samples to determine viral load were taken 5 days post infection under short anesthesia at the facial vein. All surviving animals were euthanized at day 14 and final serum samples were collected. Mice were euthanized by cervical dislocation under isoflurane anesthesia when a clinical score of 10 (e.g., weight loss>15%) or 6 on two consecutive days was reached.

Body Weight

[0573] The body weights of the animals were recorded once a week. During the period of challenge infections, the body weights of all animals were recorded daily along with observation of clinical scores.

Endpoint of Experiments/Termination Criteria

[0574] Animals were euthanized in accordance with § 4 of the German animal welfare act and the recommendation of GV-SOLAS by cervical dislocation or by exposure to carbon dioxide. The experiment was terminated after an observation period of 56 days. Additionally, termination criteria applied according to the recommendation of GV-SOLAS as listed below. Body weight losses exceeding 20%, or a high severity level in any of the other categories were on their own sufficient reason for immediate euthanasia.

[0575] Intra-muscular and intra-dermal injections: Mice were anesthetized by inhalation anaesthesia (isoflurane 2.5%) (Abbott, Ludwigshafen, Germany). Subsequently, Mice were immunized with vaccine candidates using a prime-boost vaccination strategy. Animals received vaccine candidates on study days 0 and 35 at a dose volume of 20 µL as intramuscular injections to the tibialis posterior or at a dose volume of 20 µL into the skin of the back.

Blood Sampling

[0576] Blood samples for IgG ELISA were collected from the retro-orbital sinus. 50 µL of blood were collected in heparin-coated serum tubes (BD Microtainer) from all animals on relevant study days. In addition, blood was collected from 10% of the animals on day 0 before the first immunization.

EBOV GP- or NP-Specific IgG ELISA

[0577] GP-specific and NP-specific IgGs were detected in serum samples using ELISA. Recombinant proteins from Ebola virus Zaire (strain *H. sapiens*-wt/GIN/2014/Kissidougou-15) produced in *E. coli* or Baculovirus insect cells have been used. Recombinant GP Protein (Acc. No.: AHX24649.1; Met1-Gln650; 69.3 kDa; Cat No.: 40442-V08B1; Sino Biological via LSZ Life Sciences) or recombinant Ebola virus Zaire NP Protein (Acc. No.: AHX24646.1; His630-Gln739; 15.6 kDa; Cat No. 40443-V07E1; Sino Biological via LSZ Life Sciences) was biotinylated using the EZ-Unk Sulfo-NHS-LC-Biotinylation Kit from Thermo Scientific (Cat No.: 28005) in accordance with manufacturer's instructions to enable them to bind with high affinity to streptavidin-precoated 96 well plates (Cat No.: 734-1284; Nunc). Successful biotinylation of the recombinant proteins was assessed using a HABA/Avidin assay (Biotin Quantitation Kit Thermo Scientific) directly after biotinylation of the protein stock. Streptavidin-precoated plates have been incubated overnight at 4° C. with 100 ng/100 µl (1 µg/ml) biotinylated recombinant protein or a mouse IgG isotype with known concentration (Mouse IgG-BIOT; Conc.: 0.5 mg/ml; Cat. No.: 0107-08; Southern Biotech) in serial dilution from 1:100 to 1:3200. In addition, positive and negative controls have been included, that have been likewise coated with biotinylated recombinant protein, but incubated with specific antibodies for EBOV GP a Human anti-EBOV GP mAb from IBT BIOSERVICES (done KZ52; Cat. No.: 0260-001; 1:1000 diluted) together with a Goat Anti-Human-IgG-HRP (Cat. No.: 109-035-098; Jackson ImmunoResearch; 1:5000 diluted) or with a rabbit anti-EBOV NP pAb (Cat. No.: 0301-012; IBT BIOSERVICES; 1:100 diluted) and a Goat Anti-rabbit-IgG-HRP (Cat No.: A0545; Sigma-Aldrich; 1:10000 diluted) for EBOV NP. After washing and blocking of unspecific binding sites (blocking buffer from Sigma-Aldrich, Cat No.: B6429), serum samples from immunized mice have been incubated with coated wells for 1 h at 37° C. on a shaker. Bound antibodies from the serum samples were detected using horseradish peroxidase (HRP) conjugated secondary antibodies (Goat anti-mouse IgG (POX); Cat. No.: 115-035-071; Jackson ImmunoResearch; 1:15000) and enzymatic reaction for 8 min at RT using TMB one substrate (Cat No.: 4380; Kem-En-Tec). Reaction was stopped using sulfuric acid (Cat. No.: 1.007.161.000; Merck) and extensive washing with H.sub.2O. Quantification of results was performed using an Epoch plate reader and measurement at 450 nm-620 nm. IgG concentration was determined using four-parameter logistic (4-PL) fit in GraphPad Prism against the included IgG standard curve with known concentrations.

CCHFV Gc+TM/NP-Specific Whole IgG ELISA

[0578] Gc+TM/NP-specific IgGs in serum samples were detected using ELISA. Maxisorp plates were coated with recombinant Crimean congo hemorrhagic fever virus (CCHFV) major glycoprotein Gc (NCBI accession number NP_950235.1, amino acids 1041-1586; produced in HEK293 cells, and purified from culture supernatant; Cat. No.: REC31696-100; TheNativeAntigenCompany, 100 kDa) or recombinant CCHFV nucleoprotein NP (CCHFV strain IbAr10200 (Nigeria, 1996), produced in HEK293 cells, and purified from culture supernatant; Cat. No.: REC31639-100; TheNativeAntigenCompany, 56 kDa) and bound serum antibodies were detected using horseradish peroxidase (HRP)-conjugated secondary antibodies and enzymatic reaction for 8 min at RT using TMB one substrate. Reaction was stopped using sulfuric acid and extensive washing with H.sub.2O. Quantification of results was performed using an Epoch plate reader and measurement at 450 nm.

MERS-CoV S1/NP-Specific Whole IgG ELISA

[0579] S1/NP-specific IgGs in serum samples were detected using ELISA. Maxisorp plates were coated with recombinant Middle East Respiratory Syndrome-related coronavirus (MERS-CoV) S1 protein (NCBI accession number AFS88936.1, amino acids 1-725, produced in HEK293 cells, Cat. No. 40069-V08H, Sino Biological, 94 kDa) or recombinant MERS-CoV NP protein (NCBI accession number AFS88943.1, amino acids 1-413, produced in Baculovirus-Insect cells, Cat. No. V0068-V08B, Sino Biological, 47 kDa) and bound serum antibodies were detected using horseradish peroxidase (HRP)-conjugated secondary antibodies and enzymatic reaction for 8 min at RT using TMB one substrate. Reaction was stopped using sulfuric acid and extensive washing with H.sub.2O. Quantification of results was performed using an Epoch plate reader and measurement at 450 nm.

EBOV Neutralization Assay

[0580] EBOV neutralization assay was performed as described by Erhardt et al., 2019, Nature Medicine 25:1589-1600. Briefly, mouse sera were serially diluted and incubated with 100 TCID₅₀ units of EBOV Mayinga (GenBank NC_002549). Following incubation at 37° C. for 1 h, Vero C1008 cells (ATCC CRL-1586) were added. Cytopathic effects were evaluated at day 7 post infection. Neutralization was defined as absence of CPE in serum dilutions. Neutralization titers of four replicates were calculated as geometric mean titers for sera (reciprocal value). The cut-off of the assay is determined by the first dilution of the respective serum.

Virus Titration by Plaque Assay

[0581] Vero C1008 cells were cultured to 100% confluence and infected with 10-fold serial dilutions of mouse sera starting at a dilution of 1:20 or 1:100. After 1 hour the inoculum was replaced by an overlay consisting of 2% carboxymethylcellulose (Sigma-Aldrich, C-5678) in 1× Minimum Essential Medium (Thermo Fisher Scientific, 51200-046) supplemented with 2% FCS, P/S and Q. At day 5 post infection (p.i.) cells were fixed with 4% paraformaldehyde (PFA) for two days. Cells were washed three times with PBS and permeabilized with PBS containing 0.1% Triton X-100 for 10 min. After washing three times with PBS cells were incubated with 100 mM glycine in PBS for 10 min. After a wash with PBS, the cells were incubated in blocking solution (BS, 2% bovine serum albumin, 0.2% Tween 20, 5% glycerol in PBS). The virus-induced plaques were stained with an EBOV-specific goat serum (dilution 1:200) and an AlexaFluor@488 secondary antibody from rabbit (Thermo Fisher Scientific, Cat. No. A27012; dilution 1:500). Plaques were counted using an Axiomat fluorescence microscope (Zeiss) and pfu/ml were calculated.

T-Cell Epitope Prediction

[0582] The respective peptides for stimulation of splenocytes (Table 3) were selected based on a prediction of immunodominant peptides via database research (IEDB (Immune epitope database and analysis resource)). For all predictions, protein sequences have been used that were also chosen for the production of rRNAs for the different antigens. In general, epitope prediction is based on, e.g., amphipathicity profile and recognized sequence motifs. The prediction method utilized by IEDB uses input protein amino acid sequences to identify binding cores, binding affinities and residues flanking peptides based on large scale systematic evaluation. Prediction is performed specifically for the major histocompatibility complex (MHC) alleles used by the particular mouse strain (mouse haplotype table; Affymetrix eBioscience). The generated output file included percentile ranks of all listed peptides. Low percentile ranks indicate good binders, so that peptides spanning the molecule of interest with the lowest percentile ranks are chosen for peptide synthesis. Specificity for MHC I or MHC II was predicted via the length of the synthesized peptides (8 to 11-mers for MHC I and 13 to 17-mers for MHC II).

TABLE-US-00006 TABLE 3 Name Sequence ZEBOV-GP C57BL/6 MHC-I 1 FEVDNLTIV (SEQ ID NO: 1) ZEBOV-GP C57BL/6 MHC-I 2 MASENSSAM (SEQ ID NO: 2) ZEBOV-GP C57BL/6 MHC-I 3 FSILNRKAI (SEQ ID NO: 3) ZEBOV-GP C57BL/6 MHC-I 4 DNLTIVQL (SEQ ID NO: 4) ZEBOV-GP C57BL/6 MHC-I 5 AFFLYDRL (SEQ ID NO: 5) ZEBOV-GP C57BL/6 MHC-I 6 ELRTFSIL (SEQ ID NO: 6) ZEBOV-GP C57BL/6 MHC-II 1 GLAWIPYFGPAAEGI (SEQ ID NO: 7) ZEBOV-GP C57BL/6 MHC-II 2 KRWGFRSGVPPKVVN (SEQ ID NO: 8) ZEBOV-NP C57BL/6 MHC-I 1 YQVNNLEEI (SEQ ID NO: 9) ZEBOV-NP C57BL/6 MHC-I 2 ARLLNLSGV (SEQ ID NO: 10) ZEBOV-NP C57BL/6 MHC-I 3 DGVKRLEEL (SEQ ID NO: 11) ZEBOV-NP C57BL/6 MHC-II 1 EVNSFAKALSSLAKH (SEQ ID NO: 12) ZEBOV-NP C57BL/6 MHC-II 2 TSNRTTPVAPPAPVY (SEQ ID NO: 13) CCHFV M/PreGc + TM MHC-I 1 ISGLKFAGL (SEQ ID NO: 14) CCHFV M/PreGc + TM MHC-I 2 SVKSFFYGL (SEQ ID NO: 15) CCHFV M/PreGc + TM MHC-I 3 GIFTFLTPF (SEQ ID NO: 16) CCHFV M/PreGc + TM MHC-I 4 FMFGWRILF (SEQ ID NO: 17) CCHFV M/PreGc + TM MHC-II 1 VQSTYKPTVSTANIA (SEQ ID NO: 18) CCHFV M/PreGc + TM MHC-II 2 DSTFKAFSAMPKTSI (SEQ ID NO: 19) CCHFV M/PreGc + TM MHC-II 3 FIGIFTFLTPFILLI (SEQ ID NO: 20) CCHFV M/PreGc + TM MHC-II 4 YTKTFHFHSKRVTAAH (SEQ ID NO: 21) CCHFV M/PreGc + TM MHC-II 5 DPDVVAASSSLMARK (SEQ ID NO: 22) CCHFV PreGc + TM Balb/c I-1 MYSPVFEYL (SQ ID NO: 23) CCHFV PreGc + TM Balb/c I-2 KLPPEIITL (SQ ID NO: 24) CCHFV PreGc + TM Balb/c I-3 TPFILLILF (SQ ID NO: 25) CCHFV PreGc + TM Balb/c I-4 GYRKIERL (SQ ID NO: 26) CCHFV PreGc + TM Balb/c I-5 TYGAGEITV (SQ ID NO: 27) CCHFV PreGc + TM Balb/c I-6 TYKPTVSTA (SQ ID NO: 28) CCHFV PreGc + TM Balb/c II-1 FDLMHVQKVLASTV (SQ ID NO: 29) CCHFV PreGc + TM Balb/c II-2 LFCFKCCRTRTGL (SQ ID NO: 30) CCHFV PreGc + TM Balb/c II-3 PVDVVAASSSLMA (SQ ID NO: 31) CCHFV PreGc + TM Balb/c II-5 TDYMFVKWKVEYIKTE (SQ ID NO: 32) CCHFV PreGc + TM Balb/c II-6 CPYEALVLRRLPLF (SQ ID NO: 33) CCHFV-NP Balb/c MHC-I-1 SYTELKVEV (SQ ID NO: 34) CCHFV-NP Balb/c MHC-I-2 VPNLDRFVF (SQ ID NO: 35) CCHFV-NP Balb/c MHC-I-3 FPTVSQFLF (SQ ID NO: 36) CCHFV-NP Balb/c MHC-I-4 SPFQNAVNV (SQ ID NO: 37) CCHFV-NP Balb/c MHC-I-5 VPGEIVMSV (SQ ID NO: 38) CCHFV-NP Balb/c MHC-I-6 KHIKAQEL (SQ ID NO: 39) CCHFV-NP Balb/c MHC-II-1 GTKKMKKALLSTPM (SQ ID NO: 40) CCHFV-NP Balb/c MHC-II-2 ELYKNSSALRAQGAQ (SQ ID NO: 41) CCHFV-NP Balb/c MHC-II-3 EHVEWCREFVKGKY (SQ ID NO: 42) CCHFV-NP Balb/c MHC-II-4 ALKWRKDIGFRVNA (SQ ID NO: 43) CCHFV-NP Balb/c MHC-II-5 MLSDMIRRRNLIL (SQ ID NO: 44) CCHFV-NP Balb/c MHC-II-6 EMNKWFEEFKKGNG (SQ ID NO: 45) MERS-NP Balb/c MHC-I-1 LYLDLLNRL (SQ ID NO: 46) MERS-NP Balb/c MHC-I-2 NYNKLWLELL (SQ ID NO: 47) MERS-NP Balb/c MHC-I-3 TGPEAALPF (SQ ID NO: 48) MERS-NP Balb/c MHC-I-4 AAPNNTVSW (SQ ID NO: 49) MERS-NP Balb/c MHC-I-5 PGDLQGNF (SQ ID NO: 50) MERS-NP Balb/c MHC-I-6 YKTFPKKEKK (SQ ID NO: 51) MERS-NP Balb/c MHC-II-1 NAGYWRRQDRKIN (SQ ID NO: 52) MERS-NP Balb/c MHC-II-2 VQGSITQRTRTRPS (SQ ID NO: 53) MERS-NP Balb/c MHC-II-3 AAANKMRHKRTSTK (SQ ID NO: 54) MERS-NP Balb/c MHC-II-4 DPRWPQIAELAPTA (SQ ID NO: 55) MERS-NP Balb/c MHC-II-5 RWYFYTTGTGPEA (SQ ID NO: 56) MERS-Spike Balb/c MHC-I-1 TYQNIISTNL (SQ ID NO: 57) MERS-Spike Balb/c MHC-I-2 YYSIIPHSI (SQ ID NO: 58) MERS-Spike Balb/c MHC-I-3 SYINKCSRL (SQ ID NO: 59) MERS-Spike Balb/c MHC-I-4 TYGPLQTPV (SQ ID NO: 60) MERS-Spike Balb/c MHC-I-5 YPLSMKSDL (SQ ID NO: 61) MERS-Spike Balb/c MHC-I-6 QYVAGYKVL (SQ ID NO: 62) MERS-Spike Balb/c MHC-I-7 YAPEPITSL (SQ ID NO: 63) MERS-Spike Balb/c MHC-II-1 QLVRSESAALSAQLA (SQ ID NO: 64) MERS-Spike Balb/c MHC-II-2 YEMLSLQVVKALNE (SQ ID NO: 65) MERS-Spike Balb/c MHC-II-3 TFFDKTWPRPIDVSK (SQ ID NO: 66) MERS-Spike Balb/c MHC-II-4 KAWAAFYVYKLQPLT (SQ ID NO: 67) MERS-Spike Balb/c MHC-II-5 STRSMLKRRDSTYG (SQ ID NO: 68) MERS-Spike Balb/c MHC-II-6 DFTCSQISPAAIASN (SQ ID NO: 69)

ELISpot Analysis

[0583] Splenocytes were isolated on day 49 or day 70 and ELISpot analysis was performed using the Mabtech Mouse IFN-γ ELISpotPLUS kit. Splenocytes were seeded to pre-coated ELISpot plates and stimulated with the indicated peptide pools overnight in a humidified incubator at 37° C. The respective peptide pools were composed of overlapping peptides spanning the whole GP protein of EBOV divided into two pools for analysis (overlapping 15-mers). Control measurements were performed using an irrelevant peptide pool, medium only or Concanavalin A. Spots were visualized with a biotin-conjugated anti-IFNγ antibody followed by incubation with streptavidin-alkaline phosphatase (ALP) and 5-bromo-4-chloro-3-indolyl-phosphate/nitroblue tetrazolium (BCIP/NBT) substrate. Plates were scanned using a CTL ImmunoSpot® Analyzer and analyzed by ImmunoCapture V6.3 software. All tests were performed in triplicate and spot counts were summarized as median values for each triplicate.

Quantitative Real Time RT-PCR Analysis of Virus Load in Mouse Tissue Samples

[0584] Tissue samples from liver and spleen of immunized and challenged mice were homogenized in 1 ml DMEM. RNA isolation was performed

with the RNeasy Mini Kit (Qiagen) according to the manufacturer's instructions. The RNA amount was measured using the NanoDrop ND-100 spectrophotometer. Total RNA was reverse transcribed and quantified by real-time PCR using the OneStep RT-PCR kit (Qiagen) with the primer pair EboGP1D-fwd (TGGGCTGAAAAYTGCTACAATC; Y=CMT) (SEQ ID NO:70) and EboGP1D-rev (CTTTGTGMACATASCGGCAC; M=A/C; S=G/C) (SEQ ID NO:71) and the probe Ebo1 DZ-Prb (6FAM-TTACCCCCACCgCcGgATg-BHQ1) (SEQ ID NO:72) on a StepOne high-throughput fast real-time PCR system (ThermoFisher). Quantification was carried out using a standard curve based on 10-fold serial dilutions of control RNA. Statistical Analysis

[0585] GraphPad Prism 8 Software (La Jolla, USA) was used for statistical analysis and figure generation. All groups were compared by a one-way ANOVA test with Tukey's multiple comparison post-test on each measurement day (VNT, ELISA, ELISpot).

Example 1: Analysis of Different Immunization Schemes in Relation to Antibody Production and Durability

[0586] Different prime/boost intervals were tested in order to find the optimal interval for generating large amounts of antibodies directed against EBOV GP or NP. Mice were vaccinated with saRNAs that code for GP (5 µg) or NP (2.5 µg). The effects of the immunization regimen while maintaining the dosages described, were analyzed. For this purpose, mice were formulated in polyplexes (PLXs) and immunized using a prime/boost scheme. The immunization was carried out on day 0 followed by either day 21 or by day 35. From previous studies it was known that boosting the immune response too early after an saRNA immunization could decrease the antibody response instead of increasing it (data not shown). The antibody responses were analyzed by ELISA using the same amount of recombinant protein coated on 96-well plates. The analysis of the EBOV GP-specific antibodies revealed a trend towards decreased antibody levels at later times when the booster took place on day 21 (FIG. 1C, top panel). This effect was not observed when using the longer interval of 35 days between the two immunizations. A further increase in GP-specific antibodies was found here (FIG. 1D, top panel). For NP, both immunization schemes did not result in homogeneous antibody production in all animals, and antibody levels decreased towards the end of experiment regardless of when the boost was given (FIGS. 1C and 1D bottom panels).

Example 2: EBOV GP and NP saRNAs Complexed in Lipid Nanoparticles (LNP) Induce Strong Immune Responses

[0587] To achieve optimal B-cell and T-cell mediated immune responses directed against different proteins of EBOV, mice were vaccinated with a combination of two saRNAs. In addition to the polyplex-formulation (showing unexpected big particle sized in FIG. 1B), nanoparticle-based formulation was used to deliver the EBOV-specific saRNAs. Mice received two doses (d0 followed by d35) of the LNP-formulated saRNAs encoding GP (5 µg) and NP (2.5 µg) in a ratio of GP:NP of 2:1 or with GP (5 µg) saRNA alone but added up to the final dose with a replicase-only encoding saRNA used as filler (2.5 µg) formulated by LNPs (GP+filler). The total amount of RNA administered intramuscularly was always 7.5 µg. The analysis of serum samples at different timepoints after immunization showed a strong immune response against both EBOV proteins when saRNAs were formulated with LNPs as shown by detecting high amounts of antibodies against GP and NP by ELISA (FIG. 2C) and high concentrations of GP- and NP-specific total IgG when quantified against a standard curve with known IgG concentrations (FIG. 2D). The same applies to the amount of neutralizing antibodies monitored against authentic EBOV (FIG. 2E). Using LNPs, EBOV-GP-specific antibodies with neutralizing capacity were detected in mice vaccinated with a combination of GP+NP saRNAs and in mice vaccinated with GP+filler saRNA. At the end of the experiment, the mice were sacrificed and their spleens resected to analyze the induced T-cell response after saRNA immunization by means of IFN γ ELISpot assays (FIG. 2F). CD4 T-cell response against GP and CD8 T cell response against NP were detected for all groups. Strong CD4 T-cell response against GP and CD8 T-cell response against NP were detected, indicating that the addition of NP strengthens the T-cell response, by activating CD8 positive T-cells, that GP alone was not able to induce. It should be noted that PLX formulation with the aforementioned saRNA showed significantly bigger size and dispersity (FIG. 1B), compare to saRNA PLX described in the literature, hence this could strongly correlate with the lower performance observed in the depicted experiments. Interestingly, there was no interference or competition observed between GP and NP saRNA, as the GP-specific antibody titers obtained after the immunization using GP and NP are comparable with the control group using GP and the irrelevant filler RNA instead of NP (FIGS. 2C and D).

Example 3: Intramuscular Vaccination of EBOV-Specific saRNAs is Superior to Intradermal Vaccination

[0588] The immunogenicity of the saRNAs with regard to the route of administration and the dose of the vaccine was investigated. There are some reports that vaccination using intradermal injection improves the immune response because the dermis is a tissue rich in immune cells. Therefore, the questions addressed were if a reduction of dosage would be feasible for the intramuscular route and if the immunogenicity could be increased by administering the saRNAs via the intradermal route. The GP:NP ratios were kept constant at 2:1. The total amount of saRNA was either 7.5 µg (high dose for i.m. and i.d.) as before or was reduced to 1.5 µg (low dose, i.m. only). The results of the ELISA demonstrated that there were no significant differences in the GP-specific IgG antibody titers between the high-dose and the low-dose groups, which received 5 µg or 1 µg of saRNA-GP respectively (FIG. 3C, left graph). When analyzing the neutralizing antibody titers, it turned out that these were reduced after vaccination with the low dose of GP (FIG. 3E). In contrast, 0.5 µg of the NP saRNA was insufficient to reach the antibody titers of the 2.5 µg group (FIG. 3C, right graph). Vaccination via the intradermal route induced lower levels of binding antibodies directed against GP and NP, and the neutralizing titers were also reduced (FIG. 3C to 3E). The dose effect becomes clear by calculating the total IgG concentrations induced by the different components of the LNP vaccine at the end of experiment (d50; FIG. 3D).

[0589] To further investigate, the T-cell induction after LNP-saRNA GP/NP combination, IFN γ ELISpot analyses have been performed using MHC-I or MHC-II specific peptides against both viral proteins (FIG. 3F). The previously observed pattern that EBOV GP mainly led to MHC-II/CD4.sup.+ specific and NP mainly to MHC-I/CD8.sup.+ specific responses could be reproduced and was independent of the route of administration of the vaccine. Interestingly, when the EBOV GP saRNA was administered via the intradermal route, a slightly higher CD4.sup.+ specific T-cell response was observed, compared to the intramuscular route, whereas the NP-induced CD8+-specific T-cell responses were lower. The results show a good induction of T-cell-specific responses which can be relevant for the induction of protective immune responses.

Example 4: The saRNA Vaccine Against EBOV GP and NP Protects Against Challenge Infection with the Ebola Virus

[0590] To investigate whether the induced humoral and cellular immune responses offer protection against EBOV infection, an efficacy study was performed. For this purpose, mice were immunized intramuscular with a prime/boost regimen (35 day interval) using the previously determined high dose of the saRNAs (5 µg GP, 2.5 µg NP). The LNP-formulated saRNAs for GP or NP were administered alone or in combination. The control animals received an empty replicase construct which does not code for an additional antigen following the replicase ORF. In order to vaccinate all experimental groups with the same total dose of saRNA, the groups that received individual EBOV antigens were filled up with the empty saRNA. The antibody responses before infection were analyzed using ELISA and neutralization tests. These analyses revealed that the GP-specific IgG levels were comparable between the groups that received only GP and GP+NP, while NP-specific IgG levels were significantly increased when the NP saRNA was administered together with GP (FIGS. 4C and 4D). The neutralization of EBOV was only observed by sera from mice vaccinated with the GP saRNA, but not with the NP saRNA alone (FIG. 4E).

[0591] The mice were infected with EBOV on day 56 after primary vaccination and monitored for 2 weeks. Control mice and mice that received only the NP saRNA lost weight from day 5 post infection with EBOV. The termination criteria of the experiment stipulated that the animals of both groups had to be euthanized between the 7th and 9th day after infection because of excessive weight loss. In contrast, mice that received only the GP saRNA and the combination of GP+NP did not lose weight and were protected from fatal EBOV infection (FIG. 4F).

[0592] Based on the analysis of serum samples at 5d p.i. and at 14d p.i. by means of plaque titration it was found that, in contrast to the NP and the control group, no infectious virus was detected in groups immunized with the GP saRNA or a combination of both saRNAs (FIG. 4G). In addition, EBOV-specific genome copies were detected in the liver and spleen of control animals and the NP group but not in the other two groups (FIG. 4H).

[0593] Taken together, the data show that immunization with the EBOV-specific GP saRNA alone or in combination with NP in a prime/boost regimen conferred protection against the lethal challenge with EBOV. Immunization with the NP saRNA alone did not protect mice from EBOV

infection indicating that the NP-induced CD8^{sup.}+ specific cellular immune response observed was not sufficient to achieve protection.

Example 5: A Single Dose of an saRNA Vaccine Protects Against Infection with EBOV

[0594] It was analyzed whether the EBOV-specific saRNA vaccine can elicit single-shot efficacy. For this purpose, C57Bl/6 IFNAR^{sup.}-/- mice were immunized only once either with the GP saRNA alone or in combination with NP as LNP-formulated saRNAs. The schedule for immunization and infection was adjusted so that mice were infected on d21 after the primary vaccination and not as before on d56 after the primary vaccination and d21 after the second vaccination. The antibody response at d14 post vaccination was analyzed using ELISA and it was found that nearly no protein-specific antibodies were detected in the GP only group and only very low titers were found in the GP+NP group at this early timepoint (FIG. 5C). Further, no virus neutralization titers were detected (FIG. 5D). Surprisingly, the combination of GP and NP saRNAs within LNPs still conferred protection against the lethal EBOV challenge in all animals (FIG. 5E). For the GP only group more weight loss was observed and one animal had to be sacrificed. Nevertheless, also in this group five from six animals survived the infection with EBOV. Plaque titration revealed that no infectious EBOV was detected in the serum samples from all vaccinated mice after challenge infection (FIG. 5F). In addition, EBOV-specific genome copies were detected in the liver and spleen of control animals and but to a much lesser extent in the GP only group but not in the GP plus NP group (FIG. 5G).

[0595] Taken together, these data demonstrate the high potency of saRNA-based vaccines that confer protection early after only a single dose of the vaccine was administered and although only low levels of antibodies were detected. This most likely results from a complex immune response combining neutralizing antibodies targeting GP and T-cell responses against the EBOV proteins GP and NP, as the combinatory approach seemed to be superior.

Example 6: Changing the Virus System to Further Analyze the Immune Response Induced after Vaccination with Lipid Nanoparticles (LNP) Formulating a Combination of saRNAs

[0596] To investigate if the effects observed in the described ZEBOV experiments are virus and antigen-specific, the immunogens have been changed and instead of ZEBOV antigens Crimean congo hemorrhagic fever virus (CCHFV) related antigens have been produced as saRNA, formulated using the same LNPs and vaccinated into mice. Again, the glycoprotein together with the NP have been chosen for vaccination. Highly comparable to the ZEBOV-GP in terms of size, only the cytoplasmic region of the CCHFV glycoprotein was used together with the transmembrane domain of the protein (Gc+TM). Experiments using the full M segment showed that Gc+TM induces similar antibodies (not shown).

[0597] A prime-boost scenario was again used for immunization at d0 and 28 and Gc+TM as well as NP-specific IgG was determined using protein-specific ELISA from serum samples (FIG. 68) together with IFN γ ELISpot analyses of splenocytes at the end of experiment (FIG. 6C). Again, it was possible to induce high antibody titers against both proteins, already starting early after first immunization (d14), but there was a clear drop of Gc+TM-specific antibodies observed after addition of NP saRNA to the vaccination (FIG. 68, top panels). In terms of induced T cells, the glycoprotein of CCHFV induced mostly MHC-I/CD8⁺ T cells whereas the NP induced mostly MHC-II/CD4^{sup.}+ T cells (FIG. 6C).

Example 7: Titration Ratios of saRNAs to Induce Best Combination of B- and T-Cell Responses after Prime-Boost Immunization Using LNP-Formulated saRNAs Encoding for CCHFV Proteins

[0598] Using different ratios of CCHFV Gc+TM mixed with CCHFV NP for the formulation within LNPs (ratio 1:1; 1:3; 3:1) it was evaluated which combination induced best antibody responses against using the protein-specific ELISA method. The ratio Gc:NP of 3:1 achieved highest GP-specific IgGs together with only marginal affected NP-specific IgGs (FIG. 78). This ratio also induced highest Gc specific MHC-I/CD8⁺ as well as highest MHC-II/CD4⁺ T-cells (FIG. 7C, top panel). For NP, the ratio using higher NP amounts would induce higher titers, but as antibodies against Gc are the first line in protection.

Example 8: Combination of CCHFV Gc+TM and CCHFV NP Using the Trans-Amplifying RNA System in a Second Mouse Model

[0599] To investigate if the effects observed in the saRNA studies described with EBOV and CCHFV antigens also apply to another replicating RNA platform, trans-replicons encoding CCHFV-Gc+TM or CCHFV NP as well as non-replicating VEEV replicase mRNA have been produced and formulated as LNP.

[0600] Mice were immunized in a prime-boost scenario at d0 and 28 with LNP-formulated VEEV replicase mRNA and CCHFV-Gc+TM or -NP encoding TR (1:1 molar ratio) or a combination of Gc+TM and NP TR in different ratios together with VEEV replicase mRNA (0.5:0.5:1, 0.25:0.75:1 or 0.75:0.25:1). Replication-deficient replicase (def. rep) together with both TR was used as negative control. CCHFV Gc- and NP-specific IgG was determined using protein-specific ELISA from serum samples together with IFN γ ELISpot analysis of splenocytes at d49. As observed for saRNA, TR immunization induced antibodies against both antigens (FIG. 8A). Addition of NP did not interfere with Gc-antibody induction and vice versa, however there was a trend towards lower Gc-specific antibodies with an excess of NP and vice versa. Vaccination induced Gc-specific MHC-I/CD8^{sup.}+ T cells and MHC-II/CD4^{sup.}+ T cells (FIG. 8B), while for NP only a very strong MHC-I/CD8^{sup.}+ T cell response was detected. Addition of Gc+TM reduced NP-specific CD8^{sup.}+ T cells with the 0.5:0.5:1 and 0.75:0.25:1 ratio, but not with an excess of NP using the 0.25:0.75:1 ratio.

Example 9: Vaccination with MERS-CoV Spike and NP Encoding TR Induced B and T Cell Responses Against Both Antigens

[0601] To investigate whether the effects observed in the TR study with CCHFV antigens can be transferred to other antigens, TR encoding the glycoprotein of MERS-CoV, MERS-CoV S, and NP were produced and formulated as LNP. BALB/c mice were immunized i.m. In a prime-boost scenario at d0 and 28 with LNP-formulated VEEV replicase mRNA and MERS-CoV S or -NP encoding TR (1:1 molar ratio) or a combination of S and NP TR in different ratios together with VEEV replicase mRNA (0.5:0.5:1, 0.25:0.75:1 or 0.75:0.25:1). Replication-deficient replicase (def. rep) together with both TR was used as negative control. MERS-CoV S1 and NP-specific IgG was determined using protein-specific ELISA from serum samples together with IFN γ ELISpot analysis of splenocytes at d49. As for CCHFV antigens, TR immunization induced antibodies against both antigens (FIG. 9A). T cell responses were predominantly MHC-I/CD8^{sup.}+ T cells against S and NP (FIG. 98), with a higher T cell induction observed for NP.

Example 10: Combination of EBOV GP and EBOV NP Using Trans-Amplifying RNA

[0602] To investigate another combination of antigens in the TR system, EBOV GP and NP encoding TR were produced and formulated as LNP. BALB/c mice were immunized i.m. in a prime-boost scenario at d0 and 28 with LNP-formulated VEEV replicase mRNA and EBOV GP or -NP encoding TR (1:1 molar ratio) or a combination of GP and NP TR in different ratios together with VEEV replicase mRNA (0.5:0.5:1, 0.66:0.33:1). Replication-deficient replicase (def. rep) together with both TR was used as negative control. EBOV GP and NP-specific IgG was determined using protein-specific ELISA from serum samples. As observed for CCHFV and MERS-CoV antigens, TR vaccination induced antibodies against both GP and NP (FIG. 10) with the trend towards lower NP-specific antibodies with an excess of GP.

Example 11: Vaccination with Four Different Antigens from Two Viruses Induced Antibodies and T Cells Against all Antigens

[0603] One advantage of the trans-amplifying RNA system is its use as platform for multivalent vaccines. We combined previously identified combinations for GP and NP of CCHFV and MERS in one vaccination as a proof of concept for a tetravalent vaccine. BALB/c mice were vaccinated with LNP-formulated CCHFV Gc+TM and CCHFV NP TR, MERS-CoV S and NP TR, or a combination of all four TR, together with VEEV replicase mRNA. For the tetravalent combination, 1 μ g CCHFV antigens was combined with either 0.1 μ g or 1 μ g MERS antigens. CCHFV Gc and NP-specific IgG as well as MERS-CoV S1 and NP-specific IgG was determined using protein-specific ELISA from serum samples together with IFN γ ELISpot analysis of splenocytes at d49. Vaccination induced antibodies against all four antigens (FIG. 11A). Addition of MERS-CoV antigens did not interfere with antibody responses against CCHFV antigens. However, MERS S1-specific antibodies were reduced in combination with CCHFV antigens in the 9:9:1:1:20 ratio, but could be restored by increasing the the MERS-CoV antigen dose to 1 μ g (1:1:1:4 ratio). Similarly, CCHFV-specific T cell responses (FIG. 11 B) were not influenced by addition of MERS-CoV antigens. Similar to antibody results, MERS-CoV S-

specific MHC-I/CD_{sup}.+ T cells were reduced after addition of CCHFV antigens for the 9:9:1:1:20 ratio, but no reduction was observed with the higher MERS-CoV antigen levels (1:1:1:1:4 ratio). Increasing the amount of replicase (1:1:1:1:8 ratio) resulted in increased MERS-CoV-S MHC-II/CD_{sup}.+ T cells and MERS-CoV-NP MHC-I/CD_{sup}.+ T cells.

Example 12: Confirmation of Immunogenicity of a Tetravalent Trans-Amplifying RNA Vaccine Candidate In a Second Mouse Model

[0604] To identify whether trans-amplifying RNA is a widely applicable vaccine platform for multivalent vaccines, the immunogenicity of the previously tested tetravalent trans-amplifying RNA vaccine candidate composed of CCHFV Gc+TM and NP and MERS S and NP encoding TRs as well as non-replicating VEEV replicase mRNA was confirmed in a second mouse strain. B6.129S2-Irfar1tm1Agt/Mmjax mice were vaccinated with LNP-formulated CCHFV Gc+TM and CCHFV NP TRs, or a combination of CCHFV Gc+TM, CCHFV NP and MERS-CoV S and NP TRs, together with VEEV replicase mRNA. CCHFV Gc and NP-specific IgG as well as MERS-CoV S1 and NP-specific IgG was determined using protein-specific ELISA from serum samples together with CCHFV-specific IFN γ ELISpot analysis of splenocytes at d49. As observed in BALB/c mice (FIG. 11), tetravalent vaccination with trans-amplifying RNA induced antibodies against all four antigens in B6.129S2-Irfar1tm1Agt/Mmjax mice and addition of MERS-CoV antigens did not interfere with CCHFV-specific antibody responses (FIG. 12A). Similarly, CCHFV-specific T cell responses were not influenced by addition of MERS-CoV antigens (FIG. 12B).

Discussion

[0605] The examples demonstrate that self-amplifying RNA (saRNA) vaccines as well as trans-amplifying RNA vaccines protect against infection, when a combination of different viral antigens is formulated together or formulated separately and mixed afterwards. Different administration routes, different dosing regimens and different formulations all induce an immune response. The combination of a glycoprotein (or part thereof) and a nucleoprotein results in general in an improved immune response.

Claims

1. A composition comprising: at least two replicable RNA molecules, each comprising a first open-reading frame (ORF) encoding at least one peptide or protein comprising an antigen or epitope suitable to induce an immune response against the antigen or epitope when administered to a subject; wherein the at least one peptide or protein encoded by one of the replicable RNA molecules is different from the at least one peptide or protein encoded by the other replicable RNA molecule, optionally wherein at least one of the replicable RNA molecules further comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules.
2. The composition according to claim 1, wherein one or both of the at least two replicable RNA molecules comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules.
3. The composition according to claim 1 or 2, wherein the composition further comprises a third RNA molecule encoding the replicase capable of replicating in cis or in trans the replicable RNA molecules and/or the third RNA molecule.
4. The composition according to claim 3, wherein the third RNA molecule is replicable.
5. The composition according to claim 3, wherein the third RNA molecule is a non-replicative RNA molecule.
6. The composition according to any one of claims 1 to 5, wherein the composition comprises the at least two replicable RNA molecules, each of which does not encode the replicase, and a third non-replicative RNA molecule encoding the replicase.
7. The composition according to claim 5 or 6, wherein the non-replicative RNA is a mRNA.
8. The composition according to any one of claims 1 to 7, wherein the replicable RNA molecule comprises an internal ribosome entry site (IRES) which controls expression of the first ORF encoding the protein or peptide comprising an antigen or epitope.
9. The composition according to any one of claims 1 to 8, wherein the replicable RNA molecule comprises an internal ribosome entry site (IRES) which controls expression of the second ORF encoding the replicase.
10. The composition according to claim 8 or 9, wherein the IRES is insensitive to cellular stress.
11. The composition according to any one of claims 8 to 10, wherein the IRES is insensitive to interferons, preferably type I interferons.
12. The composition according to any one of claims 8 to 11, wherein the IRES is a cellular or viral IRES, preferably a viral IRES.
13. The composition according to any one of claims 8 to 12, wherein the IRES is derived from viruses selected from the group consisting of picornaviruses, flaviviruses or dicistroviruses.
14. The composition according to any one of claims 8 to 13, wherein the IRES is derived from a picornavirus or a dicistrovirus, preferably a dicistrovirus.
15. The composition according to any one of claims 8 to 14, wherein the IRES is a type IV IRES.
16. The composition according to any one of claims 8 to 15, wherein expression controlled by the IRES is independent of IRES trans-acting factors.
17. The composition according to any one of claims 8 to 16, wherein expression controlled by the IRES is independent of cellular translation initiation factors.
18. The composition according to any one of claims 8 to 17, wherein expression controlled by the IRES is independent of phosphorylation of eukaryotic initiation factor 2 (eIF2).
19. The composition according to any one of claims 1 to 18, wherein both replicable RNA molecules comprise a second ORF encoding the replicase.
20. The composition according to any one of claims 1 to 19, wherein at least one of the replicable RNA molecules comprises a 5' cap for driving translation of the replicase or for driving translation of the peptide or protein comprising an antigen or epitope.
21. The composition according to claim 20, wherein the 5' cap is a natural 5' cap or a 5' cap analog.
22. The composition according to any one of claims 1 to 21, wherein at least one replicable RNA comprises a 5' replication recognition sequence which is characterized in that at least one initiation codon is removed compared to a native alphavirus 5' replication recognition sequence.
23. The composition according to claim 22, wherein the 5' replication recognition sequence comprises a sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus, wherein the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus is characterized in that it comprises the removal of at least one initiation codon compared to the native viral sequence.
24. The composition according to claim 23, wherein the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus is characterized in that it comprises the removal of at least the native start codon of the open reading frame of a non-structural protein from a self-replicating virus.
25. The composition according to claim 23 or 24, wherein the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus is characterized in that it comprises the removal of at least one initiation codon other than the native start codon of the open reading frame of a non-structural protein from a self-replicating virus.
26. The composition according to any one of claims 22 to 25, wherein the sequence homologous to an open reading frame of a non-structural protein or a portion thereof from a self-replicating virus is characterized in that it is free of initiation codons.
27. The composition according to any one of claims 22 to 26, which comprises at least one nucleotide change compensating for nucleotide pairing disruptions within at least one stem loop introduced by the removal of at least one initiation codon.
28. The composition according to any one of claims 22 to 27, wherein the open reading frame encoding a functional non-structural protein from a self-replicating virus does not overlap with the 5' replication recognition sequence.
29. The composition according to any one of claims 8 to 28, wherein the first ORF is downstream from the 5' replication recognition sequence and

upstream from the IRES.

30. The composition according to any one of claims 1 to 29, which replicable RNA comprises a subgenomic promotor controlling production of subgenomic RNA comprising the first ORF encoding the protein or peptide.
31. The composition according to claim 30, wherein the subgenomic RNA is a transcription product of an RNA-dependent RNA polymerase derived from the functional non-structural protein from a self-replicating virus.
32. The composition according to claim 30 or 31, wherein the protein or peptide can be expressed from the subgenomic RNA as a template.
33. The composition according to any one of claims 30 to 32, wherein the first ORF encoding the protein or peptide controlled by the subgenomic promotor is downstream from the second ORF encoding the replicase.
34. The composition according to claim 33, wherein the subgenomic promotor overlaps with the second ORF.
35. The composition according to any one of claims 1 to 34, wherein at least one of the replicable RNA molecules comprises a 3' replication recognition sequence.
36. The composition according to claim 35, wherein the second ORF encoding the replicase, the 5' and/or 3' replication recognition sequences and the subgenomic promotor are derived from a self-replicating virus, preferably the same self-replicating virus species.
37. The composition according to any one of claims 1 to 36, wherein the replicable RNA molecules can be replicated by an RNA-dependent RNA polymerase derived from the functional non-structural protein from a self-replicating virus.
38. The composition according to claim 37, wherein the self-replicating virus is an alphavirus, preferably selected from the group consisting of Venezuelan equine encephalitis complex viruses, Eastern equine encephalitis complex viruses, Western equine encephalitis complex viruses, Chikungunya virus, Semliki Forest virus complex viruses, Sindbis virus, Barmah Forest virus, Middelburg virus and Ndumu virus.
39. The composition according to claim 37 or 38, wherein the alphavirus is a Venezuelan equine encephalitis virus or Semliki Forest virus.
40. The composition according to any one of claims 1 to 39, wherein at least one of the replicable RNA molecules comprises a 3' poly(A) sequence.
41. The composition according to any one of claims 1 to 40, wherein the antigen or epitope of the encoded protein or peptide is a or is derived from a bacterial, viral, parasitological or fungal antigen.
42. The composition according to any one of claims 1 to 41, wherein the protein or peptide encoded by the first replicable RNA and the protein or peptide encoded by the second replicable RNA are both obtained or derived from the same bacterium, virus, parasite or fungus.
43. The composition according to any one of claims 1 to 42, wherein the protein or peptide encoded by the first replicable RNA and the protein or peptide encoded by the second replicable RNA are obtained or derived from different strains of the same bacterium, virus, parasite or fungus, respectively or are obtained or derived from different pathogenic organisms, for example, different viruses.
44. The composition according to any one of claims 1 to 43, wherein the protein or peptide encoded by the first replicable RNA is a surface expressed protein or peptide and wherein the protein or peptide encoded by the second replicable RNA is not a surface expressed protein or peptide, which surface expressed and non-surface expressed proteins or peptides are obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or are obtained or derived from different pathogenic organisms, for example, different viruses.
45. The composition according to any one of claims 1 to 43, wherein the protein or peptide encoded by the first replicable RNA is not a surface expressed protein or peptide and wherein the protein or peptide encoded by the second replicable RNA is not a surface expressed protein or peptide and is different from that encoded by the first replicable RNA, which different non-surface expressed proteins or peptides are obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or are obtained or derived from different pathogenic organisms, for example, different viruses.
46. The composition according to any one of claims 1 to 43, wherein the protein or peptide encoded by the first replicable RNA is a surface expressed protein or peptide, and wherein the protein or peptide encoded by the second replicable RNA is a surface expressed protein or peptide and is different from that encoded by the first replicable RNA, which surface expressed proteins or peptides are obtained or derived from the same or from different strains of the same bacterium, virus, parasite or fungus, respectively or are obtained or derived from different pathogenic organisms, for example, different viruses.
47. The composition according to claim 44 or 46, wherein the surface expressed protein is expressed on the surface of a virus/viral particle or wherein, where the virus is an enveloped virus, the surface expressed protein is expressed on the surface of the viral envelope.
48. The composition according to claim 47, wherein the surface expressed protein is a viral capsid protein or a viral envelope or glycoprotein.
49. The composition according to claim 44 or 45, wherein the protein or peptide that is not a surface expressed protein or peptide is a viral matrix protein, a viral nucleoprotein, or a viral capsid protein where the virus is an enveloped virus.
50. The composition according to any one of claims 1 to 49, wherein the protein or peptide encoded by the first replicable RNA is a viral glycoprotein and the protein or peptide encoded by the second replicable RNA is a viral nucleoprotein, wherein the glycoprotein and the nucleoprotein are obtained or derived from the same virus, optionally from the same strain of the same virus.
51. The composition according to any one of claims 1 to 50, wherein the induced immune response against the antigens or epitopes is an increase in the activity of CD4⁺ T cells and/or CD8⁺ T cells, preferably an increase in the activity of both CD4⁺ and CD8⁺ T cells.
52. The composition according to any one of claims 1 to 51, wherein the protein or peptide encoded by the first replicable RNA is an Ebola virus protein or fragment thereof or an epitope of the Ebola virus protein, and the protein or peptide encoded by the second replicable RNA is a different Ebola virus protein or fragment thereof or an epitope of the different Ebola virus protein.
53. The composition according to any one of claims 1 to 52, wherein the protein or peptide encoded by at least one replicable RNA molecule is a structural Ebola virus protein selected from the group consisting of glycoprotein (GP), nucleoprotein (NP), polymerase cofactor (VP35), VP40, transcription factor (VP30), VP24 or RNA-dependent RNA polymerase (L), or a fragment thereof or an epitope of the Ebola structural protein.
54. The composition according to any one of claims 1 to 53, wherein the protein or peptide encoded by at least one replicable RNA molecule is expressed as a fusion protein.
55. The composition according to claim 54, wherein the protein or peptide is fused to a targeting or secretory motif.
56. The composition according to any one of claims 1 to 55, wherein the protein or peptide encoded by at least one of the replicable RNA molecules is the Ebola structural GP protein or a fragment thereof, or an epitope of the GP protein.
57. The composition according to claim 56, wherein the GP protein is derived or obtained from the Ebola subtype Zaire, virus strain *H. sapiens*-wt/SLE/2014/Makona-EM095B.
58. The composition according to any one of claims 1 to 57, wherein the protein or peptide encoded by at least one of the replicable RNA molecules is the Ebola structural NP protein or a fragment thereof, or an epitope of the NP protein.
59. The composition according to claim 58, wherein the NP antigen is derived or obtained from the Ebola subtype Zaire, virus strain *H. sapiens*-wt/GIN/2014/Makona-EM096.
60. The composition according to any one of claims 1 to 59, wherein the protein or peptide encoded by the first replicable RNA molecule is the Ebola structural GP protein or a fragment thereof, or an epitope of the GP protein, and the protein or peptide encoded by the second replicable RNA molecule is the Ebola structural NP protein or a fragment thereof, or an epitope of the NP protein.
61. The composition according to any one of claims 1 to 51, wherein the protein or peptide encoded by the first replicable RNA is an CCHFV virus protein or fragment thereof or an epitope of the CCHFV virus protein, and the protein or peptide encoded by the second replicable RNA is a different CCHFV virus protein or fragment thereof or an epitope of the different CCHFV virus protein.
62. The composition according to any one of claims 1 to 51, wherein the protein or peptide encoded by the first replicable RNA is an MERS-CoV

virus protein or fragment thereof or an epitope of the MERS-CoV virus protein, and the protein or peptide encoded by the second replicable RNA is a different MERS-CoV virus protein or fragment thereof or an epitope of the different MERS-CoV virus protein.

63. The composition according to any one of claims 1 to 62, wherein the first and/or second ORF is flanked by a 5' untranslated region (UTR) and/or 3' UTR, preferably wherein said 5' UTR and/or 3' UTR is/are not native to the alphavirus from which the replicase is derived.

64. The composition according to any one of claims 1 to 63, wherein at least one of the replicable RNA molecules does not comprise an open reading frame for an intact alphavirus structural protein.

65. The composition according to any one of claims 1 to 64 further comprising a reagent capable of forming particles with the replicable RNA molecules.

66. The composition according to claim 65, wherein the reagent is a lipid or polyalkyleneimine.

67. The composition according to claim 65 or 66, wherein the reagent is a lipid comprising a cationic headgroup.

68. The composition according to any one of claims 65 to 67, wherein the reagent is a pH responsive lipid.

69. The composition according to any one of claims 65 to 68, wherein the reagent is a PEGylated-lipid.

70. The composition according to any one of claims 65 to 69, wherein the reagent is conjugated to polysarcosine.

71. The composition according to any one of claims 65 to 70, wherein the particles formed from the replicable RNA molecules and the reagent are polymer-based polyplexes (PLX) or lipid nanoparticles (LNP), wherein the LNP is preferably a lipoplex (LPX) or a liposome.

72. The composition according to any one of claims 65 to 71, wherein the particle further comprises at least one phosphatidylserine.

73. The composition according to any one of claims 65 to 72, wherein the particles are nanoparticles, in which: (i) the number of positive charges in the nanoparticles does not exceed the number of negative charges in the nanoparticles and/or (ii) the nanoparticles have a neutral or net negative charge and/or (iii) the charge ratio of positive charges to negative charges in the nanoparticles is 1.4:1 or less and/or (iv) the zeta potential of the nanoparticles is 0 or less.

74. The composition according to claim 73, wherein the charge ratio of positive charges to negative charges in the nanoparticles is between 1.4:1 and 1:8, preferably between 1.2:1 and 1:4.

75. The composition according to claim 73 or 74, wherein the nanoparticles comprise at least one lipid, preferably comprise at least one cationic lipid.

76. The composition according to claim 75, wherein the positive charges are contributed by the at least one cationic lipid and the negative charges are contributed by the replicable RNA molecules.

77. The composition according to claim 75 or 76, wherein the nanoparticles further comprise at least one helper lipid.

78. The composition according to claim 77, wherein the helper lipid is a neutral lipid.

79. The composition according to any one of claims 75 to 78, wherein the at least one cationic lipid comprises 1,2-di-1-octadecenyl-3-trimethylammonium propane (DOTMA), 1,2-dioleoyl-3-dimethylaminopropane (DODMA), and/or 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP).

80. The composition according to any one of claims 77 to 79, wherein the at least one helper lipid comprises 1,2-di-(9Z-octadecenoyl)-n-glycero-3-phosphoethanolamine (DOPE), cholesterol (Chol), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), and/or 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC).

81. The composition according to any one of claims 77 to 80, wherein the molar ratio of the at least one cationic lipid to the at least one helper lipid is from 10:0 to 3:7, preferably 9:1 to 3:7, 4:1 to 1:2, 4:1 to 2:3, 7:3 to 1:1, or 2:1 to 1:1, preferably about 1:1.

82. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DODMA and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

83. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DODMA and Cholesterol in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

84. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DODMA and DSPC in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

85. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DODMA:Cholesterol:DOPE:PEGcerC16 in a molar ratio of 40:48:10:2.

86. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DOTMA and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

87. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DOTMA and Cholesterol in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

88. The composition according to any one of claims 71 to 81, wherein the nanoparticles are lipoplexes comprising DOTAP and DOPE in a molar ratio of 10:0 to 1:9, preferably 8:2 to 3:7, and more preferably of 7:3 to 5:5 and wherein the charge ratio of positive charges in DOTMA to negative charges in the RNA is 1.8:2 to 0.8:2, more preferably 1.6:2 to 1:2, even more preferably 1.4:2 to 1.1:2 and even more preferably about 1.2:2.

89. The composition according to any one of claims 65 to 88, wherein the reagent comprises a lipid and the particles formed are LNPs which are complexed with and/or encapsulate the replicable RNA molecules.

90. The composition according to any one of claims 65 to 88, wherein the reagent comprises a lipid and the particles formed are vesicles encapsulating the replicable RNA molecules, preferably unilamellar liposomes.

91. The composition according to claim 65 or 66, wherein the reagent is polyalkyleneimine.

92. The composition according to claim 91, wherein the molar ratio of the number of nitrogen atoms (N) in the polyalkyleneimine to the number of phosphorus atoms (P) in the replicable RNA molecules (N:P ratio) is 2.0 to 15.0, preferably 6.0 to 12.0.

93. The composition according to claim 91 or 92, wherein the ionic strength of the composition is 50 mM or less, preferably wherein the concentration of monovalent cationic ions is 25 mM or less and the concentration of divalent cationic ions is 20 μ M or less.

94. The composition according to any one of claims 91 to 93, wherein the particles formed are polyplexes.

95. The composition according to any one of claims 91 to 94, wherein the polyalkyleneimine comprises the following general formula (I):

##STR00068## wherein R is H, an acyl group or a group comprising the following general formula (II): ##STR00069## wherein R.sub.1 is H or a group comprising the following general formula (III): ##STR00070## n, m, and l are independently selected from integers from 2 to 10; and p, q, and r are integers, wherein the sum of p, q, and r is such that the average molecular weight of the polymer is 1.5.Math.10.sup.2 to 10.sup.7 Da, preferably 5000 to 10.sup.5 Da, more preferably 10000 to 40000 Da, more preferably 15000 to 30000 Da, even more preferably 20000 to 25000 Da.

96. The composition according to claim 95, wherein n, m, and l are independently selected from 2, 3, 4, and 5, preferably from 2 and 3.

97. The composition according to claim 95 or 96, wherein R.sub.1 is H.

98. The composition according to any one of claims 95 to 97, wherein R is H or an acyl group.

99. The composition according to any one of claims 95 to 98, wherein the polyalkyleneimine comprises polyethylenimine and/or polypropylenimine,

preferably polyethyleneimine.

- 100.** The composition according to any one of claims 95 to 99, wherein at least 92% of the N atoms in the polyalkyleneimine are protonatable.
 - 101.** The composition according to any one of claims 1 to 100 further comprising one or more peptide-based adjuvants, wherein peptide-based adjuvants optionally comprise immune regulatory molecules, such as cytokines, lymphokines and/or co-stimulatory molecules.
 - 102.** The composition according to any one of claims 1 to 101 further comprising one or more additives, wherein the additives optionally are selected from the group consisting of buffering substances, saccharides, stabilizers, cryoprotectants, lyoprotectants, and chelating agents.
 - 103.** The composition according to claim 102, wherein the buffering substances comprise at least one selected from the group consisting of 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), 2-(N-morpholino)ethanesulfonic acid (MES), 3-morpholino-2-hydroxypropanesulfonic acid (MOPSO), acetic acid, acetate buffers and analogues, phosphoric acid and phosphate buffers, and citric acid and citrate buffers.
 - 104.** The composition according to claim 102 or 103, wherein the saccharides comprise at least one selected from the group consisting of monosaccharides, disaccharides, trisaccharides, oligosaccharides, and polysaccharides preferably from glucose, trehalose, and saccharose.
 - 105.** The composition according to any one of claims 102 to 104, wherein the cryoprotectants comprise at least one selected from the group consisting of glycols, such as ethylene glycol, propylene glycol, and glycerol.
 - 106.** The composition according to any one of claims 102 to 105, wherein the chelating agent comprises EDTA.
 - 107.** The composition according to any one of claims 1 to 106, wherein the composition is a vaccine.
 - 108.** A pharmaceutical composition comprising the composition according to any one of claims 1 to 107, and a pharmaceutically acceptable carrier.
 - 109.** The pharmaceutical composition according to claim 108, which is formulated for intradermal, subcutaneous, and/or intramuscular administration, such as by injection.
 - 110.** The pharmaceutical composition according to claim 108 or 109 for use in therapy, such as inducing an immune response or vaccination.
 - 111.** The pharmaceutical composition according to claim 108 or 109 for use in a method for inducing an immune response specific for the encoded proteins or peptides in a subject, preferably wherein the subject is a mammal, more preferably wherein the mammal is a human, said method comprising administering the pharmaceutical composition according to claim 93 or 94 to the subject.
 - 112.** The pharmaceutical composition for use according to claim 111, wherein administering the pharmaceutical composition comprises intradermal, subcutaneous, or intramuscular administration, such as by intradermal, subcutaneous or intramuscular injection.
 - 113.** The pharmaceutical composition for use according to claim 112, wherein the injection is by use of a needle or is by use of a needleless injection device.
 - 114.** The pharmaceutical composition for use according to any one of claims 111 to 113, wherein administering comprises administration by intramuscular injection, preferably with a needle.
 - 115.** A method for inducing an immune response specific for at least two antigens or epitopes in a subject comprising administering the pharmaceutical composition according to claim 108 or 109 to the subject, preferably wherein the subject is a mammal, more preferably wherein the mammal is a human.
 - 116.** The pharmaceutical composition for use according to claim 114 or the method according to claim 115, wherein the immune response comprises the activation of T cells and/or B cells, preferably wherein the activated T cells comprise T helper cells and cytotoxic T cells.
 - 117.** The pharmaceutical composition for use according to claim 114 or the method according to claim 115, wherein the immune response comprises activation of antigen specific T helper cells, optionally wherein the T helper cells proliferate, release T cell cytokines, mediate the growth and/or activation of antigen specific cytotoxic T cells.
 - 118.** The pharmaceutical composition for use according to claim 114 or the method according to claim 115, wherein the immune response comprises activation of antigen specific T helper cells, wherein the T helper cells stimulate B cell proliferation, antibody class switching, production and/or secretion of neutralizing antibodies.
 - 119.** A method for producing at least two proteins or peptides of interest in a cell comprising, inoculating the pharmaceutical composition according to claim 108 or 109 into the cell.
 - 120.** A method for producing at least two proteins or peptides of interest in a subject comprising administering the pharmaceutical composition according to claim 108 or 109 to the subject.
 - 121.** A method for the treatment or prevention of a bacterial, viral, parasitical or fungal infection in a subject, said method comprising administering to the subject a composition comprising: at least two replicable RNA molecules, each comprising a first open-reading frame (ORF) encoding at least one peptide or protein comprising an antigen or epitope suitable to induce an immune response against the bacterium, virus, parasite or fungus, respectively; wherein the at least one peptide or protein encoded by one of the replicable RNA molecules is different from the at least one peptide or protein encoded by the other replicable RNA molecule, optionally wherein at least one of the replicable RNA molecules further comprises a second ORF encoding an RNA-dependent RNA polymerase (replicase) capable of replicating in cis or in trans the replicable RNA molecules.
 - 122.** The method according to claim 121, wherein the composition further comprises a third RNA molecule encoding the replicase capable of replicating in cis or in trans the replicable RNA molecules and/or the third RNA molecule.
 - 123.** The method according to claim 121, wherein the immune response is a specific immune response against the bacterium, virus, parasite or fungus, respectively.
 - 124.** The method according to claim 121 or 123, wherein the immune response lessens the severity of one or more symptoms of the infection.
 - 125.** The method according to any one of claims 121 to 124, wherein the method involves only a single administration of the composition.
 - 126.** The method according to any one of claims 121 to 124, wherein the method comprises multiple administrations of the composition.
 - 127.** The method according to any one of claims 121 to 126, further comprising administering a booster dose of the composition.
 - 128.** The method according to any one of claims 121 to 127, wherein the infection is a viral infection, optionally wherein the infection is an Ebola virus infection.
 - 129.** The method according to any one of claims 115 and 120 to 128, wherein administering the composition comprises intradermal, subcutaneous, or intramuscular administration, such as by intradermal, subcutaneous or intramuscular injection.
 - 130.** The method according to claim 129, wherein the injection is by use of a needle or is by use of a needleless injection device.
 - 131.** The method according to any one of claims 115 and 120 to 128, wherein administering comprises administration by intramuscular injection, preferably with a needle.
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